ESE 3203 Complete Solutions

Petroleum and Natural Gas Processing · KUET · past papers 2019, 2020, 2021, 2022, 2022B, 2023 · answers from the teachers' slides (Shameem sir: refining; Muhit sir: natural gas) and the textbooks they follow (Rao, Fahim et al., Sanjay Kumar, Guo & Ghalambor, Ahmed)

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 3 h for Part A and 3 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 ...) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×6, 2022B ×5, 2022 ×5, 2021 ×3, 2020 ×2, 2019 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer first (30 minutes). This course is almost all theory with diagrams, and the same questions repeat word for word: desalting and distillation (A3), reforming reactions (A4), delayed coking and FCC (A5), hydrotreating (A6), the amine process (B5), the LNG plant (B6) and separators (B3). Learn the memory tricks and practise each diagram once. The short numericals (API gravity, viscosity blend, VI, porosity, Darcy flow) are easy marks and each has a calculator.

Section A: Petroleum refining

Block 1: very likely and easy · 1.2 h

Topic	Chance	Level	Covered
A1 Crude oil: what it is, composition, classification, impurities, why and how it is refined	Very likely 100%	Easy	19%
A6 Hydrotreating, hydrogen in the refinery, lube oil processing	Very likely 95%	Easy	30%
A2 Product properties and test numbers: octane and cetane number, API gravity, flash point, RVP, gasoline grades and pool, diesel classes	Very likely 86%	Easy	41%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
A3 Distillation curves (TBP, ASTM, EFV), desalting, atmospheric and vacuum distillation	Very likely 100%	Medium	60%
A4 Catalytic reforming and isomerization	Very likely 100%	Medium	76%

✓ Pass line: 76% of Section A marks covered, enough to choose 3 of the 4 questions (about 2.8 h)

A5 Cracking and coking: thermal cracking, hydrocracking, delayed and fluid coking, visbreaking, FCC	Very likely 86%	Medium	89%

Block 4: likely · 0.8 h

Topic	Chance	Level	Covered
A7 Product blending: objectives, blending indices (RVP, viscosity, flash point), viscosity index numericals	Likely 77%	Medium	100%

Section B: Natural gas processing

Block 1: very likely and easy · 0.4 h

Topic	Chance	Level	Covered
B5 Gas sweetening: reasons for removing H_2S and CO_2, amine process, iron sponge, molecular sieve, sulfur recovery	Very likely 95%	Easy	8%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
B6 LNG and LPG: LNG plant and refrigeration cycles, storage and carriers, LPG uses and hazards, LPG vs LNG vs CNG, NGL recovery	Very likely 100%	Medium	32%
B1 Natural gas: what it is, origin, classification, condensate, reservoirs, Bangladesh outlook	Very likely 100%	Medium	52%
B3 Gas-liquid separators: vertical, horizontal, double-barrel, spherical, three-phase; internals; sizing	Very likely 100%	Medium	68%
B4 Water in natural gas: dehydration, water content (Campbell), hydrates	Very likely 100%	Medium	82%

✔ Pass line: 82% of Section B marks covered, enough to choose 3 of the 4 questions (about 3.4 h)

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B2 Gas properties and rock properties: z-factor, pseudo-reduced properties, density, B_g, compressibility, viscosity, porosity and permeability	Very likely 95%	Hard	92%

Block 4: likely · 0.8 h

Topic	Chance	Level	Covered
B7 Reservoir flow and gas measurement: Darcy's law (linear and radial), orifice meter, choosing a meter	Likely 73%	Medium	100%

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★ One-shot notes

One-shot notes: ESE 3203 Petroleum and Natural Gas Processing

Crude oil (A1)

Fossil fuel from buried plankton/algae + heat + pressure. C 83-87, H 10-14, S 0.05-6, N 0.1-2, O 0.1-1.5 wt%; metals Ni, V; salts NaCl, $MgCl_2$, $CaCl_2$.

Families: paraffin C_nH_{2n+2} (chain), naphthene C_nH_{2n} (ring), aromatic (benzene ring); olefins only from cracking.

Classify: P/N/A/asphalt base; Mallison wax > 5% paraffinic, < 2% naphthenic; USBM key fractions 250-275 °C (atm) and 275-300 °C (40 mmHg); API > 20 conventional, < 20 heavy, < 10 bitumen; sweet < 0.5% S.

TAN = mg KOH/g (> 1 corrosive). $K_w = T_B^{1/3}/G$ (°R): 12.9 paraffinic, 10.5 naphthenic. BPSD = full-rate day; BPCD = yearly average.

Refining scheme

Separate (desalt, ADU, VDU) → convert (crack, reform, isomerize, alkylate) → treat (hydrotreat, sweeten, dewax) → blend. Complexity = conversion capacity / distillation capacity.

Energy engineer: audits, heat integration (preheat train, pinch), furnace/boiler efficiency, steam and condensate, CHP, flare gas recovery, emissions.

Fuel numbers (A2)

Octane: % iso-octane (100) in n-heptane (0) knocking the same. RON 600 rpm city, MON 900 rpm highway, sensitivity RON − MON, PON = average. Grades: regular 85-88, mid 88-90, premium > 90.

Cetane: % n-cetane (100) in α-methylnaphthalene (0) with same ignition delay; diesel 40-55. Diesel index = aniline pt (°F) × API/100.

API = 141.5/SG − 131.5: 0.802 → 44.9; 0.821 → 40.9. Flash point = lowest T vapour flashes. RVP at 37.8 °C, 4:1 air.

Octane requirement: −1 RON per 100 m altitude, per +300 rpm, per +10% humidity (0.5); +1 per +12 °C air, +6 °C coolant, 5,000-10,000 km deposits.

Pool: LSR, isomerate, reformate, FCC (38%), HC, alkylate, polymer, n-butane, oxygenates (MTBE, ETBE, TAME, ethanol: high octane, cleaner burn).

Distillation curves and desalting (A3)

TBP: 15 plates, reflux 5, atm then 40 mmHg, crude assay. ASTM D86/D1160: Engler flask, no reflux, products. EFV: flash, vapour stays with liquid, design. Sharpness TBP > ASTM > EFV.

Desalt if > 10 lb/1000 bbl: wash water 3-10% + demulsifier, heat 90-150 °C (~120), electric field 1000 V/cm (secondary) and 600 V/cm (primary), settle (Stokes). Variables: T, P, water ratio, level, injection point, demulsifier rate, water type, mixing valve Δp.

ADU and VDU (A3)

ADU: preheat 280 °C → furnace 350-400 °C → 30-50 trays, ~50 m; gas/LPG, naphtha 30-180, kerosene 180-240, diesel 240-350, residue > 350 °C; side strippers, pump-around, bottom steam.

VDU: residue would crack at > 370 °C, so 25-40 mmHg; furnace 390-450 °C; packing, wide (up to 12 m); steam ejectors (3 stages); LVGO, HVGO, lube cuts, vacuum residue. Wet/damp/dry operation.

Reforming and isomerization (A4)

Heavy naphtha RON 50-60 → reformate 90-104 + H_2. Pt 0.2-0.6% on Cl-alumina (bifunctional).

Reactions: naphthene dehydrogenation (fast, very endothermic), dehydroisomerization, paraffin dehydrocyclization (slow), isomerization (mild exo), hydrocracking (exo, bad), coke.

SR: 3-4 fixed beds + reheaters, R-1 ΔT biggest. SR 12-15 bar Pt-Re 12-24 months RON 92-100; CCR 3-10 bar Pt-Sn continuous RON 98-105, capex ×1.3.

Poisons: permanent (Pb, As, Hg, Cu, Na), coke (semi), S, N, water (temporary).

Isom.: C5-80 °C naphtha, 95-205 °C, +12-20 RON; nC5 62 → iC5 92; nC6 26 → 2,2-DMB 92-96. 2019 4c: feed 724,000 kg/h; C_3 2,520, iC_4 4,482, nC_4 12,815, C_5^+ 704,170 kg/h; isomerate RON ≈ 82.

Cracking (A5)

Thermal: heat only, free radical, olefinic.
Hydrocracking: zeolite/silica-alumina + Ni/W/Pt, high H_2 pressure, saturated S-free products, net exothermic.

Delayed coker: residue → fractionator bottom → furnace ~500 °C → coke drum (cracks there), two drums switched 16-24 h, vapours back to fractionator, coke cut by 2000-4500 psig water jets.

Visbreaking: mild, cuts viscosity/pour point. Coil 474-500 °C, quench at furnace exit; soaker 427-438 °C + soaking drum. Reactions: side chains off rings, resins → light + asphaltenes, ring cracking > 480 °C.

FCC: VGO + hot zeolite in riser 480-540 °C, 2-4 s; cyclones; spent (coked) catalyst steam-stripped → regenerator 650-815 °C burns coke → hot catalyst back (heat balance). Fixed bed = cyclic, obsolete.

Hydrotreating, H_2, lube (A6)

Feed + H_2 → heater → reactor Co-Mo/Ni-Mo on Al_2O_3, 290-400 °C, 10-100 bar → cooler → HP separator → amine scrubber, H_2 recycle → stripper.

$RSH + H_2 \rightarrow RH + H_2S$; pyridine + $5H_2 \rightarrow C_5H_{12} + NH_3$.

H_2 from reformer, steam reforming $CH_4 + H_2O \rightarrow CO + 3H_2$, shift, partial oxidation, off-gas recovery.

Lube: vacuum cuts → propane deasphalting → furfural/NMP extraction (aromatics out, VI up) → MEK dewaxing (pour pt down) → hydrofinishing (colour, stability) → additives. Lube: friction, wear, cooling, cleaning, sealing, corrosion.

Blending (A7)

Objective: meet specs and demand at least cost, max profit. Linear: SG, S, BP. Indices: $RVP^{1.25}$; flash log BI = −6.1188 + 2414/(FP − 42.6); pour $PP^{1/0.08}$.

Viscosity: BI = logv/(3 + logv), $v = 10^{3BI/(1-BI)}$.
75/100/200 cSt at 2:3:4 → BI 0.4117 → **125.8 cSt**; 72/100/201 → **124.8 cSt**.

VI = 100(L − U)/(L − H) = 100(120 − 73.3)/50 = **93.4** (table L, H give 92.4).

Natural gas and condensate (B1)

Mostly CH_4 + C_2-C_4 + N_2, CO_2, H_2S, He. Dry/wet, sweet/sour, associated/non-associated, conventional/unconventional (shale, tight, CBM, hydrates).

Origin: thermogenic (most accepted; kerogen → oil → gas with depth), biogenic (bacteria), abiogenic. Trap under cap rock.

Condensate: C_5^+ that drops out below dew point.

Inlet sep → cooling/LTS → stabilizer → splitter (LPG, naphtha, kero, gas oil).

Flare not vent: CH_4 28-34× CO_2 warming. Clean fuel: high H/C, no S, ash.

Phase envelope: bubble line, dew line, critical pt, cricondenbar (max p), cricondentherm (max T), retrograde zone. Volatile oil near T_c, GOR 2000-3300, B_o > 2.

Gas properties (B2)

pV = znRT; R = 10.73; $\gamma_g = M_a/28.97$; Kay $p_{pc} = \Sigma y p_c$. $\rho = pM/(zRT)$. $\mathbf{B_g = 0.02827zT/p}$ ft^3/scf. $c_g = 1/p - (1/z)dz/dp$.

2020: M 17.745, γ 0.6125, p_{pr} 2.96, T_{pr} 1.84, ρ 5.45, B_g 0.00858.

2021 (sour, Σy = 0.991): M 22.97, p_{pc} 680.9, T_{pc} 409.8; W-A ε 11.2 → 661.5 / 398.6; p_{pr} 2.87, T_{pr} 1.61, **z ≈ 0.83**; μ_1 0.0116 + corrections = **0.0118 cp**.

Porosity = void/bulk: 35/83 = 42.2%; 25/68 = 36.8%. Permeability = flow ability (md).

Separators (B3)

Inlet diverter (momentum), settling (gravity), mist extractor (coalescence), liquid section, LC + PC valves. LTS uses JT cooling, 0-20 °F.

Vertical: slugs, sand, small footprint; costly.
Horizontal: smaller, cheaper, high GOR, half full; poor with solids. Double barrel: liquid in lower barrel, no re-entrainment. Sphere: cheapest, little surge.
Three-phase: weir for oil, interface control for water.

De-foaming plates break foam; vortex breaker stops gas pulled into liquid outlet; centrifugal diverter; wave breakers damp sloshing.

2023 6c: ρ_g 3.845; $D^2 = 0.463/K$; V_L = 0.583 bbl.

Vertical 20 in × 6 ft; horizontal 26 in × 6.5 ft; sphere ~34 in (1/4 full; 27 in if half full).

Water and hydrates (B4)

Dehydrate: hydrates, slugging, corrosion, lower heating value; spec 6-8 lb/MMscf. Water content ↓ with p, ↑ with T, ↓ with salinity and gravity.

Campbell: W = 0.89×59 + 0.025×67 + 0.085×150 = **67 lb/MMscf**.

Hydrates: water cages + gas; need free water, low T, high p, agitation, H_2S/CO_2. Prevent: dehydrate, heat, lower p, methanol/MEG, KHI (PVP), anti-agglomerants.

Desiccant: sour → molecular sieve; cheap → alumina; 12 ppm → silica gel. Adsorption: 2 towers, down-flow drying, up-flow hot regen 230-315 °C. Glycol: TEG contactor, reboiler 175-205 °C.

Sweetening (B5)

H_2S: > 600 ppm fatal, corrosive, SO_2, catalyst poison. CO_2: no heating value, carbonic acid, freezes in LNG. Both promote hydrates.

Amine (MEA, DEA, MDEA): contactor ~38 °C high P → flash → lean/rich HX → regenerator 115-125 °C low P → acid gas to Claus.

Iron sponge: Fe_2O_3 on wood chips, < 300 ppm H_2S, pH 8-10, < 47 °C; $Fe_2O_3 + 3H_2S \rightarrow Fe_2S_3 + 3H_2O$; regen $2Fe_2S_3 + 3O_2 \rightarrow 2Fe_2O_3 + 6S$. Keeps CO_2. 3% CO_2 + 200 ppm H_2S, low budget → iron sponge.

Claus: $H_2S + 1.5O_2 \rightarrow SO_2 + H_2O$; $SO_2 + 2H_2S \rightarrow 3S + 2H_2O$.

LNG and LPG (B6)

LNG: −162 °C, 1/600 volume. Plant: inlet sep → acid gas → dehydration (< 1 ppm) → Hg removal → NGL recovery → liquefaction → N_2 rejection → storage/ship.

Cascade: propane −35, ethylene −100, methane −160 °C; each condenses the next (no single fluid spans ambient to −162).

Carrier: double hull; Moss spheres (Al, 9% Ni) or membrane (Invar); boil-off as fuel.

LPG: C_3/C_4 (BD 30/70), heavier than air. Hazards: leaks/explosion, poor storage, unattended cooking, suffocation, BLEVE. LPG Rules 2004, Explosives Act.

LPG liquid 4-15 bar; LNG −162 °C; CNG gas 200-250 bar. NGL: turbo-expander, de-ethanizer, de-propanizer, de-butanizer. Kettle: bundle under weir, vapour space above.

Reservoir flow and meters (B7)

Linear: $q = 0.001127kA\Delta p/(\mu L)$ = $0.001127\times100\times6000\times10/(2\times2000)$ = **1.69 bbl/d**; v = 0.00158 ft/d; actual v/φ = 0.0105 ft/d.

Radial: $p = 1800 + 88.28 \ln(r/0.25)$; r_e = 745 ft. Δp: 142.1, 19.7, 4.53, 0.89, 0.12 psi.

Orifice: $q = C_dA_o\sqrt{(2\Delta p/\rho(1 - \beta^4))}$, vena contracta.

Meter choice: accuracy, range, fluid, pressure loss, size, cost, output, standards.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
ADU	atmospheric distillation unit
AGA	American Gas Association
API	American Petroleum Institute
ASTM	American Society for Testing and Materials
bbl	barrel (42 US gallons, 5.615 ft^3)
BLEVE	boiling liquid expanding vapour explosion
BPCD	barrels per calendar day
BPSD	barrels per stream day
CBM	coal-bed methane
CCR	continuous catalyst regeneration
CDU	crude distillation unit
CNG	compressed natural gas
cp	centipoise
cSt	centistokes
DEA	diethanolamine
EFV	equilibrium flash vaporization
ETBE	ethyl tertiary-butyl ether
FCC	fluid catalytic cracking
FSRU	floating storage and regasification unit
GLR	gas-liquid ratio
GOR	gas-oil ratio
HDN	hydrodenitrogenation
HDS	hydrodesulfurization
HHV	higher heating value
KHI	kinetic hydrate inhibitor
LNG	liquefied natural gas
LPG	liquefied petroleum gas
LSR	light straight-run (naphtha)
LTS	low-temperature separator
md	millidarcy
MDEA	methyldiethanolamine
MEA	monoethanolamine
MEK	methyl ethyl ketone
MMscfd	million standard cubic feet per day
MON	motor octane number
MTBE	methyl tertiary-butyl ether

NGC	natural gas condensate
NGL	natural gas liquids
NMP	N-methyl pyrrolidone
PON	posted (pump) octane number
PSA	pressure swing adsorption
psia	pounds per square inch absolute
psig	pounds per square inch gauge
RON	research octane number
RVP	Reid vapour pressure
scf	standard cubic foot
SR	semi-regenerative (reformer); also straight-run
STB	stock-tank barrel
TAME	tertiary-amyl methyl ether
TAN	total acid number
TBP	true boiling point
TEG	triethylene glycol
VDU	vacuum distillation unit
VI	viscosity index

Section A: Petroleum refining

A1 Crude oil: what it is, composition, classification, impurities, why and how it is refined Pass set Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q1(a)** [6] What is crude oil? Explain its classification based on composition.
- **2023 Q1(c)** [7] What are the key properties and characteristics of petroleum that need to be evaluated?
- **2022B Q1(a)** [10] Briefly discuss the composition of petroleum with figures of commonly found hydrocarbon compounds in crude oil.
- **2022B Q1(b)** [6] Why does crude oil require refining, and what are the essential processes involved?
- **2022 Q1(a)** [10] Briefly describe crude oil compositions. What are the impurities associated with a crude oil?
- **2022 Q1(c)** [7] Explain briefly why crude oil needs to be refined and how this is done.
- **2021 Q5(a)** [15] Why is petroleum called a "fossil fuel"? Briefly discuss the composition of petroleum with figures of commonly found hydrocarbon compounds.
- **2021 Q5(b)** [8] What is meant by "heat capacity"? Why is dehydration needed in crude oil refining?
- **2021 Q5(c)** [12] Adverse effects of sulfur compounds in refineries; test methods for sulfur; types of stabilizers for crude oil stabilization.
- **2021 Q8(c)** [15] Describe the function of an energy engineer in a petroleum refinery.
- **2020 Q1(a)** [10] Classify and briefly describe the basic types of processes and operations in petroleum refining.
- **2020 Q1(b)** [10] Explain: BPSD (barrels per stream day), BPCD (barrels per calendar day), refinery complexity, °API, K_w.
- **2019 Q1(a)** [15] Briefly describe crude oil compositions. What are the impurities associated with crude oil? Give examples of each.
- **2019 Q3(a)** [9] Describe in brief: total acid number, TBP (true boiling point) distillation, natural gas condensate.

LEARN IT FROM ZERO

Crude oil and refining, from zero

Crude oil is a mixture of thousands of hydrocarbons (molecules of only carbon and hydrogen) formed from ancient sea plankton buried and heated for millions of years: that is why it is a **fossil** fuel. Four families: **paraffins** (straight or branched chains, C_nH_{2n+2}), **naphthenes** (rings, C_nH_{2n}), **aromatics** (benzene rings) and, in products only, **olefins** (double bonds, made by cracking). Impurities: sulfur, nitrogen, oxygen compounds, metals (Ni, V), salt and water.

Crude as it comes out of the ground is useless in an engine. A refinery does four kinds of job: **separate** (distillation by boiling range), **convert** (crack big molecules into small, or reshape small ones into high-octane ones), **treat** (remove sulfur and other impurities with hydrogen) and **blend** (mix streams to meet product specifications).

★ REMEMBER IT

- Impurities: **S-N-O-M-W**: Sulfur (H_2S, mercaptans: corrosion, SO_2, catalyst poison), Nitrogen, Oxygen (naphthenic acids, high TAN (total acid number)), Metals (Ni, V poison FCC (fluid catalytic cracking) catalyst), Water and salt (corrosion, fouling).
- BPSD = barrels per **stream** day (when running); BPCD = barrels per **calendar** day (yearly average, smaller). $K_W = (T_B)^{1/3}/SG$: 12.5+ paraffinic, 10–11 aromatic/naphthenic.

Before you start: what petroleum is

Petroleum (Latin *petra* = rock, *oleum* = oil) is a naturally occurring yellowish-black liquid, mainly hydrocarbons (molecules of only carbon and hydrogen), found in rock formations underground. "Crude oil" is petroleum as it comes out of the ground, before any processing.

Every answer in this topic builds on one picture: crude is a mixture of thousands of molecules of different **size** (boiling point) and **shape** (chain, ring, benzene ring), plus a few unwanted atoms (S, N, O, metals, salt, water). A refinery sorts the molecules by size, changes their shape, removes the unwanted atoms and blends the result. Source: Shameem sir's slides "Introduction" and "Refinery raw materials" (which follow Rao, *Modern Petroleum Refining Processes*).

Why petroleum is called a fossil fuel (2021 5a)

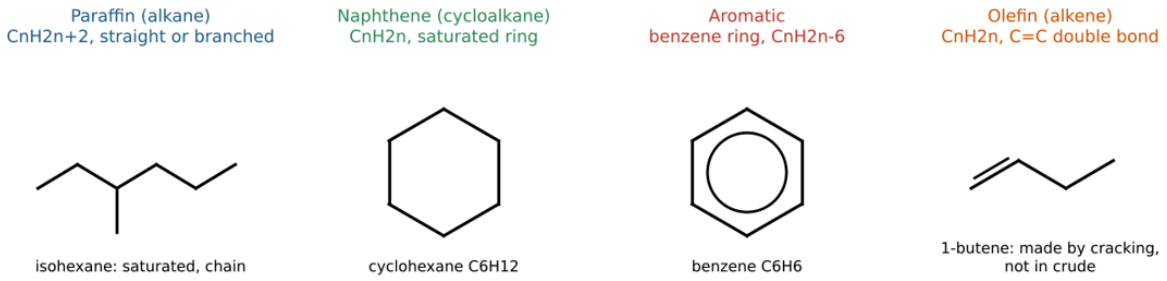
Petroleum formed from the **fossilized remains of dead organisms**, mostly zooplankton and algae, that were buried under sedimentary rock and subjected to prolonged heat and pressure over millions of years. Because it comes from ancient (fossil) life and is burnt for energy, it is a **fossil fuel**. It takes millions of years to form and is being used up much faster, so it is **non-renewable**.

Crude oil and its composition (2023 1a, 2022 1a, 2022B 1a, 2021 5a, 2019 1a)

Crude oil is the unprocessed oil that comes out of the ground: a very complex mixture of hydrocarbons with small amounts of compounds of sulfur, nitrogen, oxygen and metals. Colour ranges from clear to tar-black and viscosity from water-like to almost solid. Carbon and hydrogen make up about 98% of it. The elemental composition falls in narrow ranges whatever the source:

Element	wt %	Present as
Carbon	83-87	hydrocarbons
Hydrogen	10-14	hydrocarbons
Sulfur	0.05-6	H_2S, mercaptans (R-SH), sulfides, disulfides, thiophenes, elemental S
Nitrogen	0.1 to < 2	basic (pyridine) and non-basic (pyrrole) compounds
Oxygen	0.1-1.5	carboxylic (naphthenic) acids, phenols, ketones, CO_2
Metals	< 0.1	Ni, V, Fe, Cu, As (organometallic porphyrins)
Salts	< 1	NaCl, $MgCl_2$, $CaCl_2$ dissolved in water droplets

Hydrocarbon compounds commonly found (draw the four structures):



The four hydrocarbon families: paraffin, naphthene, aromatic, olefin

Family	Formula and structure	Notes
Paraffins (alkanes)	C_nH_{2n+2}; saturated straight (normal) or branched (iso) chains, e.g. methane CH_4, n-hexane, isooctane	chemically most stable; natural crude has mainly straight chains, refinery processes make branched ones
Naphthenes (cycloalkanes)	C_nH_{2n}; saturated rings of 5 or 6 carbons, e.g. cyclopentane, cyclohexane	higher boiling point and density than paraffins of the same carbon number; good reformer feed (become aromatics)
Aromatics	one or more benzene rings, e.g. benzene C_6H_6, toluene, xylene; fused rings = polynuclear aromatics	raise gasoline octane; heavy polynuclear aromatics cause coke and catalyst deactivation
Olefins (alkenes)	C_nH_{2n} with a C=C double bond, e.g. ethylene, propylene	**not present in crude**; formed by cracking; reactive, form gums on storage

Side note: The hydrogen-to-carbon ratio decides the physical properties: as H/C falls, density and boiling point rise. A feed with high H/C is worth more because it needs less hydrogen in processing.

Impurities in crude oil and examples (2022 1a, 2019 1a)

Impurity	Examples	Why it is a problem
Sulfur compounds	H_2S, mercaptans (thiols) R-SH, sulfides R-S-R, disulfides R-S-S-R, thiophenes	corrosion, bad odour, SO_2 on burning, poisons catalysts (see 2021 5c below)
Nitrogen compounds	pyridine (basic), pyrrole (non-basic), amines	poison cracking catalysts, form gums; harder to remove than S; crude above 0.25 wt% N needs N removal
Oxygen compounds	naphthenic (carboxylic) acids, phenols, ketones, esters	acidity → corrosion of distillation columns (measured as TAN)
Metals	Ni, V, Fe, Cu, As (porphyrins)	deactivate hydrotreating and cracking catalysts; concentrate in the residue
Salts	NaCl, $MgCl_2$, $CaCl_2$	$MgCl_2$ and $CaCl_2$ hydrolyse to HCl at 150-200 °C → overhead corrosion; deposits foul furnace tubes; poison FCC catalyst
Water and sediment	emulsified water; clay, sand, rust, FeS	carries salt, causes foaming and furnace upsets

Classification of crude oil (2023 1a)

1. **By predominant hydrocarbon (chemical composition):** paraffin base (saturated chains, no rings), naphthene base (saturated rings), aromatic base (benzene rings), and asphalt or mixed

base.

2. **Mallison (wax in the distillation residue):** residue with more than 5% paraffin wax = paraffinic crude; less than 2% = naphthenic (asphaltic) base; 2-5% = mixed base.
3. **U.S. Bureau of Mines (key fractions):** distil the crude; key fraction 1 boils 250-275 °C at atmospheric pressure, key fraction 2 boils 275-300 °C at 40 mmHg. Their API gravities classify the crude: paraffin if cut 1 ≥ 40 and cut 2 ≥ 30 °API; naphthene if cut 1 < 33 and cut 2 < 20; intermediate in between (combinations like paraffin-intermediate).
4. **By API gravity:** conventional (light/medium) crude API > 20, heavy crude API < 20, bitumen API < 10.
5. **By sulfur:** sweet < 0.5 wt% S, sour > 0.5 wt% S.
6. **By Watson characterization factor K_W:** about 12.9 highly paraffinic, about 10.5 highly naphthenic; aromatic about 10.

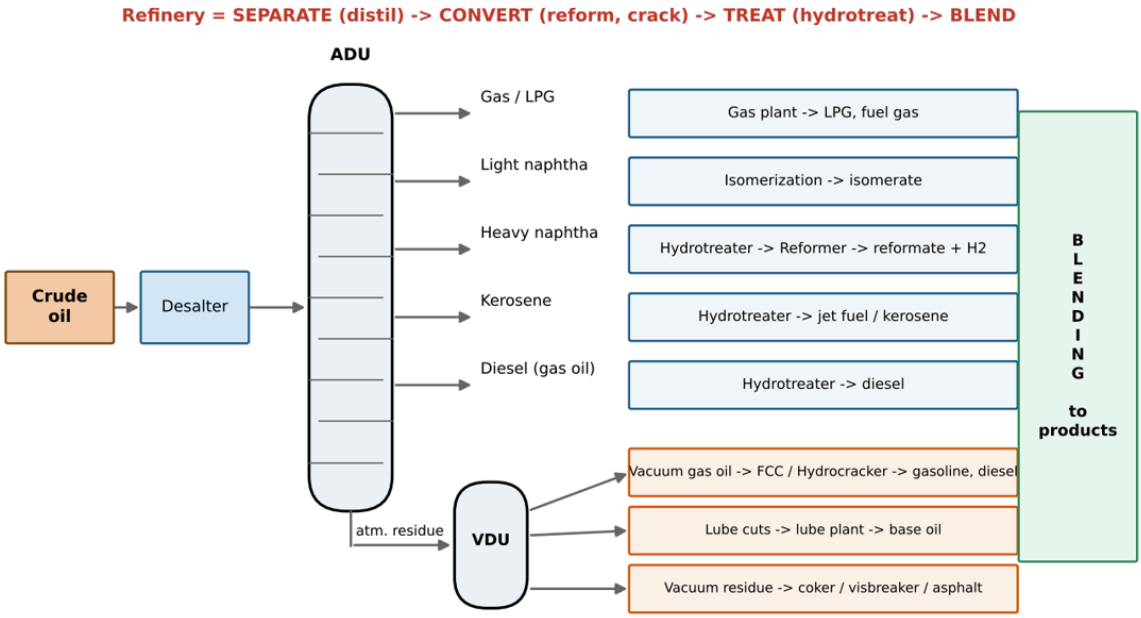
> **Memory trick: P-N-A**: Paraffin (chain), Naphthene (ring), Aromatic (benzene ring) are in crude; **O**lefins are **O**nly made in the refinery.

Key properties of crude that must be evaluated (2023 1c)

Property	Meaning and why it matters
API gravity	API = 141.5/SG − 131.5; light crude (high API) gives more gasoline and is worth more
Sulfur content	wt% S; sour crude needs more hydrotreating
Total acid number (TAN)	mg KOH needed to neutralise 1 g of oil; TAN > 1 = corrosive "high-TAN" crude
Pour point	lowest temperature at which the oil still flows; high pour point = waxy (paraffinic); matters for pipelines
Salt content	lb NaCl per 1000 bbl; > 10 lb/1000 bbl must be desalted to about 0.5
Nitrogen content	wt%; > 0.25% needs nitrogen removal
Carbon residue	Ramsbottom (ASTM D524) or Conradson (ASTM D189); indicates asphalt content and C/H; lower is better
Metal content	Ni, V, Cu in ppm; poison catalysts
Distillation (TBP) curve	how much of each product the crude will give
Characterization factor K_W, correlation index CI, viscosity and VI	type of hydrocarbon present

Why crude oil must be refined, and how (2022 1c, 2022B 1b)

Why: unprocessed crude can only be burnt for heating. It is a mixture boiling from below 0 °C to above 550 °C, contains sulfur, salt, water and metals, and its natural fractions do not have the octane, cetane, volatility or sulfur limits that engines and environmental laws demand. Refining separates it into products of the right boiling range, upgrades their quality (octane, stability), removes impurities and blends products to specification.



Overall refinery scheme: separate, convert, treat, blend

How (essential processes):

1. **Separation:** desalting, then atmospheric distillation (ADU) into gas, naphtha, kerosene, diesel and residue; vacuum distillation (VDU) of the residue into gas oils, lube cuts and vacuum residue.
2. **Conversion:** decomposition (thermal cracking, FCC, hydrocracking, coking, visbreaking) breaks big molecules; unification (alkylation, polymerization) joins small ones; reforming and isomerization change their shape to raise octane.
3. **Treatment:** desalting, hydrotreating (remove S, N, metals), solvent extraction, dewaxing, sweetening.
4. **Blending:** mix the semi-finished streams with additives into products that meet specifications.
5. **Other operations:** light-end recovery, sour-water treatment, sulfur recovery, waste water treatment, storage.

Basic types of processes and operations (2020 1a)

Write the same five groups as above (distillation; conversion = decomposition, unification, reforming; treatment; blending; auxiliary operations), each with two examples and one line on what it does.

Definitions: BPSD, BPCD, refinery complexity, °API, K_W (2020 1b)

Term	Meaning
BPSD (barrels per stream day)	maximum throughput a unit can process in 24 h when running at full rate (on-stream day); refinery capacity is quoted this way
BPCD (barrels per calendar day)	average throughput over a whole year including shutdowns: BPCD = annual throughput/365; always below BPSD (BPCD ≈ 0.9-0.95 BPSD)
Refinery complexity	measure of how much conversion capacity a refinery has compared with its distillation capacity (Nelson index); simple (distillation, reforming, HDS (hydrodesulfurization)) < cracking < coking refinery
°API	API gravity = $141.5/SG_{60°F} - 131.5$; water = 10; higher API = lighter oil
K_W (Watson/UOP characterization factor)	$K_W = T_B^{1/3}/G$, T_B = mean average boiling point in °R (= $(T_{20} + T_{50} + T_{80})/3$), G = SG at 60 °F; 10 aromatic, 10.5 naphthenic, 12.9-15 paraffinic

Heat capacity; why dehydration is needed (2021 5b)

Heat capacity is the amount of heat needed to raise the temperature of a given amount of matter by 1 °C. For 1 g it is the **specific heat** (J/g·°C); for 1 mol, the **molar heat capacity**. Petroleum fractions have specific heats of about 1.7-2.5 kJ/kg·K, rising with temperature and with H/C ratio.

Dehydration is needed because crude may arrive with up to 15% water, much of it as a fine water-in-oil emulsion carrying dissolved salts. Water (i) carries chloride salts that hydrolyse to HCl and corrode the overhead system; (ii) flashes to steam in the furnace and column, causing pressure surges, foaming and upsets; (iii) wastes furnace fuel (water's latent heat is large); (iv) deposits salts that foul exchangers and furnace tubes and poison catalysts.

Sulfur: adverse effects, test methods, stabilizers (2021 5c)

- **Adverse effects:** corrosion of equipment (H_2S, mercaptans); poisoning of reforming (Pt + S → PtS) and other catalysts; foul odour; SO_2/SO_3 emissions and acid rain when products burn; gum formation and poor stability; lower octane response; product specifications (ultra-low sulfur diesel).
- **Test methods:** total sulfur by lamp/bomb combustion (ASTM D1266, D129) or X-ray fluorescence (ASTM D4294); mercaptans by the Doctor test (sodium plumbite); corrosive sulfur by the copper strip corrosion test (ASTM D130).
- **Stabilization:** crude is "stabilized" by removing dissolved light gases (C_1-C_4, H_2S) so it can be stored and shipped safely. Types: multistage flash separation, stabilizer (reboiled stripping) columns, and heating plus flash; product stabilizers in a refinery are the debutanizer/stabilizer columns after the reformer and FCC. Chemical stabilizers (antioxidants, metal deactivators, H_2S scavengers) are added to products.

Total acid number, TBP distillation, natural gas condensate (2019 3a)

- **TAN**: mg of KOH required to neutralise the acids in 1 g of oil; crude with TAN > 1 mg KOH/g is corrosive (high-TAN crude).
- **TBP distillation**: batch distillation in a column of 15 theoretical plates at reflux ratio 5 ("15/5", ASTM (American Society for Testing and Materials) D2892), first at atmospheric pressure then at 40 mmHg; gives the true composition curve of the crude (see A3).
- **Natural gas condensate (NGC)**: ultra-light liquid hydrocarbons (up to 98% boiling below 360 °C) that are gaseous in the reservoir and condense when the raw gas is cooled below its

hydrocarbon dew point (see B1).

Function of an energy engineer in a refinery (2021 8c)

- Energy audits and benchmarking: measure fuel, steam and power use per barrel and find losses (refining is very energy intensive).
- Heat integration: design and maintain the crude preheat train and pinch analysis; use pump-around heat; recover heat from products.
- Furnace and boiler efficiency: excess-air control, air preheat, burner tuning, reducing flue gas temperature.
- Steam system: traps, condensate recovery, cogeneration (combined heat and power), flare gas recovery.
- Electric drives and utilities: variable-speed drives, compressor and pump efficiency, cooling water optimisation.
- Emission and environmental compliance: CO_2, SO_x, NO_x reduction; energy management systems (ISO 50001); economic evaluation of energy projects.

Remember: Crude = C 83-87, H 10-14, S up to 6%; families P-N-A (+ olefins made by cracking); impurities S, N, O, metals, salt, water; refinery = separate → convert → treat → blend.

A6 Hydrotreating, hydrogen in the refinery, lube oil processing Pass set

Very likely · 95% Easy

PAST-PAPER QUESTIONS

- **2023 Q4(d)** [12] Explain the hydrotreating process with a neat labeled diagram.
- **2023 Q4(b)** [8] Describe the hydrofinishing process in lube oil processing with a labeled diagram.
- **2022B Q4(c)** [12] Same as 2023 4(d).
- **2022 Q4(d)** [9] Same as 2023 4(d).
- **2022 Q4(b)** [8] Detailed description of lube oil processing with a diagram.
- **2021 Q8(a)** [8] Clarify the necessity of using lube oil in vehicles.
- **2020 Q3(b)** [10] With a neat sketch explain the hydrotreating process.
- **2020 Q3(c)** [10] Why is hydrogen required in a refinery? From which sources can it be obtained?

★ REMEMBER IT

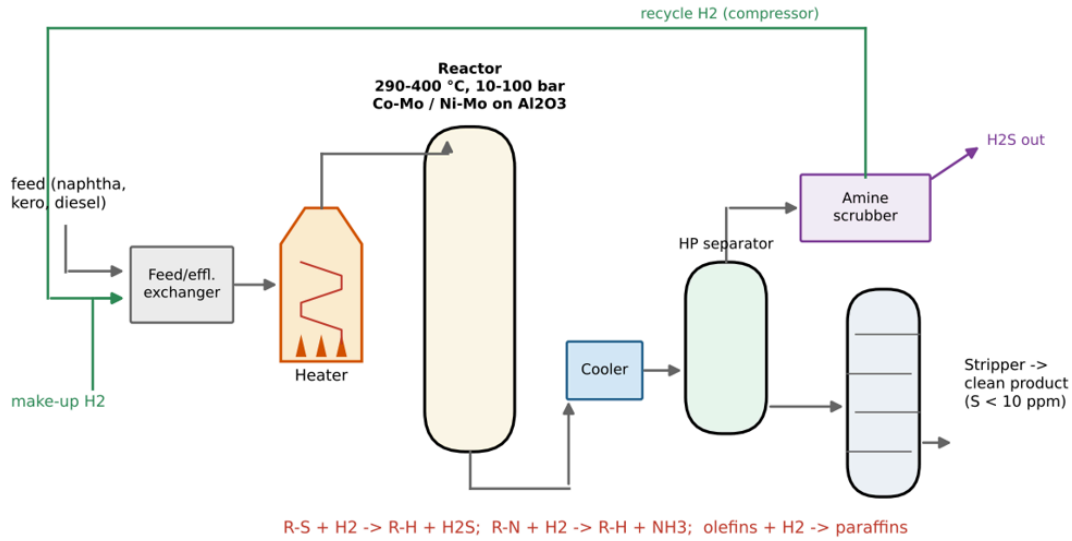
- **Hydrotreating**: feed + H_2 → heater → reactor (Co-Mo or Ni-Mo on alumina, 300–400 °C) → separator (H_2 recycled) → stripper. Removes S as H_2S, N as NH_3, saturates olefins, removes metals.
- Hydrogen sources: catalytic reformer (by-product), steam methane reforming, partial oxidation of heavy oil, recovery from off-gases (PSA, membranes).
- Lube oil chain: vacuum distillate → solvent extraction (remove aromatics, raise VI) → dewaxing (lower pour point) → **hydrofinishing** (mild hydrotreating for colour and stability).

Before you start: why hydrogen is used in a refinery

Sulfur and nitrogen atoms in fuels poison catalysts and pollute the air. Reacting them with hydrogen over a catalyst turns them into H_2S and NH_3, which are easy to remove, and turns unstable olefins into stable paraffins. This is **hydrotreating**. Lube oil is made from vacuum distillates by removing asphalt, aromatics and wax, then polishing with hydrogen. Source: Shameem sir's slides "Hydrotreating" and "Lubricating oil".

Hydrotreating process with a labelled diagram (2023 4d, 2022 4d, 2022B 4c, 2020 3b)

Hydrotreating is a catalytic process with hydrogen that stabilises products (saturates olefins and some aromatics) and removes heteroatoms: sulfur (hydrodesulfurization, HDS (hydrodesulfurization)), nitrogen (hydrodenitrogenation, HDN (hydrodenitrogenation)), oxygen (HDO) and metals, without much change in boiling range. It is placed before the reformer, hydrocracker and FCC (fluid catalytic cracking) to protect their catalysts, and on product streams (naphtha, kerosene, diesel) to meet sulfur specs.



Hydrotreater flow diagram

1. Liquid feed is mixed with make-up and recycle hydrogen, preheated against reactor effluent and heated in a furnace to reaction temperature (290-400 °C).
2. The mixture flows down through a fixed-bed **reactor** of Co-Mo (desulfurization) or Ni-Mo (nitrogen removal, aromatic saturation) on porous alumina, at 10-100 bar. Reactions are exothermic; cold H_2 quench between beds controls temperature.
3. Effluent is cooled (wash water injected to dissolve ammonium salts) and goes to a **high-pressure separator**: hydrogen-rich gas off the top, liquid at the bottom.
4. The gas passes through an **amine scrubber** to remove H_2S, is compressed and recycled; some is purged to stop C_1-C_4 build-up.
5. The liquid goes to a **stripper/fractionator** that removes dissolved H_2S and light ends, giving the clean product (e.g. diesel with < 10 ppm S).

Reactions: RSH + H_2 → RH + H_2S; R-S-R + $2H_2$ → 2RH + H_2S; thiophene C_4H_4S + $4H_2$ → C_4H_{10} + H_2S; pyridine C_5H_5N + $5H_2$ → C_5H_{12} + NH_3; phenol C_6H_5OH + H_2 → C_6H_6 + H_2O; olefins + H_2 → paraffins; RCl + H_2 → RH + HCl. Hydrogen use: about 70 scf/bbl per % S, 320 per % N, 180 per % O. Severity rises from olefin saturation (mild) to HDS (severe) to HDN (very severe).

Why hydrogen is required and its sources (2020 3c)

Required for: hydrotreating of naphtha to vacuum gas oil (0.6-10 kg H_2/t) to remove S, N and metals; hydrocracking and residue hydroconversion (15-35 kg H_2/t); saturating olefins and aromatics (stable fuels, low-aromatic diesel and jet); lube hydrofinishing; isomerization; protecting downstream catalysts. Demand is rising because crudes are heavier and more sour and fuel sulfur limits are tighter, so most refineries run at a hydrogen deficit. It must be > 99% pure and at high pressure.

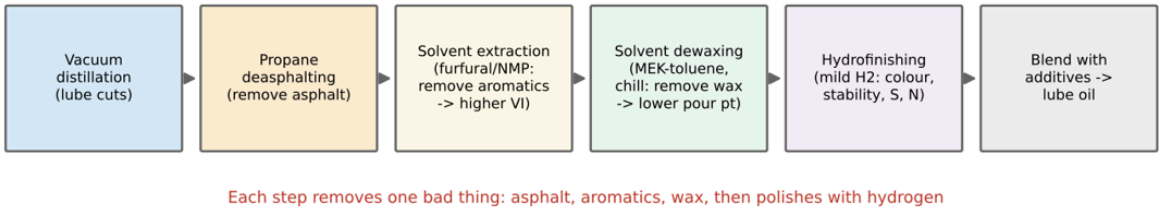
Sources:

1. **Catalytic reformer** (by-product of dehydrogenation and dehydrocyclization): the cheapest internal source.
2. **Steam methane reforming** (main purpose-built source): CH_4 + H_2O → CO + $3H_2$ (Ni catalyst, 800-900 °C), then water-gas shift CO + H_2O → CO_2 + H_2, then PSA (pressure swing adsorption) purification.

3. **Partial oxidation** (gasification) of heavy residue or coke: $C_nH_m + (n/2)O_2 \rightarrow nCO + (m/2)H_2$.
4. **Recovery** from refinery off-gases (hydrotreater purge, FCC off-gas) by PSA (pressure swing adsorption) or membranes; naphtha cracking (ethylene plant) off-gas.

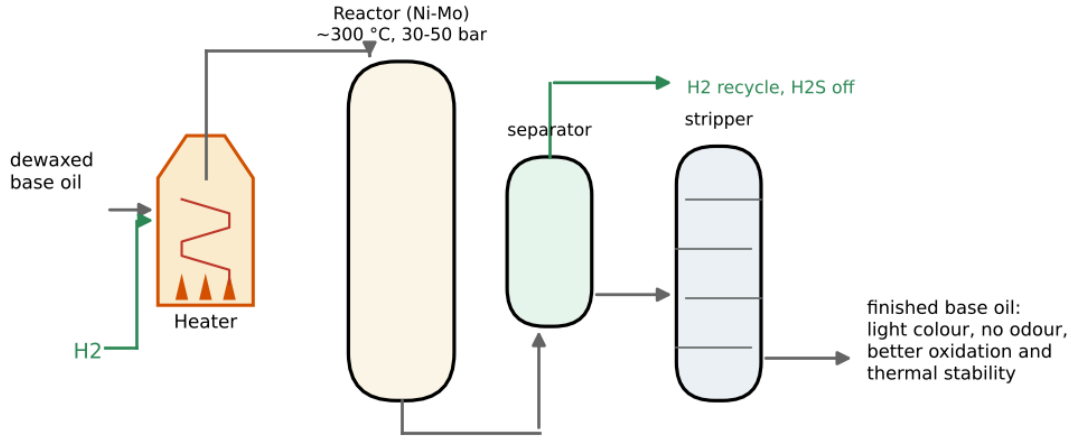
Lube oil processing and hydrofinishing (2023 4b, 2022 4b)

Lubricating oil reduces friction, heat and wear between moving parts. Mineral lube base oils are made from vacuum distillates by removing the components that spoil viscosity index and stability:



Lube oil processing by the solvent route

1. **Vacuum distillation** of atmospheric residue into lube cuts (light, medium, heavy neutral) with sharp fractionation (controls flash point and viscosity); vacuum residue to deasphalting.
2. **Propane deasphalting** of the vacuum residue: liquid propane dissolves the oil and rejects resins and asphaltenes, giving bright stock.
3. **Solvent extraction** with furfural, phenol or NMP (N-methyl pyrrolidone): removes aromatics, mainly polyaromatics, which have low VI and poor stability; most S and N leave with them. Product = raffinate.
4. **Solvent dewaxing**: raffinate mixed with MEK-toluene and chilled; wax crystallises and is filtered off, lowering the pour point.
5. **Hydrofinishing** of the dewaxed oil (below).
6. **Blending** of base oils with additives (VI improvers, pour-point depressants, antioxidants, detergents, dispersants, anti-wear/EP, rust and corrosion inhibitors).



Hydrofinishing of dewaxed base oil

Hydrofinishing is a mild hydrotreatment of the dewaxed base oil: oil + hydrogen heated to about 250-320 °C, 30-50 bar, over Ni-Mo or Co-Mo on alumina in a fixed-bed reactor; then separator (H_2 recycled, H_2S off) and stripper (removes light ends, H_2S, water). It saturates olefins and some aromatics and removes traces of S, N and O compounds and solvent, improving **colour, odour,**

oxidation and thermal stability and demulsibility. It replaced the old clay treatment. (Process conditions from Rao; the teacher's slides list it as the finishing step.)

Need for lube oil in vehicles (2021 8a)

- Reduces **friction** between pistons, rings, cylinder walls and bearings, so the engine needs less work to move, uses less fuel and runs cooler.
- Reduces **wear** (mechanical and corrosive), lengthening engine life.
- **Cools** parts by carrying heat away from hot zones.
- **Cleans**: carries soot, metal particles and contaminants to the filter; detergents keep deposits off.
- **Seals** the gap between piston rings and cylinder, keeping compression.
- Protects against **corrosion** and rust; cushions shock loads in gears and bearings.
- Motor oil is a blend: low-viscosity oil for easy cold starting with high-viscosity oil (or VI improver) for protection at running temperature.

Remember: Hydrotreating: Co-Mo/Ni-Mo on alumina, 290-400 °C, 10-100 bar; S → H_2S, N → NH_3; amine scrubber; H_2 from reformer + steam reforming. Lube: vacuum cuts → deasphalt → extract aromatics → dewax → hydrofinish → additives.

A2 Product properties and test numbers: octane and cetane number, API gravity, flash point, RVP, gasoline grades and pool, diesel classes Pass set

Very likely · 86% Easy

PAST-PAPER QUESTIONS

- **2023 Q1(b)** [6] What is the octane number, and how is it used to evaluate the anti-knock performance of gasoline?
- **2023 Q1(e)** [8] Define: cetane number, API gravity, flash point, RVP (Reid vapour pressure).
- **2022B Q1(c)** [7] What is the anti-knock performance of gasoline, and how does it relate to octane numbers?
- **2022B Q1(d)** [7] What is the cetane number, and how is it used to evaluate diesel fuel quality?
- **2022B Q1(e)** [5] The specific gravity of crude oil is 0.821; determine its API gravity.
- **2022 Q1(b)** [7] Same question as 2023 1(b).
- **2022 Q1(e)** [6] Define: octane number, API gravity, flash point.
- **2022 Q1(d)** [5] The specific gravity of NGC (natural gas condensate) is 0.802; determine its API gravity.
- **2020 Q2(a)** [6] Classify different grades of gasoline based on octane number.
- **2020 Q2(b)** [10] Components of the gasoline pool? Why are oxygenates added to gasoline? Name a few oxygenates.
- **2019 Q1(c)** [7] What is octane number? Explain the effect of variables on an engine's octane rating.
- **2019 Q1(b)** [8] Classify petroleum products based on boiling point and carbon numbers.
- **2019 Q1(d)** [5] Classify diesel fuel in terms of engine speed.

★ REMEMBER IT

- **Octane** (petrol, spark ignition) = resistance to **knock** (self-ignition before the spark). Scale: iso-octane = 100, n-heptane = 0. RON (mild test) > MON (severe test); pump AKI = (RON + MON (motor octane number))/2. Branched, aromatic, olefin = high octane; straight chains = low.
- **Cetane** (diesel, compression ignition) = the *opposite*: ease of self-ignition. Cetane (n-hexadecane) = 100, α-methylnaphthalene = 0. Straight chains = high cetane. Memory: **Octane resists, Cetane ignites**.
- Flash point = lowest temperature at which vapour above the liquid flashes with a flame (storage safety). RVP = vapour pressure at 100 °F (evaporation loss, vapour lock, cold start).

CALCULATOR API gravity and density (2022 1(d), 2022B 1(e))

Change any number; every step is recalculated and shown with the numbers put in.

1. API gravity: °API = 141.5/SG − 131.5 = 141.5 / **0.802** - 131.5 = **44.9** °API
2. Density: **0.802** × 999 = **801** kg/m³
3. Class: **light (> 31.1)**

The smaller the SG, the bigger the API. Water (SG 1) is exactly 10 °API.

Before you start: what a fuel specification is

An engine needs a fuel that evaporates the right amount (volatility), burns at the right moment (octane for petrol engines, cetane for diesel engines) and is safe to store (flash point). Each property has a standard test and a number. Learn the definition, how it is measured and what high or low means. Source: Shameem sir's slides "Petroleum products" and "Refinery raw materials".

Octane number and anti-knock performance (2023 1b, 2022 1b, 2022B 1c, 2019 1c)

Knocking is untimely (premature) self-ignition of the fuel-air mixture in a spark-ignition engine before the flame front arrives. It causes loss of power, overheating, overloading of the cooling system, and mechanical damage (erosion of cylinder head and piston, damage to valves and spark plugs).

Octane number (ON) measures a gasoline's resistance to knock. It is the volume percent of **iso-octane** (2,2,4-trimethylpentane, ON = 100) in a mixture with **n-heptane** (ON = 0) that knocks with the same intensity as the fuel, tested in a standard single-cylinder CFR engine with variable compression ratio.

Rating	Test condition	Represents
RON (research octane number)	600 rpm, lower intake temperature	low-speed city driving
MON (motor octane number)	900 rpm, hotter mixture	high-speed highway driving
Sensitivity = RON (research octane number) − MON	–	how much the fuel suffers under severe conditions
PON (posted/pump octane) = (RON + MON)/2	–	number shown at the pump (anti-knock index)

How it is used: the higher the ON, the higher the compression ratio the engine can use without knock, so more power and efficiency. Grades are sold by ON; refiners raise ON by reforming, isomerization, alkylation and oxygenates. Molecule ranking: aromatics and iso-paraffins very high, olefins high, naphthenes medium, n-paraffins very low.

Effect of variables on the engine's octane requirement (2019 1c):

- Altitude: every 100 m rise allows a fuel 1 RON lower (thinner air, lower compression pressure).
- Humidity: +10% humidity at 25 °C allows 0.5 RON less.
- Engine speed: +300 rpm allows 1 RON less.
- Air temperature: +12 °C intake air needs 1 RON more (hot countries need higher RON than cold).
- Coolant temperature: every +6 °C needs 1 RON more.
- Deposits: after 5,000-10,000 km of running, deposits raise the requirement by about 1 RON.

> **Memory trick:** Octane = **O**ppose ignition (petrol must NOT self-ignite); Cetane = **C**atch fire quickly (diesel MUST self-ignite).

Cetane number and diesel quality (2022B 1d, 2023 1e)

Cetane number (CN) measures the ignition quality of diesel fuel, i.e. its **ignition delay** (time between injection and start of combustion). CN = volume percent of **n-cetane** (n-hexadecane

$C_{16}H_{34}$, CN = 100, ignites quickly) in a mixture with **α-methylnaphthalene** ($C_{11}H_{10}$, CN = 0) that gives the same ignition delay as the fuel. Diesel fuels have CN 40-55. High CN = short delay = smooth combustion, easy cold start, less diesel knock, less smoke. Straight-chain paraffins have high CN; aromatics low. Without an engine, the **cetane index** is calculated from API gravity and the D86 mid-boiling point; the **diesel index** = aniline point (°F) × API/100 is another indicator.

Definitions: API gravity, flash point, RVP (2023 1e, 2022 1e)

Property	Definition
API gravity	API = 141.5/SG − 131.5, SG at 60 °F (15.6 °C); water = 10; API > 10 floats on water; higher API = lighter
Flash point (ASTM D93)	lowest temperature at which the vapour above a liquid forms a flammable mixture with air that briefly flashes when a flame is applied; tells the maximum safe storage temperature
Reid vapour pressure (ASTM D323)	pressure of the vapour of a gasoline in a standard Reid bomb at 37.8 °C (100 °F), vapour-to-liquid volume ratio 4:1; indicates easy starting (high RVP), vapour lock and evaporation losses (too high)
Octane number	see above
Cetane number	see above

API gravity numericals (2022 1d, 2022B 1e)

Step (write this)	Why / what it means
`API = 141.5/SG - 131.5`	definition, SG at 60 °F
`NGC, SG 0.802: 141.5/0.802 = 176.43`	divide first
`API = 176.43 - 131.5 =` **`44.9 °API`**	a light liquid, as condensate should be
`Crude, SG 0.821: 141.5/0.821 = 172.35; API = 172.35 - 131.5 =` **`40.9 °API`**	light crude (API > 31.1)

> **Side note:** Check: lighter than water so the answer must be above 10; the smaller the SG, the bigger the API.

Grades of gasoline (2020 2a)

Grade	Octane rating (anti-knock index)
Regular	≥ 85 and < 88
Mid-grade	≥ 88 and ≤ 90
Premium	> 90

Components of the gasoline pool; oxygenates (2020 2b)

Component	Source and character
LSR (light straight-run (naphtha)) gasoline	C_5-88 °C straight-run naphtha; ON 70-80, high RVP
Isomerate	isomerization of LSR; ON 82-92, no aromatics, benzene, sulfur or olefins
Reformate	catalytic reforming of heavy naphtha; ON 90-104, high aromatics
FCC (fluid catalytic cracking) gasoline	catalytic cracking; largest share (about 38%), olefins and aromatics, most of the sulfur
Hydrocracked gasoline	hydrocracker; low sulfur
Alkylate	isobutane + C_3-C_5 olefins; high ON, low sensitivity, no aromatics
Polymer gasoline	polymerization of C_3/C_4 olefins; highly olefinic
n-Butane	added to adjust RVP; ON in the 90s, cheap
Oxygenates	ethers and alcohols (below)

Why oxygenates are added: they have high octane (RON 115-126), so they replace toxic lead and reduce aromatics; the oxygen they carry gives more complete combustion, cutting CO and unburnt hydrocarbon emissions ("reformulated gasoline"). **Examples:** MTBE (methyl tertiary-butyl ether, RON 118), ETBE (ethyl tertiary-butyl ether, RON 118), TAME (tertiary-amyl methyl ether, RON 115), methanol (RON 126), ethanol (RON 120; E85 = 70-85% ethanol).

Classification of products by boiling point and carbon number (2019 1b)

Product	Carbon number	Boiling range, °C
Refinery gas (C_1, C_2)	C_1-C_2	< −88
LPG (liquefied petroleum gas)	C_3-C_4	−42 to 0
Gasoline (naphtha)	C_5-C_{12}	38-205
Kerosene / jet fuel	C_{10}-C_{16}	175-275
Diesel (gas oil)	C_{14}-C_{20}	250-360
Lube oil / vacuum gas oil	C_{20}-C_{50}	350-550
Residual fuel oil, asphalt	> C_{50}	> 550

Grouped by volatility as: low-boiling products (gases, LPG), gasoline, middle distillates (kerosene, jet, diesel, heating oil) and residue (fuel oil, bitumen, coke).

Diesel fuel by engine speed (2019 1d)

Class	Speed	Use
Low-speed diesel	< 300 rpm	heavy loads at constant speed (ships, stationary)
Medium-speed diesel	300-1000 rpm	fairly heavy loads, moderately constant speed (rail, generators)
High-speed diesel (HSD)	> 1000 rpm	varying load and speed (cars, trucks, buses)

Remember: API = 141.5/SG − 131.5. Octane: iso-octane vs n-heptane, RON 600 rpm city, MON 900 rpm highway. Cetane: n-cetane vs α-methylnaphthalene, ignition delay. RVP at 37.8 °C. Grades 85-88-90.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2022 1(d) NGC, SG = 0.802

API gravity: °API = 141.5/SG − 131.5

= 141.5/0.802 − 131.5 = 176.43 − 131.5

= **44.9 °API (a light liquid)**

A3 Distillation curves (TBP, ASTM, EFV), desalting, atmospheric and vacuum distillation Pass set Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q1(d)** [8] What are distillation curves? How does ASTM (American Society for Testing and Materials) differ from EFV (equilibrium flash vaporization)?
- **2023 Q2(a)** [5] What is meant by True Boiling Point (TBP) distillation?
- **2023 Q2(b)** [10] Describe the dehydration and desalting of crude oil with a diagram.
- **2023 Q2(c)** [12] Explain atmospheric and vacuum distillation of crude oil showing the arrangement of towers to obtain principal fractions.
- **2022B Q3(a)** [8] Describe in brief: TBP (true boiling point) distillation, ASTM distillation, API gravity.
- **2022B Q2(a)** [10] Describe crude oil desalting with a diagram. Operating variables of the desalter?
- **2022B Q2(b)** [10] What are ADU (atmospheric distillation unit) and VDU (vacuum distillation unit)? Schematic of a distillation tower with products and their temperature ranges.
- **2022 Q2(d)** [7] What do you understand by TBP distillation? Explain.
- **2022 Q3(a)** [8] Discuss ASTM distillation characteristics with a neat diagram.
- **2022 Q2(a)** [10] Describe the crude oil desalting process with a diagram.
- **2022 Q2(b)** [10] Same as 2023 2(c).
- **2021 Q6(a)** [15] Briefly explain the crude oil distillation process.
- **2020 Q2(c)** [14] Explain vacuum distillation with a flow diagram.
- **2019 Q4(d)** [12] Same as 2022B 2(a).

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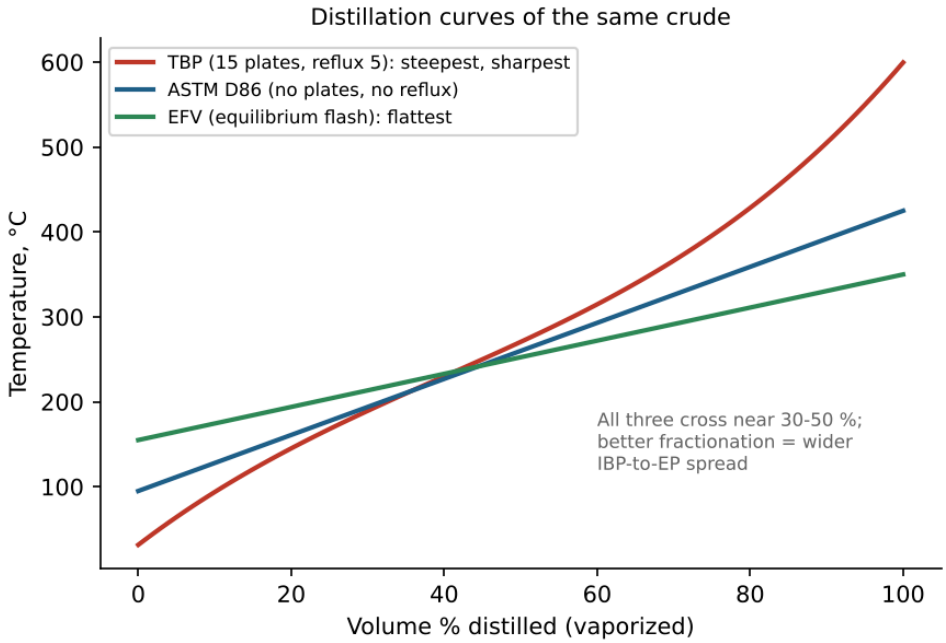
- Distillation curves: **TBP** (true boiling point: many plates, high reflux, sharp separation, used for crude assay), **ASTM D86** (quick lab test, no reflux, poor separation, for product specs), **EFV** (equilibrium flash vaporization: liquid and vapour kept together, flattest curve). Slope: TBP steepest > ASTM > EFV.
- Order of units: **desalter** (wash water + electric field, 120–150 °C, removes salt to avoid HCl corrosion) → furnace → **atmospheric tower (ADU)**: gases, LPG (liquefied petroleum gas), naphtha, kerosene, diesel, atmospheric residue → **vacuum tower (VDU)** on the residue (low pressure lowers boiling points so heavy oil does not crack): vacuum gas oils, vacuum residue.
- Desalter variables: temperature, wash-water rate, mixing (pressure drop across the mixing valve), demulsifier dose, electric field, pH.

Before you start: how distillation separates crude

Heat a liquid mixture and the light (low boiling) molecules vaporise first. In a tall column with trays, rising vapour meets falling liquid on every tray; heavy molecules condense and fall, light molecules rise. Product streams of different boiling range are drawn off at different heights. Before that, the salt and water must be washed out. Source: Shameem sir's slides "Crude distillation" and "Refinery raw materials".

Distillation curves; TBP; ASTM vs EFV (2023 1d, 2023 2a, 2022 2d, 2022 3a, 2022B 3a)

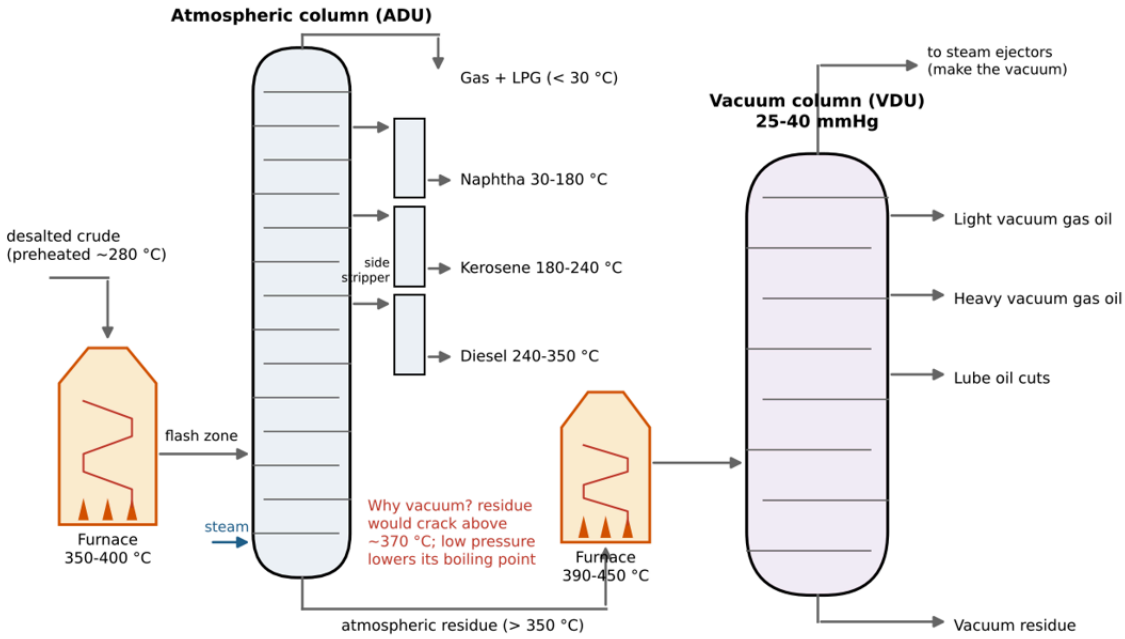
A **distillation curve** is a plot of boiling (vapour) temperature against the volume percent of the sample distilled. It shows how much of each product a crude or fraction contains. Three types are used:



TBP, ASTM and EFV curves of the same crude

TBP (true boiling point) distillation: a batch distillation in a column with about 15 theoretical plates at a reflux ratio of 5 ("15/5 distillation", ASTM D2892). Done first at atmospheric pressure (up to about 275-300 °C) and then under vacuum at 40 mmHg (to avoid cracking); vacuum temperatures are converted to atmospheric equivalents (300 °C at 40 mmHg ≈ 421 °C at 760 mmHg). Because separation is nearly perfect, each point is the true boiling point of the material at that percent: it gives the actual composition and is the core of the crude assay.

ASTM distillation (D86, D1160): a simple, non-fractionating batch distillation in an Engler flask; no packing or trays, no reflux except heat loss from the neck. D86 at atmospheric pressure for light products (gasoline, kerosene, diesel); D1160 at 1-50 mmHg for heavy products. Simpler, cheaper, needs less sample and about one-tenth the time of TBP, so it is used for product specifications.



Desalted crude to atmospheric column, residue to vacuum column

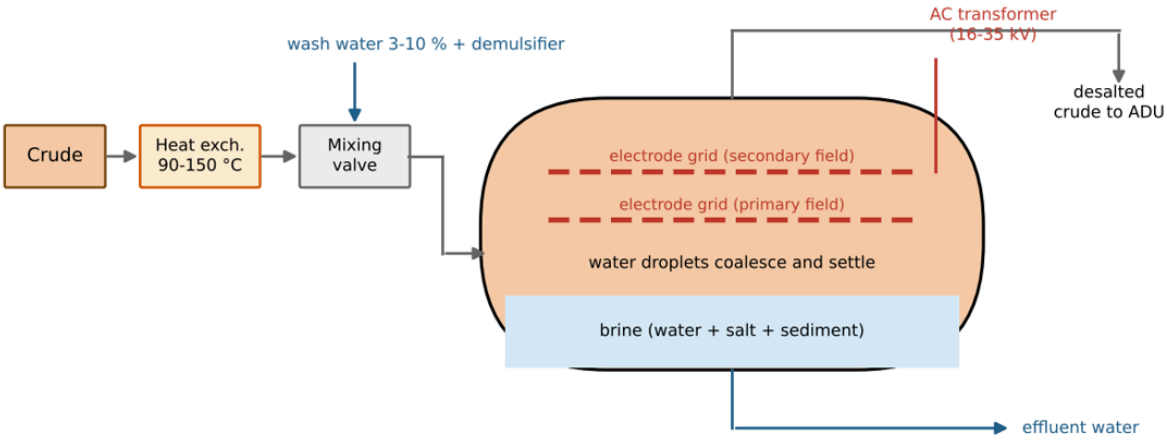
EFV (equilibrium flash vaporization): the sample is heated continuously without letting the vapour separate from the liquid, then flashed; at each heater outlet temperature the vapour fraction is measured. EFV initial point = bubble point, final point = dew point. Used by designers to size flash zones and furnaces.

	TBP	ASTM	EFV
Operation	batch, fractionating	batch, non-fractionating	continuous flash, vapour stays with liquid
Plates / reflux	15 plates, reflux 5	none, no reflux	none
Pressure	atmospheric + vacuum (40 mmHg)	atmospheric (D86), vacuum (D1160)	atmospheric, above, or vacuum
Used for	crude oil assay	petroleum products, specs	design (phase behaviour)
Separation	sharpest (steepest curve)	medium	poorest (flattest curve)
IBP / FBP	lowest IBP, highest FBP	in between	highest IBP, lowest FBP

> **Memory trick:** Sharpness falls **T-A-E**: TBP > ASTM > EFV, like the letters getting further from a "True" answer.

Dehydration and desalting of crude oil (2023 2b, 2022 2a, 2022B 2a, 2019 4d)

Why: crude with more than 10 lb salt per 1000 bbl must be desalted (to about 0.5 lb/1000 bbl). Salts (NaCl, $MgCl_2$, $CaCl_2$) are dissolved in tiny water droplets emulsified in the oil, shielded by asphaltene films so they cannot settle. If not removed they deposit in furnace tubes and exchangers (fouling), hydrolyse to HCl at 150-200 °C (overhead corrosion) and poison catalysts.



Electrostatic desalter

Process (four steps):

1. **Water washing:** 3-10 vol% fresh water is mixed into the crude through a mixing valve; demulsifier is injected to strip asphaltenes off the droplet surfaces.
2. **Heating:** the crude is heated to 90-150 °C (about 120 °C) by exchange with hot products, lowering oil viscosity so droplets move easily.
3. **Coalescence:** in the desalter drum a high-voltage AC field polarises the droplets so they attract and merge into large drops. Secondary field about 1000 V/cm between the two electrodes (ionisation and coalescence); primary field about 600 V/cm between the water-oil interface and the lower electrode (helps drops settle).
4. **Settling:** large drops settle by gravity (Stokes' law, $v = gd^2(\rho_w - \rho_o)/18\mu$); brine with salt and sediment leaves from the bottom to effluent treatment, desalted crude from the top to the preheat train.

Dehydration works the same way without the wash water: free water is removed by gravity, and the emulsion is broken by heat, chemicals (demulsifiers) and the electric field. A dehydrator followed by a desalter is used when salt specs are tight. **Operating variables of the desalter** (2022B 2a): desalting temperature, desalter pressure (kept above vapour pressure so no vaporisation), wash water ratio, water (interface) level, wash water injection point, demulsifier injection rate, type/quality of wash water, and pressure drop across the mixing valve.

Atmospheric and vacuum distillation of crude (2023 2c, 2022 2b, 2022B 2b, 2021 6a)

Atmospheric distillation unit (ADU, CDU (crude distillation unit), topping unit):

1. Desalted crude is pumped through a train of heat exchangers to about 280 °C; sometimes a preflash drum removes light ends.
2. A fired furnace (pipestill) heats it to 350-400 °C, enough to vaporise all products above the flash zone plus 10-20% "overflash" of the bottoms, without coking.
3. The two-phase feed enters the flash zone of a column about 50 m tall with 30-50 trays. Vapour rises through the rectifying section; stripping steam is injected at the bottom to strip light ends from the residue.
4. Products are drawn off: overhead gas and LPG and light naphtha (condensed, part returned as reflux), heavy naphtha, kerosene, diesel/gas oil as side streams; each side stream passes

through a small steam **side stripper** (4-10 trays) to remove light ends and correct its flash point. **Pump-around reflux** removes heat at intermediate points.
5. Atmospheric residue (> 350 °C) leaves from the bottom.

Fraction	Approx. TBP range, °C	Goes to
Gas + LPG	< 30	gas plant
Light naphtha	30-90	isomerization, gasoline
Heavy naphtha	90-180	hydrotreater + reformer
Kerosene	180-240	jet fuel, kerosene
Diesel (gas oil)	240-350	hydrotreater → diesel
Atmospheric residue	> 350	vacuum distillation

Vacuum distillation unit (VDU) (2020 2c): heating residue above about 370-400 °C at atmospheric pressure would thermally crack it (coke, product loss). Lowering the pressure lowers boiling points, so the heavy oils distil without cracking.

1. Atmospheric residue is heated in a furnace to 390-450 °C; steam is added to the furnace coil (raises velocity, less coke, lowers hydrocarbon partial pressure).
2. It flashes into a large-diameter vacuum column (up to 12 m wide, because vapour volume is huge at low pressure); flash-zone pressure 25-40 mmHg.
3. Packing (low pressure drop) instead of trays in the upper sections; steam ejectors with barometric condensers (three stages for 25 mmHg) pull the vacuum.
4. Products: light vacuum gas oil and heavy vacuum gas oil (FCC/hydrocracker feed), lube oil cuts, and vacuum residue (asphalt, coker or visbreaker feed).
5. Operation can be wet (steam to furnace and column, lowest furnace temperature), damp (steam to furnace only) or dry (no steam, highest furnace temperature).

Remember: Desalt: water 3-10 %, 90-150 °C, 1000/600 V/cm, settle. ADU: 350-400 °C furnace, 30-50 trays, side strippers. VDU: 390-450 °C, 25-40 mmHg, ejectors, wide column, no cracking.

A4 Catalytic reforming and isomerization Pass set Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q2(d)** [8] What is catalytic reforming? Briefly describe the major reactions.
- **2022B Q2(c)** [9] Same as 2023 2(d).
- **2022B Q2(d)** [6] Distinguish between CCR (continuous catalyst regeneration) and SR reforming.
- **2022B Q4(d)** [5] Main reactions of an isomerization process.
- **2022 Q2(c)** [8] Same as 2023 2(d).
- **2021 Q6(c)** [12] Write down the catalytic reforming reactions.
- **2020 Q3(a)** [10] What is catalyst poisoning? Examples of catalyst poisons.
- **2019 Q2(a)** [10] Same as 2023 2(d).
- **2019 Q2(b)** [12] Process diagram of the SR reforming process; variation of temperature and effluent composition in the reactors.
- **2019 Q2(c)** [8] Distinguish between CCR and SR reforming.
- **2019 Q2(d)** [5] Write down C_5 and C_6 isomers and their RON (research octane number) values.
- **2019 Q4(a)** [5] What are the objectives of light naphtha isomerization?
- **2019 Q4(b)** [10] Main and side reactions of an isomerization process.
- **2019 Q4(c)** [8] Light naphtha (SG 0.724, 1000 m^3/h) to isomerization; converted yields C_3 0.348, iC_4 0.619, nC_4 1.770, C_5^+ 97.261 wt%: product composition.

★ REMEMBER IT

- Catalytic reforming turns low-octane naphtha into high-octane **reformate** plus hydrogen. Four reactions: **D-I-D-H**: **D**ehydrogenation of naphthenes to aromatics (fast, very endothermic, makes H_2), **I**somerization, **D**ehydrocyclization of paraffins to aromatics (slow), **H**ydrocracking (unwanted, uses H_2). Catalyst Pt (+ Re) on chlorided alumina.
- Temperature drops sharply in reactor 1 (endothermic dehydrogenation), less in 2 and 3: hence heaters between reactors.
- **SR** (semi-regenerative): fixed beds, shut down every 6–24 months to regenerate. **CCR** (continuous catalyst regeneration): catalyst moves slowly through stacked reactors to a regenerator, so it runs at lower pressure with higher octane and H_2 yield.
- Isomerization: n-C_5/n-C_6 (RON 62/26) → i-C_5/2,2-DMB (RON 92/96). Catalyst poisons: S, N, water, metals, CO.

CALCULATOR **Isomerization product rates (2019 4(c))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Feed mass: m = V × SG × 1000 = **1000** × **0.724** × 1000 = **724000** kg/h
2. C_3: **0.348** / 100 × **724000** = **2520** kg/h
3. iC_4: **0.619** / 100 × **724000** = **4482** kg/h
4. nC_4: **1.77** / 100 × **724000** = **12815** kg/h
5. C_5+ isomerate: **97.26** / 100 × **724000** = **704170** kg/h

6. Yield check: **0.348** + **0.619** + **1.77** + **97.26** = **99.998** wt%

Before you start: why change the shape of a molecule

Straight-chain paraffins knock badly (n-heptane has octane 0). The same atoms rearranged into a branched chain or a benzene ring have a much higher octane. **Catalytic reforming** turns the rings and chains of heavy naphtha into aromatics; **isomerization** turns the straight C_5/C_6 chains of light naphtha into branched ones. Source: Shameem sir's slides "Catalytic reforming and isomerization".

What catalytic reforming is (2023 2d, 2022 2c, 2022B 2c, 2019 2a)

Catalytic reforming is a catalytic process that converts low-octane heavy naphtha (SR, coker or hydrocracker naphtha, RON 50-60) into high-octane **reformate** (RON 90-104) by rearranging molecules into aromatics and iso-paraffins, with little change in boiling range. It contributes 30-40% of the gasoline pool, supplies aromatics (BTX) for petrochemicals, and produces **hydrogen** as a by-product used in hydrotreating.

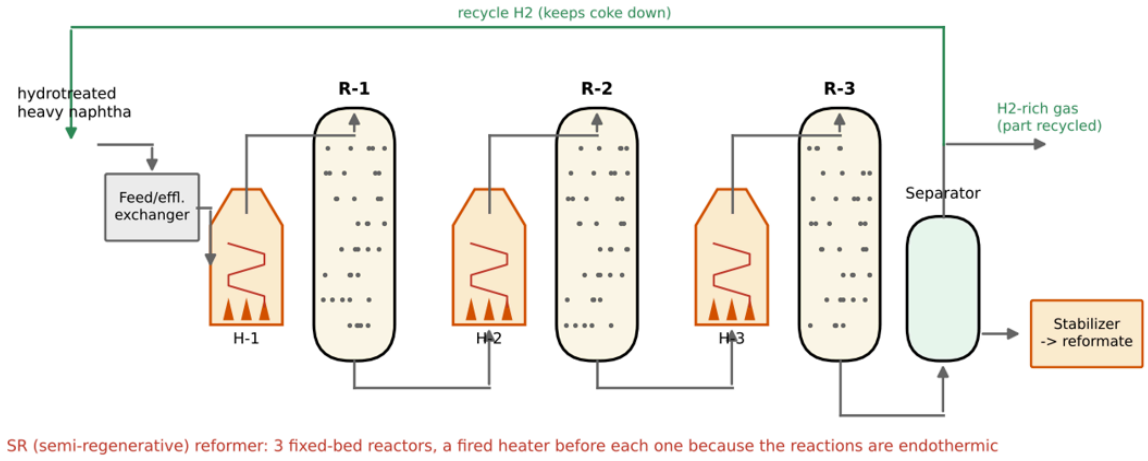
Catalyst: bifunctional, 0.2-0.6 wt% platinum on chlorinated alumina (about 1% Cl). Pt (metal function) does dehydrogenation; the acidic chlorided alumina does isomerization and cyclization. Promoters: Re (lower pressure, SR), Sn (yield at low pressure, CCR), Ir (activity). Feed must be hydrotreated first (S < 1 ppm).

Major reactions (2021 6c: write them as equations):

Reaction	Example	Heat / rate
1. Dehydrogenation of naphthenes (alkylcyclohexanes) to aromatics	methylcyclohexane C_7H_{14} → toluene $C_7H_8 + 3H_2$	highly endothermic, fastest
2. Dehydroisomerization of alkylcyclopentanes to aromatics	methylcyclopentane C_6H_{12} → cyclohexane → benzene $C_6H_6 + 3H_2$	endothermic, rapid
3. Dehydrocyclization of paraffins to aromatics	n-heptane C_7H_{16} → toluene $C_7H_8 + 4H_2$	endothermic, slow
4. Isomerization	n-hexane → 2-methylpentane; n-paraffins → iso-paraffins	mildly exothermic, rapid
5. Hydrocracking (undesired)	$C_{10}H_{22} + H_2 \rightarrow C_3H_8 + C_7H_{16}$	highly exothermic, slow, consumes H_2, makes gas
6. Coke formation (undesired)	polymerisation of olefins/aromatics on the catalyst	deactivates catalyst

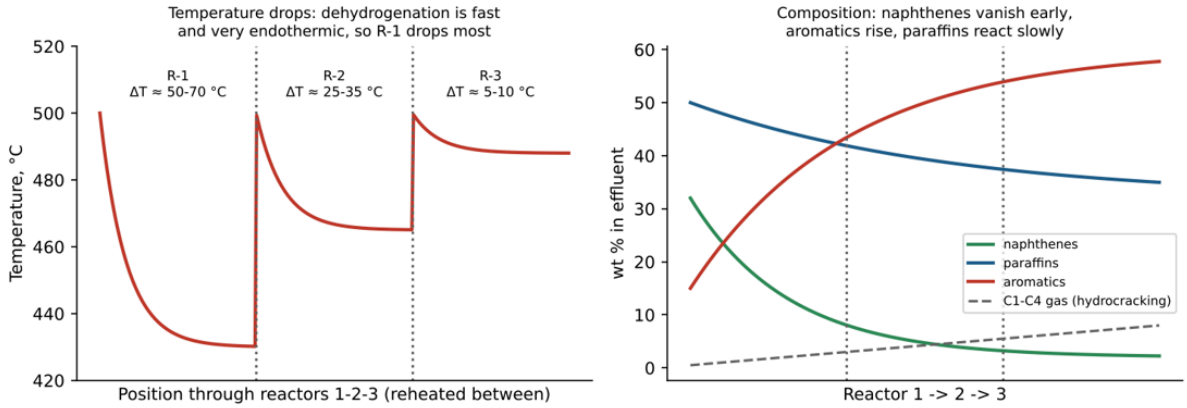
Aromatics yield is favoured by high temperature, low pressure, low space velocity and low H_2/HC ratio; but low H_2 pressure increases coke, so pressure is kept at 5-25 atm.

SR (semi-regenerative) reforming: process, temperature and composition profiles (2019 2b)



Semi-regenerative fixed-bed reformer with interheaters

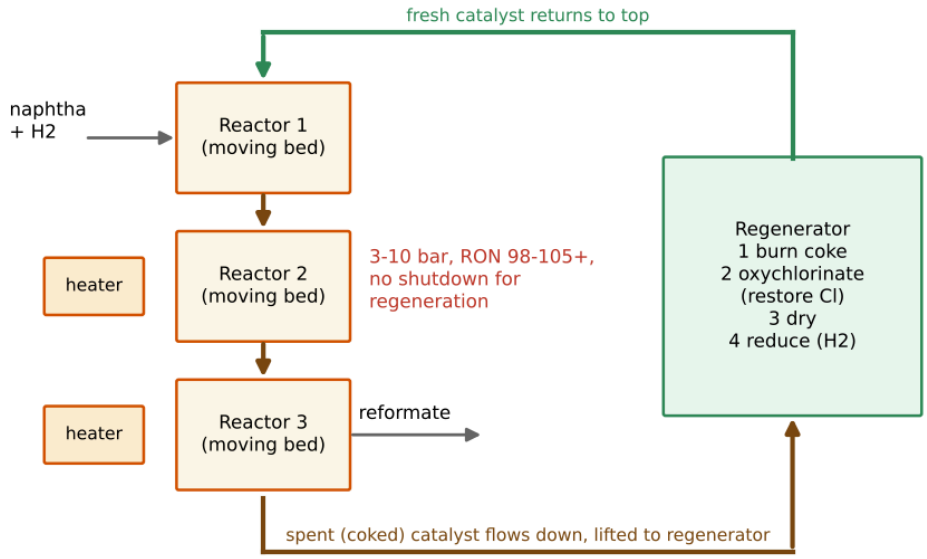
Hydrotreated naphtha is mixed with recycle hydrogen, heated in a furnace and passed through 3-4 fixed-bed reactors in series (500-525 °C, 12-15 bar, H_2/HC 5-7). Because the reactions are endothermic, the stream is reheated in a furnace before each reactor. The effluent is cooled; the separator gives H_2-rich gas (part recycled, the rest exported) and liquid that is stabilised (C_4− removed) to give reformate. The unit is shut down every 12-24 months to regenerate the catalyst (burn off coke).



Temperature drop and effluent composition through the three reactors

- **Reactor 1:** fast, very endothermic naphthene dehydrogenation: the largest temperature drop (50-70 °C); naphthenes almost disappear.
- **Reactor 2:** paraffin isomerization and naphthene dehydroisomerization: moderate drop.
- **Reactor 3:** slow dehydrocyclization (endothermic) and hydrocracking (exothermic) roughly cancel: small drop; reactors get larger down the train. Aromatics rise steadily; paraffins fall slowly; light gas rises.

CCR vs SR reforming (2019 2c, 2022B 2d)



CCR: moving-bed reactors stacked, catalyst regenerated continuously

Parameter	SR (semi-regenerative)	CCR (continuous catalyst regeneration)
Catalyst bed	fixed beds	moving bed, catalyst flows down and is lifted to a regenerator
Pressure, bar	12-15	3-10
H_2/HC, mol/mol	5-7	1.5-4
WHSV, h^{-1}	1-2	2-3
Catalyst	Pt-Re	Pt-Sn
Cycle	12-24 months	3-10 days (each particle loops in 5-7 days)
Temperature, °C	500-525	500-525
RON	92-100	98-105+
C_5^+ yield, wt%	75-84	85-92
H_2 yield, wt%	1.5-2.5	2.5-4.0
Regeneration	about 2 weeks shutdown	continuous, no shutdown
Capital cost	base	base × 1.3

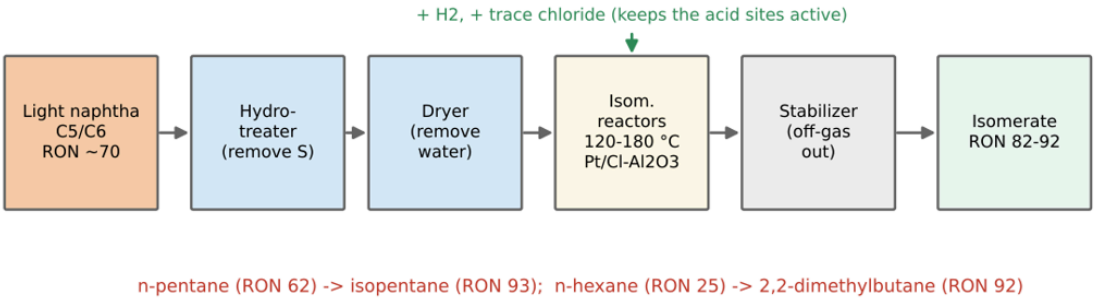
Catalyst poisoning (2020 3a)

Catalyst poisoning is the loss of catalyst activity when impurities in the feed adsorb strongly on (or react with) the active sites, so reactants can no longer reach them. Types and examples for reforming catalyst:

- **Permanent poisons** (cannot be removed; catalyst must be replaced): Pb, As, Hg, Cu, Na, Fe, Ni, Si.
- **Semi-permanent:** coke, removed by burning in regeneration.
- **Temporary poisons** (activity returns when the poison is removed from the feed): sulfur (Pt + S → PtS: lower reforming activity, more cracking; keep S < 1 ppm), organic nitrogen (forms NH_3 that neutralises the acid sites), water and chlorine imbalance.

- Other catalysts: Ni, V on FCC (fluid catalytic cracking) catalyst; organic N on hydrocracking catalyst; H_2S and CO_2 on others.

Isomerization: objectives (2019 4a)



Light naphtha isomerization unit

- Upgrade the octane of light straight-run naphtha (C_5-80 °C, RON about 70) by 12-20 numbers, to an isomerate of RON 82-92.
- Make a clean blending component: no aromatics, no benzene, no sulfur, no olefins.
- Saturate benzene and remove benzene precursors (C_6 would form benzene in the reformer).
- Light C_5/C_6 would only crack to gas in the reformer, so they are isomerized instead.
- n-butane → isobutane for alkylation feed and for oxygenates (MTBE).

Main and side reactions of isomerization (2019 4b, 2022B 4d)

Catalyst: Pt on chlorinated alumina (with organic chloride injection, very active, feed must be dry), Pt on zeolite (tolerates water) or Pt on metal oxides. Temperature 95-205 °C (low temperature favours the branched isomers at equilibrium), hydrogen atmosphere to limit coke. Reactions are slightly exothermic with no change in moles, so pressure has no effect on equilibrium.

Main reactions	Side reactions
n-pentane → isopentane (2-methylbutane): RON 62 → 92	Hydrocracking: $C_6H_{14} + H_2 \rightarrow C_3H_8 + C_3H_8$ (makes LPG (liquefied petroleum gas), loses yield)
n-hexane → 2-methylpentane, 3-methylpentane → 2,3-dimethylbutane → 2,2-dimethylbutane (RON 26 → 74 → 84 → 92-96)	Benzene saturation: $C_6H_6 + 3H_2 \rightarrow$ cyclohexane (exothermic; wanted for benzene limits)
Naphthene ring isomerization: methylcyclopentane ↔ cyclohexane	Ring opening: cyclohexane + H_2 → n-hexane
n-butane → isobutane	Coke formation, disproportionation ($2C_5 \rightarrow C_4 + C_6$)

C_5 and C_6 isomers and their RON (2019 2d)

Isomer	RON	MON (motor octane number)
n-pentane	61.7	61.9
isopentane (2-methylbutane)	92.3	90.3
n-hexane	24.8	26.0
2-methylpentane	73.4	73.5
3-methylpentane	74.5	74.3
2,2-dimethylbutane	91.8	93.4
2,3-dimethylbutane	100.5 (some tables 103)	94.3

Side note: Read from the teacher's table; the more branched the molecule, the higher the octane. Teacher's rounded RON values: nC5 62, iC5 92, nC6 26, 2,2-DMB 96, 2,3-DMB 84, 2-MP 74, 3-MP 74 (used below).

Isomerization numerical (2019 4c)

Given: light naphtha SG 0.724, 1000 m^3/h; yields: C_3 0.348, iC_4 0.619, nC_4 1.770, C_5^+ 97.261 wt%.

Find: product rates and C_5^+ composition.

Step (write this)	Why / what it means
`Feed mass = 1000 m³/h × 724 kg/m³ = 724,000 kg/h`	density = SG × 1000 kg/m^3
`C3 = 0.00348 × 724,000 = 2,520 kg/h`	each product = wt fraction × feed
`iC4 = 0.00619 × 724,000 = 4,482 kg/h`	
`nC4 = 0.01770 × 724,000 = 12,815 kg/h`	
`C5+ isomerate = 0.97261 × 724,000 = 704,170 kg/h`	yields add to 99.998%: rounding in the paper
`Split C5+ with the typical product composition (teacher's table)`	assumption: the question gives no split, so use the teacher's isomerate composition

Component	wt % of C_5^+	kg/h	RON
n-pentane	13	91,540	62
isopentane	42	295,750	92
n-hexane	3	21,130	26
2,2-dimethylbutane	15	105,630	96
2,3-dimethylbutane	5	35,210	84
2-methylpentane	15	105,630	74
3-methylpentane	7	49,290	74
Total	100	704,170	

Isomerate RON (linear volume blending, approximate) = $\Sigma x_i RON_i$ = 0.13×62 + 0.42×92 + 0.03×26 + 0.15×96 + 0.05×84 + 0.15×74 + 0.07×74 = **82.4**

This matches the 82-84 expected from a once-through isomerization unit, against a feed RON of about 70.

Remember: Reforming: Pt/Cl-alumina, 3 reactors with reheaters, dehydrogenation (fast, R-1 big ΔT), dehydroisomerization, dehydrocyclization (slow), isomerization, hydrocracking (bad). SR 12-15 bar Pt-Re vs CCR 3-10 bar Pt-Sn. Isomerization 95-205 °C, +12-20 RON.

A5 Cracking and coking: thermal cracking, hydrocracking, delayed and fluid coking, visbreaking, FCC

Pass set | Very likely · 86% | Medium

PAST-PAPER QUESTIONS

- **2023 Q3(a)** [6] Distinguish between thermal cracking and catalytic hydrocracking.
- **2023 Q3(b)** [10] Describe the delayed coking process with a neat sketch.
- **2023 Q3(c)** [8] Define visbreaking. Principal reactions during visbreaking.
- **2023 Q3(d)** [11] With neat labeled diagrams explain the working of an FCC (fluid catalytic cracking) unit.
- **2022B Q3(b)** [10] Same as 2023 3(b).
- **2022B Q3(c)** [8] Same as 2022 3(c).
- **2022B Q3(d)** [9] What distinguishes fluidized-bed catalytic cracking from fixed-bed catalytic cracking?
- **2022 Q3(b)** [10] Same as 2023 3(b).
- **2022 Q3(c)** [7] What is visbreaking? Differentiate between coil and soaker visbreaker.
- **2022 Q3(d)** [10] Same as 2023 3(d).
- **2021 Q6(b)** [8] Basic functions and types of catalytic cracking. What is "spent catalyst"?

★ REMEMBER IT

- **Thermal cracking**: heat alone, free radicals, more olefins and coke. **Hydrocracking**: catalyst + high-pressure H_2, saturated clean products (jet, diesel), no coke.
- **Delayed coking**: residue heated to ~500 °C in a furnace, then sent to a big **coke drum** where it cracks for ~24 h; vapours go to a fractionator, solid coke stays. Two drums: one filling, one being cut out with water jets.
- **Visbreaking**: mild cracking to cut the viscosity of residue. **Coil** type: high temperature, short time, cracking inside the furnace coil. **Soaker** type: lower temperature, longer time in a soaker drum (cheaper fuel, longer run).
- **FCC**: hot powdered zeolite catalyst flows like a liquid; feed cracks in seconds in the **riser**; spent catalyst (coked) is separated by cyclones, steam-stripped and burnt clean in the **regenerator**, which supplies the heat back to the riser.

Before you start: cracking

Demand is for light fuels (gasoline, diesel, jet) but every crude leaves a heavy "bottom of the barrel". **Cracking** breaks big molecules into smaller ones: by heat alone (thermal cracking, coking, visbreaking), by heat and a catalyst (FCC) or by a catalyst with hydrogen (hydrocracking). Source: Shameem sir's slides "Upgrading processes: coking and thermal processes; catalytic cracking; hydrocracking".

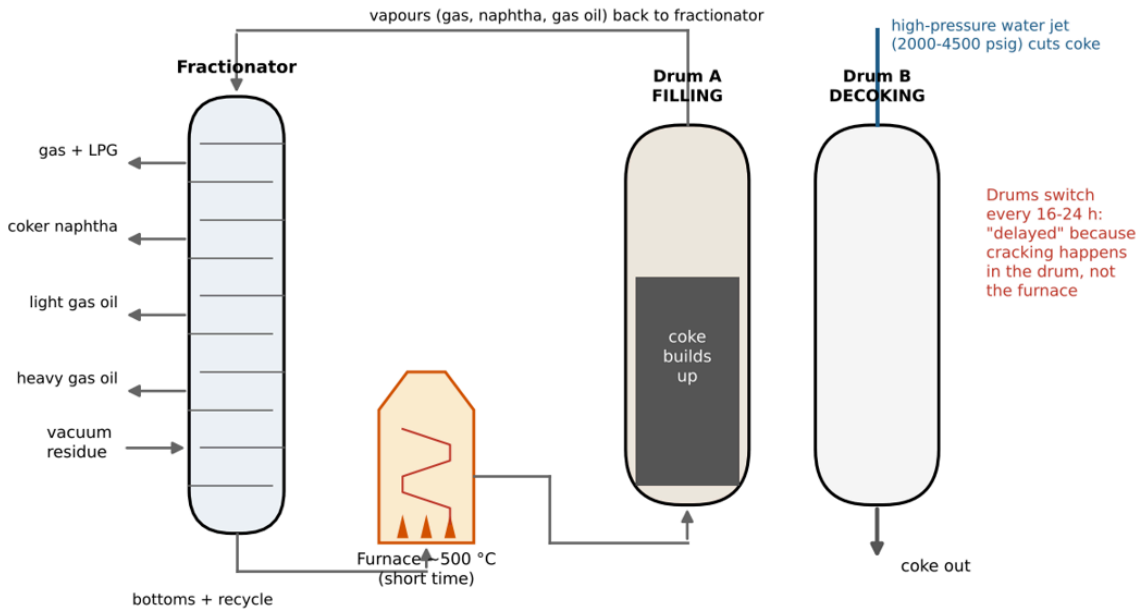
Thermal cracking vs catalytic hydrocracking (2023 3a)

Thermal cracking is the thermal decomposition, under pressure, of large hydrocarbon molecules into smaller ones, using heat only. **Catalytic hydrocracking** is catalytic cracking with hydrogenation superimposed, at high hydrogen pressure.

Thermal cracking	Catalytic hydrocracking
heat only, no catalyst	bifunctional catalyst: silica-alumina/zeolite (cracking) + Ni, W, Pt, Pd (hydrogenation)
450-750 °C, moderate pressure	about 260-430 °C, very high H_2 pressure (70-200 bar)
free-radical mechanism	carbonium-ion (acid) mechanism + hydrogenation
products olefinic, unstable, low octane; much gas and coke	products saturated, stable, sulfur- and nitrogen-free; little coke
cheap, simple; used for residue (coking, visbreaking)	expensive (H_2, high-pressure equipment); flexible, maximises jet/diesel
no hydrogen consumed	consumes 15-35 kg H_2 per tonne; net exothermic (cold H_2 quench)
feed: residues, heavy gas oils	feed: refractory cycle oils, coker and vacuum gas oils

Delayed coking (2023 3b, 2022 3b, 2022B 3b)

Delayed coking is a semi-continuous severe thermal cracking process that turns vacuum residue into gas, naphtha, gas oils and solid petroleum coke. It is "delayed" because the feed is heated quickly in the furnace but the cracking happens later, in large coke drums that provide the long residence time.



Delayed coking unit: fractionator, furnace and two coke drums

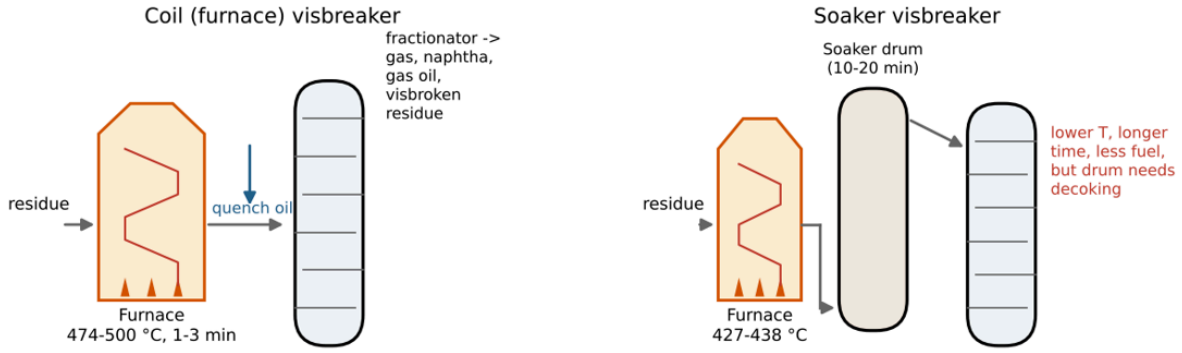
1. Fresh hot residue enters the bottom of the **fractionator** 2-4 trays above the vapour zone: it quenches the hot drum vapours, picks up recycle heavy ends and is preheated.
2. Fractionator bottoms (fresh feed + recycle) are heated in the **furnace** to about 480-500 °C with a short residence time (steam injected to prevent coking in the tubes).
3. The hot feed enters the bottom of one **coke drum** (filling). Cracking and coking take place there over many hours; coke builds up in the drum.
4. Vapours (gas, naphtha, gas oils, steam) leave the top of the drum and go back to the fractionator, where gas, LPG (liquefied petroleum gas), coker naphtha, light and heavy coker gas oil are drawn off (gas oil side stripper with 6-8 trays).

5. When the drum is nearly full (about every 16-24 h) the feed is switched to the empty drum. The full drum is steamed, cooled with water, opened, and the coke is cut out with **high-pressure water jets** (2000-4500 psig) on a rotating drill stem, after first boring an 18-24 in “rat hole”. Coke falls out of the bottom.

Visbreaking; principal reactions; coil vs soaker (2023 3c, 2022 3c, 2022B 3c)

Visbreaking (viscosity breaking) is a mild thermal cracking of vacuum residue to reduce its viscosity and pour point so it meets fuel oil specs with less expensive cutter stock (cuts heavy fuel oil 20-35% and cutter stock 20-30%); the gas oil made is extra FCC feed. Severity is limited because over-cracked tar becomes unstable and forms sludge.

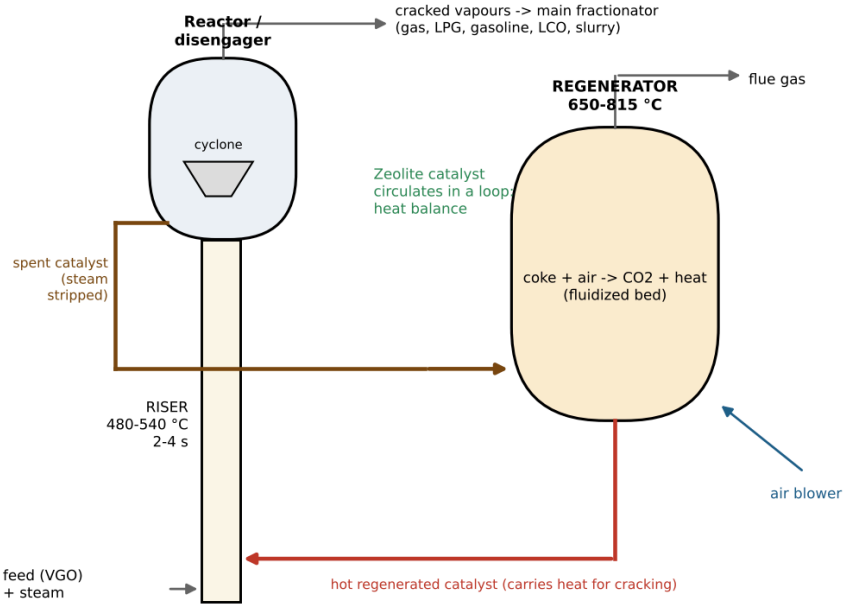
Principal reactions: (i) cracking of the long paraffinic side chains attached to naphthene and aromatic rings, at or near the ring, leaving short methyl/ethyl groups; (ii) cracking of resins to light hydrocarbons (mainly olefins) and to compounds that become asphaltenes; (iii) above 480 °C, some cracking of naphthene rings (little below 480 °C).



Coil (furnace) visbreaker and soaker visbreaker

Coil (furnace) visbreaker	Soaker visbreaker
cracking takes place in the furnace coil	cracking mainly in a soaking drum after the furnace
furnace outlet 474-500 °C, short time (1-3 min)	furnace outlet 427-438 °C, longer time (10-20 min) in the soaker
quenched at the furnace exit with gas oil	quenched after the soaker
more fuel used, higher temperature; easier decoking of the coil	less fuel, cheaper furnace, longer run; soaker drum needs decoking
produces slightly more stable products	higher selectivity to middle distillates

FCC: working with a labelled diagram (2023 3d, 2022 3d)



Fluid catalytic cracking: riser reactor and regenerator

1. **Feed:** vacuum gas oil (260-425 °C) with dispersion steam is injected at the bottom of the **riser**, where it meets hot regenerated zeolite catalyst (about 650-700 °C).
2. The oil vaporises and cracks in 2-4 s as catalyst and vapour rise together; average reaction temperature 480-540 °C. Cracking is endothermic; the heat comes from the hot catalyst.
3. In the **reactor/disengager**, cyclones separate catalyst from the cracked vapours. Vapours go to the main fractionator: gas, LPG (rich in propylene/butylene), FCC gasoline, light cycle oil, slurry oil.
4. The **spent catalyst** (coated with coke) is steam-stripped of oil and flows to the **regenerator**, where air blown in burns the coke to CO/CO_2 at 650-815 °C in a fluidised bed. This exothermic burning reheats the catalyst.
5. Hot **regenerated catalyst** flows back to the riser, carrying the heat for vaporisation and cracking (**heat balance**). Flue gas leaves through cyclones and electrostatic precipitators.

Fluidised-bed vs fixed-bed catalytic cracking (2022B 3d)

Fixed-bed (Houdry) catalytic cracking	Fluidised-bed catalytic cracking (FCC)
catalyst pellets stay in fixed reactors; several reactors cycle between cracking, purging and regeneration	fine catalyst powder (zeolite, 20-100 µm) behaves like a fluid and circulates continuously
batch/cyclic operation, complex valves	continuous operation
heat for cracking supplied separately; regeneration heat hard to use	regeneration heat carried by the catalyst to the reactor (heat balance)
poor temperature uniformity, hot spots	uniform temperature (good mixing)
catalyst activity drops during each cycle	constant catalyst activity (continuous regeneration)
small capacity, obsolete	large capacity, the main conversion unit today

Functions and types of catalytic cracking; spent catalyst (2021 6b)

Functions: convert heavy gas oils into high-octane gasoline, LPG (propylene, butylene for alkylation and petrochemicals) and light cycle oil; reduce heavy fuel oil. **Types:** fixed-bed (Houdry), moving-bed (TCC, Thermofor catalytic cracking) and fluidised-bed (FCC, including residue FCC). **Spent catalyst** is catalyst that has lost activity because coke (and metals Ni, V) has deposited on its surface during cracking. In FCC it is steam-stripped and sent to the regenerator, where the coke is burnt off and activity restored.

Remember: Thermal = heat only, olefinic products; hydrocracking = catalyst + H_2, saturated clean products. Delayed coker: furnace 500 °C, two drums, switch 16-24 h, water jets. Visbreaker coil 474-500 °C vs soaker 427-438 °C. FCC: riser 480-540 °C, regenerator 650-815 °C, heat balance.

A7 Product blending: objectives, blending indices (RVP, viscosity, flash point), viscosity index numericals Pass set Likely · 77% Medium

PAST-PAPER QUESTIONS

- **2023 Q4(a)** [7] Objectives of product blending? How to calculate viscosity blending.
- **2023 Q4(c)** [8] What is viscosity index? Calculate VI: v_{40} = 73.3 cSt, v_{100} = 8.86 cSt, L = 120, H = 70.
- **2022B Q4(a)** [8] Same as 2022 4(a).
- **2022B Q4(b)** [10] Same with 75, 100 and 200 cSt.
- **2022 Q4(a)** [8] Objectives of product blending? How to calculate RVP (Reid vapour pressure) blending.
- **2022 Q4(c)** [10] Estimate the viscosity of a blend: 2000 bbl at 72 cSt, 3000 bbl at 100 cSt, 4000 bbl at 201 cSt (130 °F).
- **2019 Q3(b)** [8] Same as 2022 4(a).
- **2019 Q3(c)** [18] Same with 75, 100 and 200 cSt.

UNDERSTAND **Read this to understand; you do not need to write it**

Most properties do **not** mix in proportion to volume. The trick is to convert each component's property into a **blending index** that does add linearly, average the indices by volume, then convert back. Viscosity: BI = logv/(3 + logv), back with $v = 10^{3BI/(1-BI)}$. RVP: $BI = RVP^{1.25}$, back with $BI^{0.8}$.

CALCULATOR **Viscosity blend of three oils (2022 4(c), 2022B 4(b), 2019 3(c))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Total volume: **2000** + **3000** + **4000** = **9000** bbl
2. BI_1: BI = logv/(3 + logv) = log10(**75**) / (3 + log10(**75**)) = **0.3846**
3. BI_2: log10(**100**) / (3 + log10(**100**)) = **0.4000**
4. BI_3: log10(**200**) / (3 + log10(**200**)) = **0.4341**
5. Blend index: $BI_{blend} = \Sigma x_i BI_i$ = (**2000** × **0.3846** + **3000** × **0.4** + **4000** × **0.4341**) / **9000** = **0.41173**
6. Blend viscosity: $v = 10^{3BI/(1-BI)}$ = pow(10, 3 × **0.4117** / (1 - **0.4117**)) = **125.8** cSt
7. Simple average (wrong): (**2000** × **75** + **3000** × **100** + **4000** × **200**) / **9000** = **138.9** cSt

2022 version: 72, 100, 201 cSt gives 124.8 cSt.

CALCULATOR **RVP blend of two streams**

Change any number; every step is recalculated and shown with the numbers put in.

1. BI_1: $BI = RVP^{1.25}$ = pow(**4**, 1.25) = **5.66**
2. BI_2: pow(**12**, 1.25) = **22.33**
3. Blend BI: **0.6** × **5.657** + (1 - **0.6**) × **22.33** = **12.33**

4. Blend RVP: RVP = $BI^{0.8}$ = pow(**12.33**, 0.8) = **7.46** psi

CALCULATOR **Viscosity index (2023 4(c))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Viscosity index: VI = 100(L − U)/(L − H) = 100 × (**120** - **73.3**) / (**120** - **70**) = **93.4**

L and H are read from the ASTM D2270 table at the oil's 100 °C viscosity (8.86 cSt here).

Before you start: blending indices

A refinery unit does not make a finished product; it makes components. Blending mixes them in the right proportions. Some properties mix in a straight line (by volume or weight fraction): specific gravity, sulfur, boiling point. Others do not (viscosity, RVP, flash point, pour point), so each is first converted to a **blending index** that does mix linearly, the indices are averaged, and the result converted back. Source: Shameem sir's slides "Product blending" (following Fahim, *Fundamentals of Petroleum Refining*, ch. 9).

Objectives of product blending (2023 4a, 2022 4a, 2022B 4a, 2019 3b)

- Refining processes give semi-finished streams; blending combines them (with additives) into products that meet **specifications** (octane, RVP, flash point, pour point, viscosity, sulfur).
- Allocate the available components to meet product **demands** at the **lowest cost** and to make extra products that **maximise profit** (often by linear programming).
- Use low-value streams (e.g. n-butane, cutter stock) where specs allow; give flexibility to change product slate with the season (winter vs summer RVP).
- Final quality is checked by laboratory tests: gasoline (octane, RVP, volatility), kerosene (flash point, volatility), gas oil (diesel index, flash point, pour point, viscosity).

Linear mixing rule: $P_{blend} = \Sigma\ x_i P_i$ (x_i = volume or weight fraction)

How to calculate RVP blending (2022 4a, 2022B 4a, 2019 3b)

RVP is not additive. Use the Chevron RVP blending index:

Step (write this)	Why / what it means
$BI_{RVP,i} = RVP_i^{1.25}$ (RVP in psi)	convert each component to its index
$BI_{RVP,blend} = \Sigma\ x_{vi}\ BI_{RVP,i}$	average the indices by volume fraction
$RVP_{blend} = (BI_{RVP,blend})^{1/1.25} = (BI)^{0.8}$	convert back

Example (to show the method): 60 vol% reformate RVP 4 psi + 40 vol% LSR (light straight-run (naphtha)) RVP 12 psi. BI = $4^{1.25}$ = 5.66 and $12^{1.25}$ = 22.33; blend BI = 0.6×5.66 + 0.4×22.33 = 12.33; RVP = $12.33^{0.8}$ = 7.5 psi. To find the n-butane (RVP 52 psi) needed to reach a target RVP, solve $x_b \cdot 52^{1.25} + (1 - x_b)BI_{rest} = target^{1.25}$ for x_b.

Other indices: flash point $\log_{10} BI_{FP} = -6.1188 + 2414/(FP - 42.6)$, FP in °F (blended by weight); pour point $BI_{PP} = PP^{1/0.08}$, PP in °R; octane often blended linearly by volume (approximate).

How to calculate viscosity blending (2023 4a)

Step (write this)	Why / what it means
$BI_{vis,i} = \log_{10}v_i/(3 + \log_{10}v_i)$ (v in cSt)	viscosity index of each component, same temperature
$BI_{blend} = \Sigma\ x_{vi}\ BI_{vis,i}$	volume-fraction average
$\log_{10}v_{blend} = 3BI_{blend}/(1 - BI_{blend})$, so $v_{blend} = 10^{3BI/(1-BI)}$	rearrange BI = L/(3 + L) ⇒ L = 3BI/(1 − BI)

Viscosity blend numericals (2022 4c; 2022B 4b and 2019 3c)

Given (2022B, 2019): 2000 bbl at 75 cSt, 3000 bbl at 100 cSt, 4000 bbl at 200 cSt, all at 130 °F. Total 9000 bbl.

Step (write this)	Why / what it means
$x_v = 2000/9000 = 0.2222,\ 3000/9000 = 0.3333,\ 4000/9000 = 0.4444$	volume fractions
$BI_1 = \log 75/(3 + \log 75) = 1.8751/4.8751 = 0.3846$	$\log_{10}75 = 1.8751$
$BI_2 = 2/(3 + 2) = 0.4000$	$\log_{10}100 = 2$
$BI_3 = 2.3010/5.3010 = 0.4341$	$\log_{10}200 = 2.3010$
$BI_{blend} = 0.2222\times0.3846 + 0.3333\times0.4000 + 0.4444\times0.4341 = 0.08547 + 0.13333 + 0.19292 = 0.41173$	
$\log v = 3\times0.41173/(1 - 0.41173) = 1.23518/0.58827 = 2.0997$	back-convert
$v_{blend} = 10^{2.0997} =$ **125.8 cSt at 130 °F**	less than the simple volume average 138.9 cSt: viscosity blends non-linearly

2022 version (72, 100 and 201 cSt, same volumes): BI = 0.3824, 0.4000, 0.4343; BI_{blend} = 0.2222×0.3824 + 0.3333×0.4 + 0.4444×0.4343 = 0.41133; log v = 1.23399/0.58867 = 2.0962; **v_{blend} = 124.8 cSt**.

Viscosity index and VI numerical (2023 4c)

Viscosity index (VI) is a number (originally 0-100) showing how much an oil's viscosity changes with temperature. The smaller the change, the higher the VI. Paraffinic oils VI about 100, naphthenic about 40 or less. It compares the oil with two reference oils that have the **same viscosity at 100 °C**: one of VI 0 (viscosity L at 40 °C, changes a lot) and one of VI 100 (viscosity H at 40 °C).

$$VI = 100\ (L - U)/(L - H)$$

Given: v_{40} = U = 73.3 cSt, v_{100} = 8.86 cSt, L = 120, H = 70 cSt.

Step (write this)	Why / what it means
$VI = 100(120 - 73.3)/(120 - 70)$	substitute
$= 100 \times 46.7/50$	subtract
$=$ **93.4 ≈ 93**	a good, mostly paraffinic oil

Side note: The teacher's worked example uses the same oil with L = 119.94 and H = 69.48 read from the ASTM (American Society for Testing and Materials) D2270 table by interpolation, giving VI = 100 × 46.64/50.46 = 92.4. The exam rounds L and H to 120 and 70, so write 93.4 and mention that the table values give 92.

Remember: Linear: SG, S, boiling point. Non-linear → index: $RVP^{1.25}$; viscosity logv/(3 + logv), back by $10^{3BI/(1-BI)}$; VI = 100(L − U)/(L − H). Blend of 75/100/200 cSt in 2:3:4 = 125.8 cSt.

Section B: Natural gas processing

B5 Gas sweetening: reasons for removing H_2S and CO_2, amine process, iron sponge, molecular sieve, sulfur recovery

Pass set · Very likely · 95% · Easy

PAST-PAPER QUESTIONS

- **2023 Q8(a)** [12] Explain the alkaline amine process.
- **2023 Q7(b)** [8] Reasons for removing H_2S and CO_2 from natural gas.
- **2022B Q7(b)** [8] Short notes: desulphurization and regeneration.
- **2022 Q7(a)** [12] Alkaline amine process for gas sweetening with a block diagram.
- **2021 Q3(c)** [10] Iron sponge unit for removal of H_2S with a schematic.
- **2021 Q4(c)** [10] Iron sponge vs solid bed (molecular sieve) sweetening for 3% CO_2, 0.02% H_2S on a low budget: which and why.
- **2020 Q4(c)** [10] Same as 2023 7(b).

★ REMEMBER IT

- Remove H_2S and CO_2 because: H_2S is toxic and corrosive, burns to SO_2; CO_2 forms carbonic acid with water and lowers heating value; both freeze in LNG (liquefied natural gas) plants. Sweet gas spec: 4 ppm H_2S.
- **Amine process**: sour gas up through the **absorber (contactor)**, lean amine (MEA, DEA (diethanolamine), MDEA (methyldiethanolamine)) down; rich amine → flash tank → lean/rich heat exchanger → **regenerator (stripper)** with reboiler boils off H_2S and CO_2 → lean amine cooled and pumped back. Acid gas goes to a Claus unit for sulfur.
- **Iron sponge**: wood chips coated with Fe_2O_3; $Fe_2O_3 + 3H_2S \rightarrow Fe_2S_3 + 3H_2O$. Cheap for small amounts of H_2S, does not remove CO_2. For CO_2 too, use molecular sieve or amine.

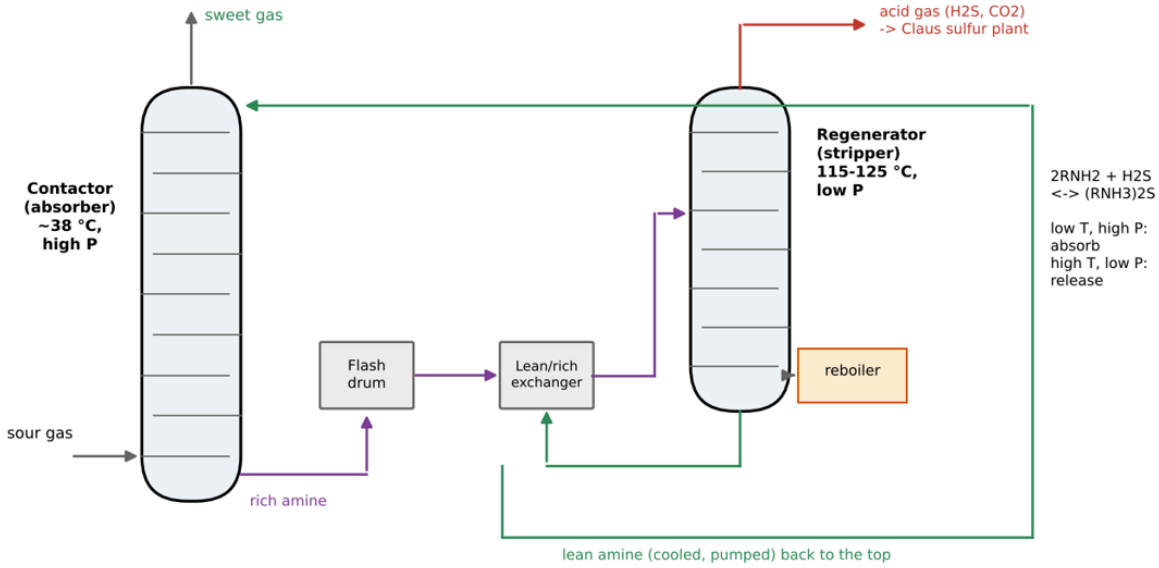
Before you start: acid gases

H_2S and CO_2 are called **acid gases** because they form acids with water. They have no heating value and cause toxicity, corrosion and freezing problems. Removing them is **gas sweetening**: sour gas in, sweet gas out. Liquids absorb them (amines, carbonate, physical solvents) or solids adsorb or react with them (iron sponge, molecular sieve). Source: Muhit sir's lectures L11-L12 (desulfurization).

Reasons for removing H_2S and CO_2 (2023 7b, 2020 4c)

H_2S	CO_2
deadly poisonous: above 600 ppm fatal in 3-5 min (as toxic as cyanide); foul odour even at low concentration; cannot be in domestic gas	no heating value: removing it raises the energy content per unit volume (sales spec usually < 2-3%)
corrosive to all metals in gas systems; sulfide stress cracking	dissolves in water/condensate to carbonic acid H_2CO_3, corroding carbon steel (iron carbonate scale)
burns to SO_2, toxic and corrosive, acid rain	freezes (solid CO_2) in cryogenic and LNG plants, plugging exchangers
poisons catalysts in downstream refinery processes	promotes hydrate formation
promotes hydrate formation; sales spec < 4 ppm (0.25 grain/100 scf)	takes up pipeline capacity; greenhouse gas

Alkaline amine process (2023 8a, 2022 7a)



Amine gas sweetening unit (contactor and regenerator)

The most widely used process for H_2S and CO_2 removal: good selectivity at low cost, flexible. Solvents are aqueous solutions of alkanolamines: MEA (monoethanolamine), DEA (diethanolamine), TEA, MDEA (methyldiethanolamine, selective for H_2S), weak bases with pH 9-11.

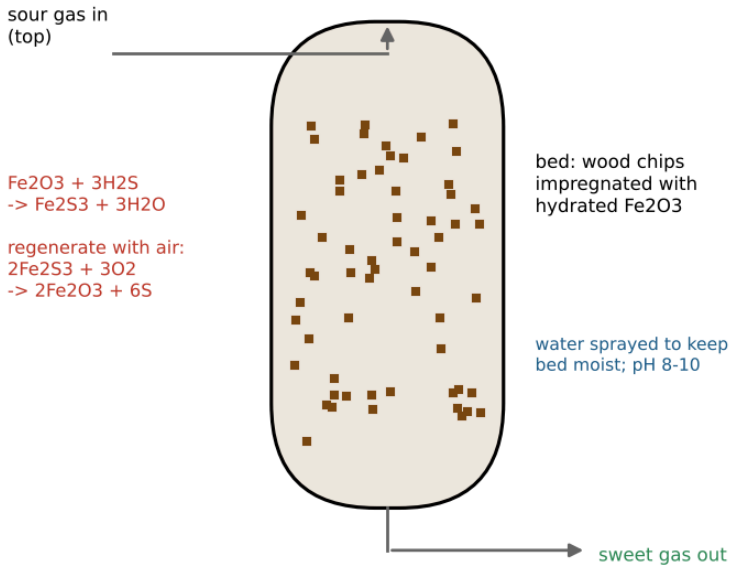
1. **Contactor (absorber):** sour gas enters the bottom of a trayed or packed column and flows up against lean amine flowing down, at about 38 °C and high pressure. The amine reacts with the acid gases: $2RNH_2 + H_2S \leftrightarrow (RNH_3)_2S$; $2RNH_2 + CO_2 + H_2O \leftrightarrow (RNH_3)_2CO_3$. Sweet gas leaves the top (to dehydration).
2. **Flash drum:** rich amine from the bottom is let down in pressure to release dissolved hydrocarbons (used as fuel).
3. **Lean/rich exchanger:** rich amine is heated by hot lean amine (energy saving).
4. **Regenerator (stripper):** at low pressure and 115-125 °C, with a steam-heated reboiler, the reactions reverse; acid gas (H_2S, CO_2, water vapour) goes overhead through a reflux condenser to a Claus sulfur unit or incinerator.

5. **Lean amine** from the reboiler is cooled in the exchanger and a cooler, filtered and pumped back to the top of the contactor.

> **Memory trick:** Amine likes **Cold and High pressure** to absorb, **Hot and Low pressure** to release: "CH in, HL out".

MEA (monoethanolamine) vs DEA: MEA has the greater H_2S cleaning efficiency; DEA has lower vapour pressure (much smaller solvent losses), is less corrosive, needs a lower circulation rate and less heat per volume to regenerate, and resists degradation by COS and CS_2.

Iron sponge unit (2021 3c)



Iron sponge vessel

The oldest and still widely used batch process for gases with small H_2S concentrations (usually < 300 ppm) at 50-500 psig. Hydrated ferric oxide is impregnated on wood chips (grades 6.5-20 lb Fe_2O_3 per bushel) giving a large reactive surface, packed in a vessel. Sour gas flows down through the bed; H_2S reacts:

$$Fe_2O_3 + 3H_2S \rightarrow Fe_2S_3 + 3H_2O; \quad FeO + H_2S \rightarrow FeS + H_2O$$

The bed needs slightly alkaline water (pH 8-10, by injecting caustic soda or soda ash with water) and temperature below 47 °C (110 °F). **Regeneration** with air, slowly because it is exothermic and can ignite the bed: $2Fe_2S_3 + 3O_2 \rightarrow 2Fe_2O_3 + 6S$ + heat. Sulfur builds up as a cake; after about 10 cycles the bed is replaced (keep spent chips wet so they do not smoulder). Does **not** remove CO_2.

Advantages: complete removal of small/moderate H_2S; low investment for small volumes; works at any pressure; removes some mercaptans. **Disadvantages:** batch (needs two vessels or interrupts flow); hydrates at high pressure; removes the ethyl mercaptan odorant; oil coating shortens bed life; disposal of spent bed.

Desulphurization and regeneration (short notes) (2022B 7b)

- **Desulphurization (sweetening)**: removal of H_2S (and other sulfur compounds, with CO_2) from natural gas. Types: adsorption on a solid (dry: iron sponge, molecular sieve, zinc oxide) and

absorption into a liquid (wet: amines, hot carbonate, physical solvents like Selexol, water wash), each physical or chemical. Also classed as non-regenerative (Chemsweet, Slurrisweet), regenerative with H_2S recovery (amines, carbonate, molecular sieve) and regenerative with elemental sulfur recovery (Stretford, liquid redox). Selection depends on the type and amount of impurity, selectivity, p and T, gas volume, CO_2/H_2S ratio, economics and sulfur recovery needs.

- **Regeneration**: restoring the activity of a solvent or solid adsorbent that is saturated with impurities so it can be reused. Solid beds (molecular sieve, silica gel, activated carbon) are heated with a hot gas stream (400-600 °F sweet gas, steam or N_2) that desorbs the impurities, then cooled; liquids (amine, glycol) are heated and stripped at low pressure; iron sponge is regenerated by controlled air oxidation.

Iron sponge or solid-bed (molecular sieve) for 3% CO_2, 0.02% H_2S on a low budget? (2021 4c)

0.02% H_2S = 200 ppm, below the 300 ppm iron-sponge limit. Iron sponge removes H_2S only, leaving the 3% CO_2, which is acceptable for most pipelines (or blended); it has the lowest capital and chemical cost, is simple, and works at any pressure. A molecular sieve bed would also remove the H_2S (and dry the gas) but needs hot regeneration gas (400-600 °F), loses 1-2% of the gas, and its H_2S capacity falls when CO_2 is high (low H_2S/CO_2 ratio, CO_2 competes for sites); it costs much more. **Choice: iron sponge.** If the 3% CO_2 must also be cut to a 2% spec, add an amine unit later.

Molecular sieve, carbonate and Claus processes (for completeness)

- **Molecular sieve:** synthetic crystalline sodium aluminosilicate; physical adsorption by molecular size on charged sites; dehydrates and desulfurizes together; best near 450 psia; capacity rises with H_2S/CO_2 ratio and falls with temperature; regenerated with 400-600 °F sweet gas.
- **Hot carbonate:** K_2CO_3 solution (pH 8.5-9.5); very corrosive, needs inhibitors; not for very low H_2S specs.
- **Claus sulfur recovery** of the amine acid gas: $H_2S + 3/2O_2 \rightarrow SO_2 + H_2O$ (thermal stage), then $SO_2 + 2H_2S \rightarrow 3S + 2H_2O$ (thermal and catalytic stages).

Remember: H_2S: toxic (600 ppm fatal), corrosive, SO_2, catalyst poison. CO_2: no heating value, carbonic acid, freezes in LNG. Amine: absorb cold/high P, strip hot/low P. Iron sponge: $Fe_2O_3 + 3H_2S \rightarrow Fe_2S_3 + 3H_2O$, < 300 ppm, keeps CO_2.

B6 LNG and LPG: LNG plant and refrigeration cycles, storage and carriers, LPG uses and hazards, LPG vs LNG vs CNG, NGL recovery Pass set

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q7(a)** [12] With a schematic of an LNG (liquefied natural gas) plant, describe the overall LNG production process.
- **2023 Q8(b)** [8] Commercial and industrial uses of LPG (liquefied petroleum gas).
- **2022B Q7(a)** [12] Same as 2023 7(a).
- **2022B Q8(d)** [10] Same as 2021 8(b).
- **2022 Q7(c)** [15] What is LNG? Properties; schematic of an LNG plant and overall production process.
- **2022 Q8(b)** [15] Potential hazards and remedies of LPG.
- **2021 Q7(a)** [10] Construction of an LNG carrier with a figure.
- **2021 Q7(b)** [10] Same as 2023 8(b).
- **2021 Q7(c)** [15] Potential hazards and safety legislation of LPG.
- **2021 Q8(b)** [12] Tabulate major differences among LPG, LNG and CNG (compressed natural gas).
- **2020 Q4(b)** [5] Short note on LPG and LNG.
- **2019 Q5(a)** [20] Same as 2022 7(c).
- **2019 Q5(b)** [6] Different types of LNG plants.
- **2019 Q5(c)** [9] Why at least three refrigerants are needed to liquefy natural gas.
- **2019 Q6(a)** [20] Cascade LNG process with schematic and T-s diagrams of the refrigerants.
- **2019 Q8(c)** [10] LPG and NGL (natural gas liquids) extraction process with a flow diagram.
- **2019 Q7(a)** [13] Kettle-type shell-and-tube heat exchanger with a schematic.

★ REMEMBER IT

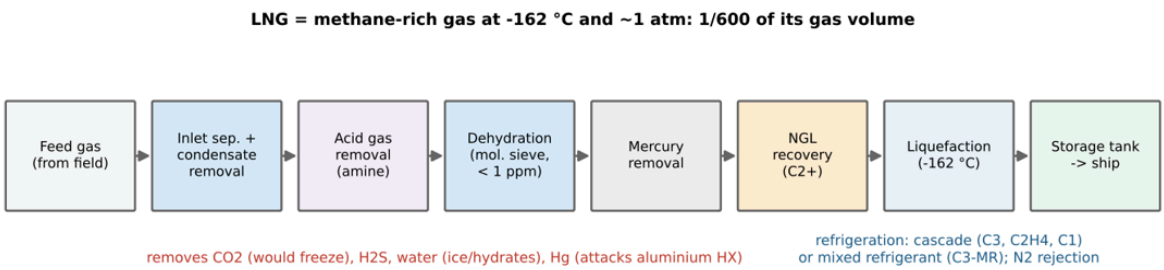
- LNG = methane cooled to −162 °C at 1 atm, 1/600 of the gas volume. Plant chain: **pre-treat** (remove CO_2, H_2S, water, mercury) → **NGL removal** (heavies) → **liquefaction** (refrigeration) → storage tanks → ship.
- Why three refrigerants (cascade): one refrigerant cannot span from ambient to −162 °C. **Propane** cools to about −35 °C, **ethylene** to about −100 °C, **methane** to −160 °C; each condenses against the one before.
- **LPG** = propane + butane, liquid at ~8 bar room temperature (cooking, autogas, industry). **CNG** = methane at 200–250 bar (vehicles). LPG hazards: heavier than air (pools in low spots), BLEVE (boiling liquid expanding vapour explosion), leaks; remedies: ventilation, odorant, relief valves, leak detectors.

Before you start: liquefied gases

Gas takes a lot of space. Cool methane to about −162 °C at atmospheric pressure and it becomes **LNG**, 1/600 of its gas volume, so it can be shipped across oceans. Propane and butane liquefy under modest pressure at room temperature: that is **LPG**, sold in cylinders. **CNG** is methane compressed to 200-250 bar, still a gas. Source: Muhit sir's lectures L13-L16; cascade and NGL details from Guo & Ghalambor (Natural Gas Engineering Handbook) and Mokhatab.

LNG: properties; schematic of an LNG plant and the production process (2023 7a, 2022B 7a, 2022 7c, 2019 5a)

LNG (liquefied natural gas) is natural gas (mainly methane, 85-99%) cooled to about −160 to −162 °C so it becomes a liquid at near-atmospheric pressure. **Properties:** volume about 1/600 of the gas; density about 430-470 kg/m^3 (less than half of water, floats); colourless, odourless, non-toxic, non-corrosive; boils off if warmed; vapour is flammable only at 5-15% in air; heating value about 50-55 MJ/kg. Uses: power generation, industry, transport, residential (after regasification).



Block diagram of an LNG liquefaction plant

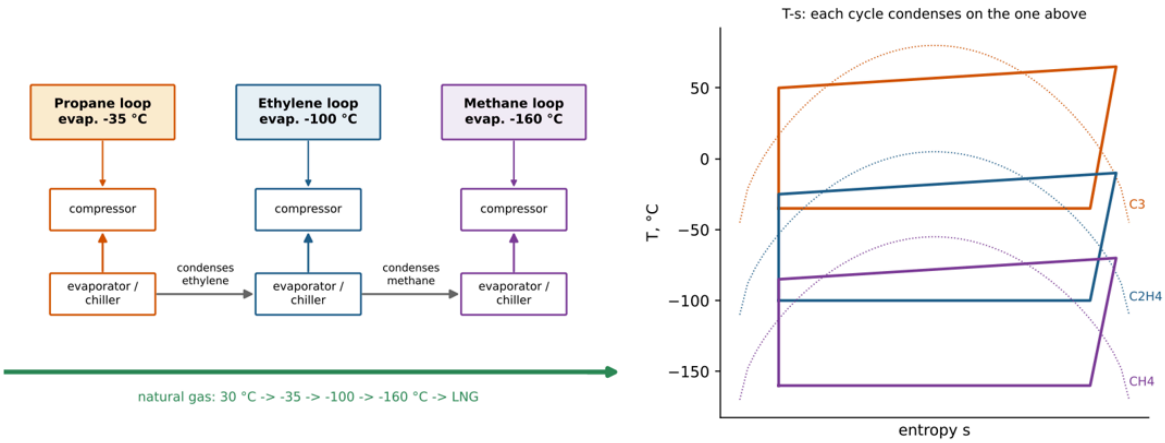
1. **Inlet receiving:** slug catcher and separator remove liquids (condensate, water) from the feed gas.
2. **Acid gas removal** (amine unit) takes CO_2 below about 50 ppm (it would freeze) and H_2S below 4 ppm; a sulfur recovery unit treats the acid gas.
3. **Dehydration** with molecular sieves to below 1 ppm water (ice and hydrates).
4. **Mercury removal** beds (sulfur-impregnated carbon) protect the aluminium heat exchangers from mercury attack.
5. **NGL recovery**: C_2^+ and C_3^+ are recovered (scrub column), fractionated into ethane, propane, butane, condensate for sale; this also prevents heavy components freezing.
6. **Liquefaction**: the lean gas is chilled and liquefied in a refrigeration process: cascade, mixed refrigerant, propane-precooled mixed refrigerant (C3-MR, most common) or gas expander cycle.
7. **Nitrogen rejection** (cryogenic) if N_2 exceeds about 1 mol%; flash/end-flash gas used as fuel.
8. **Storage** in full-containment tanks (9% Ni steel inner tank, pre-stressed concrete outer tank, insulated) at −160 °C, then loading onto LNG carriers. At the receiving end a regasification terminal warms it back to gas.

Types of LNG plants (2019 5b)

- **Base-load (export) plants**: very large trains (3-8 MTPA each) at gas fields, feeding ships (Qatar, Australia).
- **Peak-shaving plants**: liquefy pipeline gas in low-demand seasons, store it and regasify at peak demand.
- **Small-scale / mid-scale plants**: for remote fields, trucking, marine bunkering.
- **Floating LNG (FLNG)** liquefaction vessels offshore; **FSRU** (floating storage and regasification unit) (floating storage and regasification units), as used by Bangladesh at Moheshkhali.
- **Receiving and regasification terminals**: unload, store and vaporise LNG.
- By refrigeration cycle: cascade, mixed-refrigerant, C3-MR, expander plants.

Why at least three refrigerants (2019 5c); the cascade process (2019 6a)

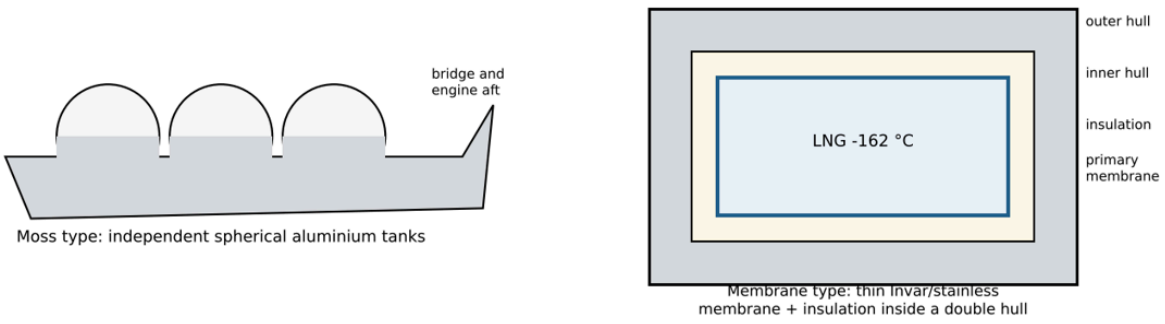
A refrigerant can only cool effectively over the range between its own boiling point (at the evaporator pressure) and the temperature at which it can be condensed. No single pure refrigerant can be condensed by ambient air/water and also boil at −160 °C. Propane boils at about −42 °C (evaporates at −35 to −40 °C) and condenses with cooling water; ethylene can reach −100 °C but its critical temperature (9.5 °C) is too low to condense with water, so propane must condense it; methane reaches −160 °C but must be condensed by ethylene. So at least three refrigerants are "cascaded", each condensing on the cold side of the one above. Using stages also keeps the temperature difference to the gas small at every level, so less work is wasted.



Classical cascade: propane, ethylene and methane cycles and their T-s diagrams

1. **Propane cycle:** propane compressed, condensed by cooling water, expanded and evaporated at about −35 °C in chillers that precool the natural gas and condense the ethylene.
2. **Ethylene cycle:** ethylene compressed, condensed by the propane chiller, expanded and evaporated at about −100 °C, cooling and partly liquefying the gas and condensing methane.
3. **Methane cycle:** methane (open or closed loop) compressed, condensed against ethylene, expanded and evaporated at about −160 °C, subcooling the LNG.
4. The gas is cooled 30 → −35 → −100 → −160 °C and flashed to storage pressure. Each cycle on a T-s diagram is a vapour-compression cycle (compression, condensation, throttling, evaporation) at its own temperature level, stacked one under the other.
5. Advantages: high efficiency, simple pure refrigerants, flexible; disadvantage: many compressors and exchangers (high capital cost). Used in the ConocoPhillips Optimized Cascade process.

Construction of an LNG carrier (2021 7a)



Moss-type and membrane-type LNG carriers

LNG carriers are special **double-hull** ships: the space between inner and outer hull protects the cargo tanks in a collision and reduces heat transfer, and the insulated tanks sit inside the inner hull so the hull steel never gets cold enough to become brittle. Two cargo containment types:

- **Moss (Type B spherical) tanks**: self-supporting spheres of aluminium alloy or 9% Ni steel, held at the equator by a skirt, with external insulation; the domes show above deck.
- **Membrane tanks** (GTT): thin primary and secondary membranes of Invar (36% Ni steel) or corrugated stainless steel, supported by the hull through load-bearing insulation; use the hull space efficiently.
- Cargo tanks are separated from each other by cofferdams (dry compartments); boil-off gas (0.1-0.15%/day) is burnt as fuel or reliquefied; gas and fire detection and suppression systems; unloading takes 14-16 h with the cargo kept at −160 °C.

Commercial and industrial uses of LPG (2023 8b, 2021 7b)

LPG is a mixture of propane and butane (in Bangladesh about 30% propane, 70% butane), colourless, odourless (odorant added), non-toxic, highly flammable and **heavier than air**.

Commercial uses	Industrial uses
cooking in homes, hotels, restaurants	ceramics: drying and firing tiles, pottery, sanitary ware (clean, controllable heat)
refrigerant replacing CFCs (fridges, car air conditioners)	food processing: bakeries, biscuits, dairy pasteurisation, sterilisation (no contamination)
autogas: vehicle fuel (burns cleaner than petrol/diesel)	metal processing: heat treatment, cutting, melting aluminium, galvanising, paint drying
agriculture: crop drying, poultry heating	textiles: dye baths, drying, singeing
central heating of buildings	printing and paper; chemical feedstock (petrochemicals); forklift trucks; power generation; road construction (bitumen heating)

Potential hazards of LPG, remedies and safety legislation (2022 8b, 2021 7c)

Hazard	Remedy
Fire/explosion from leakage: worn or holed hoses, poor stove maintenance, valve left open; gas (heavier than air) collects at floor level and explodes when ignited	use approved hoses and regulators, replace expired tubes, close cylinder valve after use, clean stove, gas leak detectors, check leaks with soap solution (never a flame), ventilate low areas
Careless storage or handling: cylinder lying down, below ground level, near combustibles	store upright, at ground level, ventilated, away from wood/paper/cloth, not two cylinders together
Unattended cooking leading to fire	never leave cooking unattended, keep a fire extinguisher
Suffocation/asphyxiation by inhaling LPG; cold burns from liquid contact	ventilation, leak sensors, medical help; rinse eyes/skin
BLEVE (boiling liquid expanding vapour explosion) of a cylinder heated by fire	pressure relief valves, cylinder testing, keep cylinders cool

Safety legislation (Bangladesh): Explosives Act 1884 and the LP Gas Rules 2004 (licensing by the Department of Explosives for storage, filling, transport and cylinder testing); Gas Cylinder Rules

1991; BSTI standards for cylinders, regulators and valves; Bangladesh Energy Regulatory Commission (BERC) Act 2003 for pricing and licensing; fire safety under the Fire Prevention and Extinguishing Act 2003. Cylinders must be periodically tested and fitted with safety valves.

Short note on LPG and LNG; LPG vs LNG vs CNG (2020 4b, 2021 8b, 2022B 8d)

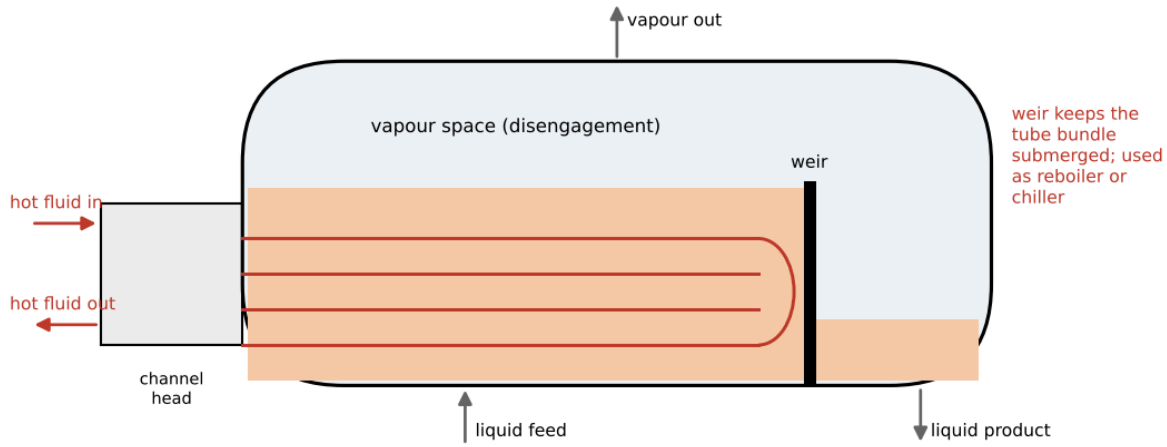
	LPG	LNG	CNG
Full name	liquefied petroleum gas	liquefied natural gas	compressed natural gas
Main components	propane + butane	methane (85-99%)	methane
State and conditions	liquid under 4-15 bar at ambient temperature	liquid at −162 °C, about 1 atm	gas compressed to 200-250 bar
Source	refinery and gas processing (NGL)	natural gas liquefaction	natural gas
Volume reduction	about 1/250	about 1/600	about 1/200-1/250
Density relative to air	heavier (collects at floor)	vapour lighter when warm (cold vapour first heavier)	lighter (disperses upward)
Storage	steel cylinders/bullets	cryogenic insulated tanks	high-pressure cylinders
Use	cooking, autogas, industry	long-distance ocean transport, power, regasified for grid	vehicle fuel, local supply
Heating value	about 46-50 MJ/kg (highest per volume of gas)	about 50-55 MJ/kg	about 50 MJ/kg
Cost of infrastructure	low	very high	medium

LPG and NGL extraction process (2019 8c)

NGL (natural gas liquids) are the ethane, propane, butanes and natural gasoline (C_5^+) in wet gas. Extracting them makes the gas meet heating-value/dew-point specs and gives valuable products (LPG = propane + butane).

1. **Pre-treatment:** separation of free liquids, sweetening and dehydration of the wet gas.
2. **Extraction** (recovery of NGL from the gas) by one of: **absorption** (lean oil absorbs C_3^+, refrigerated lean oil); **refrigeration** (propane chilling to −20 to −40 °C); **Joule-Thomson expansion**; **cryogenic turbo-expander** (expansion to −85 to −100 °C, recovers 90%+ of ethane): the most common modern method. Residue (lean) gas goes to the pipeline after recompression.
3. **Fractionation** of the mixed NGL in a train of columns: **de-ethanizer** (ethane overhead), **de-propanizer** (propane), **de-butanizer** (butane overhead, natural gasoline bottoms), optionally a butane splitter (iso/normal).
4. **LPG product**: propane and butane are blended (e.g. 30:70), treated (caustic/molecular sieve to remove mercaptans and water), odorized, stored in bullets/spheres and filled into cylinders.

Kettle-type shell-and-tube heat exchanger (2019 7a)



Kettle-type reboiler/chiller

A shell-and-tube exchanger with an enlarged shell: the U-tube (or straight) bundle sits in the lower part of a large-diameter shell, and the space above forms a vapour disengagement zone. The heating medium (steam, hot oil, or in a chiller the process gas) flows inside the tubes. The liquid to be boiled enters the shell bottom, boils on the outside of the submerged tubes (pool boiling), vapour rises through the vapour space and leaves at the top, nearly free of droplets. A **weir** at the end keeps the bundle submerged; excess liquid overflows the weir and leaves as liquid product. Used as reboilers in distillation (e.g. de-propanizers, glycol and amine regenerators) and as propane **chillers** in gas plants and LNG refrigeration, where refrigerant boils in the shell. Advantages: easy to control, handles large vapour fractions, no separate drum; disadvantages: large shell, higher cost, long residence time (fouling of heavy liquids).

Remember: LNG: −162 °C, 1/600 volume; plant = inlet sep → acid gas → dehydration → Hg → NGL → liquefaction → N_2 rejection → storage. Cascade C3 (−35) → C2H4 (−100) → CH4 (−160). LPG = C3 + C4, heavier than air, BD 30/70.

B1 Natural gas: what it is, origin, classification, condensate, reservoirs, Bangladesh outlook Pass set Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q5(a)** [14] What is natural gas condensate? Flow diagram of separation of condensate from raw natural gas.
- **2023 Q5(b)** [11] Classify natural gas by chemical composition. Potential of unconventional gas reserves in Bangladesh.
- **2022B Q5(a)** [10] Same as 2023 5(a).
- **2022B Q5(c)** [10] Explain the origin of natural gas with a neat sketch.
- **2022 Q5(a)** [12] Same as 2023 5(a).
- **2022 Q5(b)** [13] Describe the most prevalent theory of natural gas formation with a neat sketch.
- **2021 Q1(b)** [6] Recommendations if an unconventional reservoir is found in Bangladesh, from an engineering perspective.
- **2021 Q1(c)** [8] Briefly describe the theories regarding the origin of natural gas.
- **2021 Q1(a)** [5] Is it better to burn natural gas or release it to the atmosphere without burning? Reasons.
- **2021 Q1(d)** [16] You have found a condensate gas reservoir: concerns about production and how to solve them.
- **2020 Q1(c)** [10] Same as 2023 5(a).
- **2020 Q5(a)** [6] "Natural gas is a clean burning fuel": explain.
- **2020 Q5(b)** [18] Compare saturated black oil, undersaturated black oil and volatile oil reservoirs, with figures.
- **2019 Q7(b)** [12] Phase envelope: bubble point line, dew point line, critical point, cricondenbar, cricondentherm, retrograde region.
- **2019 Q6(b)** [15] Short notes: gas quality aspect of LNG (liquefied natural gas), Wobbe index, associated and non-associated gas.

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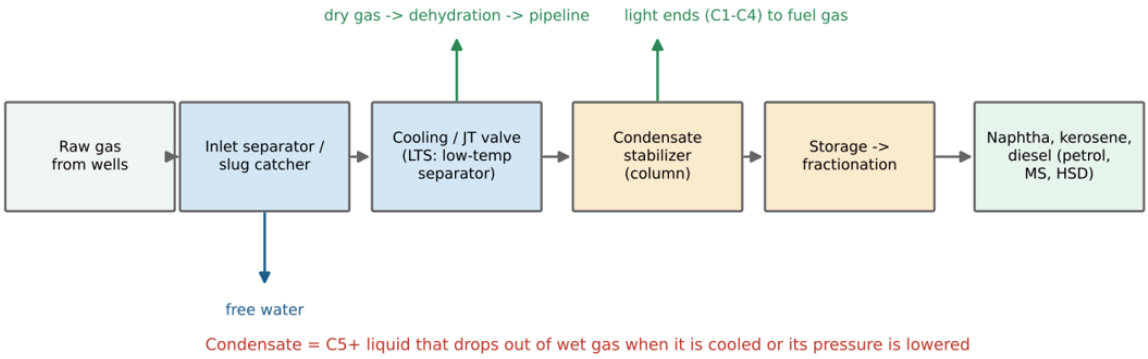
- Natural gas is mostly methane. **Associated** gas comes with oil; **non-associated** from gas-only reservoirs. **Condensate** = the heavier hydrocarbons (C_5+) that are gas in the reservoir but condense to liquid when pressure and temperature fall at the surface.
- Origin: **biogenic** (bacteria, shallow, cool, almost pure methane) and **thermogenic** (buried organic matter cooked at depth, the main theory) plus abiogenic (deep inorganic, minority view).
- Phase envelope: bubble point line (first bubble of gas), dew point line (first drop of liquid), critical point (they meet), **cricondenbar** (highest pressure with two phases), **cricondentherm** (highest temperature with two phases). Retrograde region: between critical point and cricondentherm, lowering pressure makes *more* liquid.
- Burn rather than vent: methane is ~28 times stronger a greenhouse gas than the CO_2 it burns to.

Before you start: what natural gas is and where it comes from

Natural gas is the gaseous member of the petroleum family: mostly methane CH_4, with smaller amounts of ethane, propane, butanes and heavier hydrocarbons, plus N_2, CO_2, H_2S and helium. It forms where buried organic matter is cooked deeper and hotter than the oil window, and it collects in porous rock under a seal. Source: Muhit sir's lectures L1-L2 (origin, properties, Bangladesh industry) and the condensate slides in Shameem sir's "Refinery raw materials"; reservoir types and phase envelope from Guo & Ghalambor and Ahmed (Reservoir Engineering Handbook), as the lectures do not cover them.

Natural gas condensate and its separation (2023 5a, 2022 5a, 2022B 5a, 2020 1c)

Natural gas condensate (also called condensate, gas condensate or natural gasoline) is a low-density mixture of hydrocarbon liquids (mostly C_5^+, in the gasoline range; up to 98% boils below 360 °C) that exists as gas in the raw natural gas in the reservoir and condenses out when the gas is cooled below its **hydrocarbon dew point** or its pressure is lowered. Raw gas comes from oil wells (associated gas), dry gas wells (non-associated, no liquids) and **condensate wells** (gas with natural gas liquids, "wet" gas).



Separation of condensate from raw natural gas

1. Raw gas from the wells enters an **inlet separator / slug catcher**, where free water and bulk liquid hydrocarbons are removed.
2. The gas is cooled (heat exchanger, or expansion through a choke/Joule-Thomson valve in a **low-temperature separator**, LTS (low-temperature separator)): more condensate drops out.
3. The gas goes on to dehydration and sweetening and then to the sales pipeline.
4. The liquids go to a **condensate stabilizer** (a reboiled column) that strips methane to butane (to fuel gas or LPG (liquefied petroleum gas)), leaving stable condensate with low vapour pressure.
5. Stable condensate is stored and sent to a **condensate splitter** (fractionation): LPG 3-8%, naphtha 25-75% (petrol/MS, petrochemical feed), kerosene 20-40%, gas oil (diesel/HSD) 10-30%. Water from the separators goes to treatment.

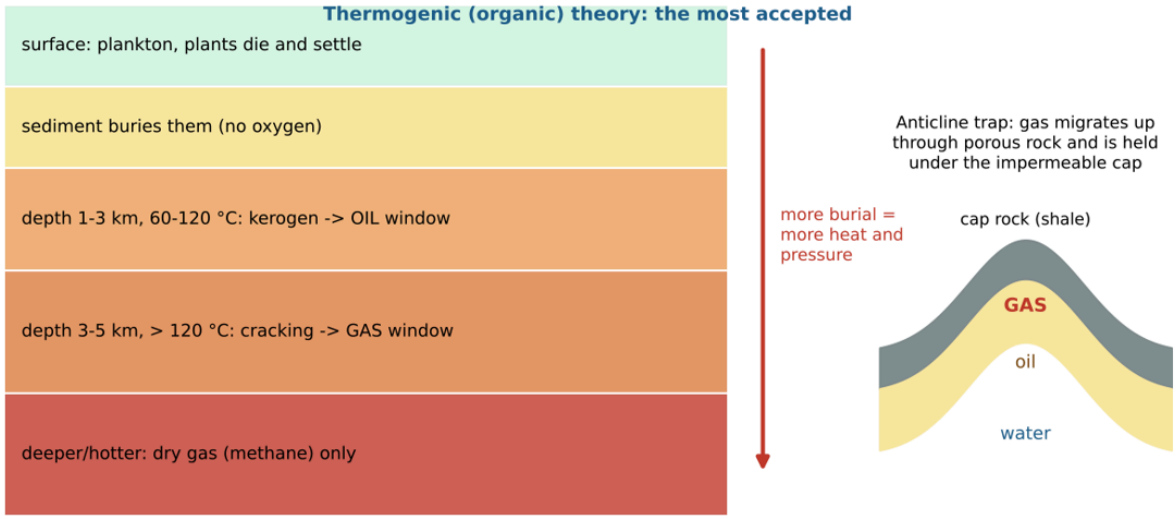
Classification of natural gas by composition; unconventional gas in Bangladesh (2023 5b, 2021 1b)

Basis	Types
Liquid content	**Dry (lean) gas**: almost pure methane, little C_2^+, no liquid on cooling. **Wet (rich) gas**: significant C_2^+ (NGL) that can be condensed.
Acid gas content	**Sweet gas**: negligible H_2S (< 4 ppm). **Sour gas**: contains H_2S (and CO_2) that must be removed.
Occurrence	**Associated gas** (with oil: gas cap or dissolved); **non-associated gas** (gas reservoir only); gas hydrates.
Reservoir type	**Conventional** (porous, permeable trap); **unconventional**: shale gas, tight gas, coal-bed methane (CBM), gas hydrates.

Unconventional gas potential in Bangladesh: the Bengal basin has thick deltaic sediments with shale layers in the north-east (Sylhet trough) and Bhola/Hatiya areas (possible shale and tight gas); coal-bed methane in the Gondwana coal of Barapukuria, Phulbari, Khalashpir and Jamalganj (deep coal, estimated trillions of cubic feet of CBM (coal-bed methane)); possible gas hydrates in the Bay of Bengal offshore. Conventional reserves are falling while demand grows, so these are seen as future resources but remain largely unexplored.

Recommendations if an unconventional reservoir is found (engineering view): first do a resource assessment (cores, logs, seismic, well tests); pilot horizontal wells with multistage hydraulic fracturing (shale/tight) or dewatering (CBM); plan water sourcing and produced-water treatment; protect aquifers (casing and cementing quality, monitoring); manage land use in a densely populated country; build gathering and processing links to the existing GTCL grid; use a production-sharing contract with experienced companies plus BAPEX capacity building; study economics against LNG import cost; control methane leaks and flaring.

Origin of natural gas (2022 5b, 2022B 5c, 2021 1c)



Thermogenic origin of natural gas and an anticline trap

- **Thermogenic (organic) theory, the most prevalent:** plants, animals and micro-organisms (plankton) die, settle and are covered by sediment without oxygen. Over millions of years they are buried deeper; temperature and pressure rise. The organic matter turns into kerogen; at about 60-120 °C (1-3 km) it cracks to oil (oil window); deeper and hotter (above about 120-150

°C) the carbon bonds break further into gas, finally dry methane: **thermogenic methane**. The gas migrates up through porous rock until trapped under an impermeable cap rock (anticline, fault, stratigraphic trap), with oil and water below it.
- **Biogenic theory:** methanogenic micro-organisms break down organic matter at shallow depth and low temperature (swamps, landfills, shallow sediments), producing **biogenic methane** (dry gas).
- **Abiogenic (inorganic) theory:** methane formed deep in the Earth's mantle from carbon and hydrogen without living matter (e.g. reaction of CO_2 with hydrogen, serpentinisation) and migrated up; a minority view.

Burn it or vent it? Natural gas as a clean fuel (2021 1a, 2020 5a)

It is better to burn (flare) unwanted gas than release it. Methane is a greenhouse gas about 28-34 times stronger than CO_2 over 100 years (over 80 times over 20 years); burning turns one CH_4 into one CO_2 ($CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$), cutting the warming effect greatly. Venting also creates a fire and explosion risk around the site and releases toxic H_2S. Best of all is to use or re-inject the gas rather than flare.

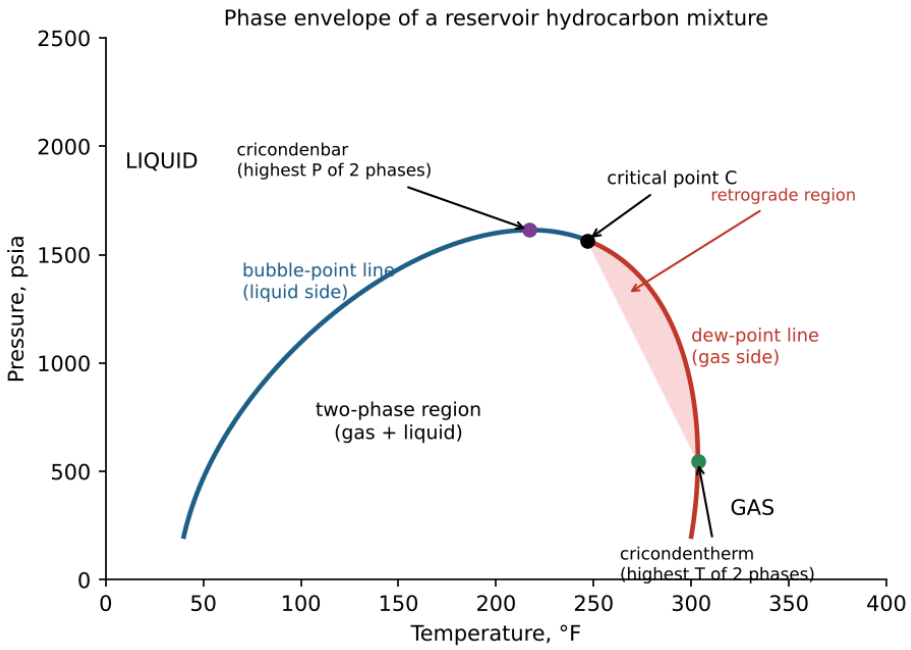
Natural gas is a clean burning fuel because methane has the highest H/C ratio of any hydrocarbon, so it gives the least CO_2 per unit energy (about 50 kg CO_2/GJ vs about 95 for coal); after processing it contains almost no sulfur (negligible SO_2), no ash or particulates, little NO_x, and burns completely with little CO or soot.

Producing a condensate gas reservoir: concerns and solutions (2021 1d)

A gas-condensate reservoir lies between the critical point and the cricondentherm on the phase envelope (see the figure below). When pressure falls below the dew point during production, liquid drops out in the rock: **retrograde condensation**.

Concern	Solution as an energy engineer
Condensate banking near the well: liquid blocks pores, gas relative permeability falls, well productivity drops	keep bottom-hole pressure above the dew point; hydraulic fracturing; solvent or methanol treatments; gas injection to revaporise
Valuable condensate left in the reservoir (lost liquid recovery)	**gas cycling**: strip liquids at surface, re-inject the dry gas to maintain pressure above the dew point; water flooding
Changing gas-condensate ratio and liquid surges at surface	properly sized separators and slug catchers, low-temperature separation, stabilizer
Hydrates and water in flowlines	dehydration, methanol/glycol injection, heating
Corrosion from CO_2/H_2S with water	corrosion-resistant alloys, inhibitors, sweetening

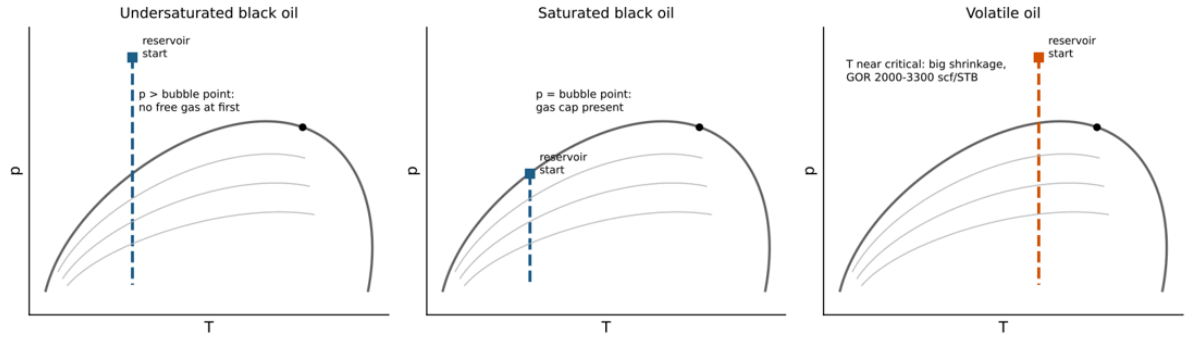
Phase envelope (2019 7b)



Pressure-temperature phase envelope of a hydrocarbon mixture

Term	Meaning
Bubble-point line	line on which the first bubble of gas appears when pressure is reduced on a liquid; left (liquid) side of the envelope
Dew-point line	line on which the first drop of liquid appears from a gas; right (gas) side
Critical point C	where the bubble-point and dew-point lines meet; liquid and gas properties become identical
Cricondenbar	the highest pressure at which liquid and vapour can coexist
Cricondentherm	the highest temperature at which liquid and vapour can coexist
Retrograde region	the area between the critical point and the cricondentherm, inside the envelope, where lowering the pressure at constant temperature **condenses** liquid (the opposite of normal behaviour)
Quality lines	lines inside the envelope of constant liquid volume percent

Saturated black oil vs undersaturated black oil vs volatile oil reservoirs (2020 5b)



Reservoir conditions relative to the phase envelope

	Undersaturated black oil	Saturated black oil	Volatile oil
Initial pressure	above the bubble point	equal to the bubble point	usually above the bubble point
Reservoir temperature	well below the critical temperature	well below the critical temperature	close to (just below) the critical temperature
Free gas initially	none, all gas dissolved	gas cap may be present	none
GOR (gas-oil ratio), scf/STB	< about 2000	< about 2000	2000-3300
Stock-tank oil	dark, API < 45	dark, API < 45	lighter colour, API > 40
B_o	< 2 bbl/STB	< 2 bbl/STB	> 2 bbl/STB (high shrinkage)
Behaviour on production	pressure falls fast (only rock and liquid expansion) until p_b, then gas comes out	gas comes out as soon as pressure drops; gas cap expands	large amounts of gas come out just below p_b; quality lines crowded near the bubble-point line

Short notes: gas quality aspect of LNG, Wobbe index, associated and non-associated gas (2019 6b)

- **Gas quality of LNG:** LNG must meet buyer specs: heating value (GHV) and Wobbe index within a band, methane number (knock resistance for engines), limits on N_2 (< 1 mol%), CO_2 (< 50 ppm, would freeze), H_2S (< 4 ppm), total sulfur, mercury (< 0.01 µg/Nm^3), water (< 1 ppm) and C_5^+. LNG "ages" in storage because methane and N_2 boil off first, so its heating value rises; rich LNG may need N_2 ballasting or LPG removal.
- **Wobbe index** = higher heating value / √(specific gravity of the gas): WI = HHV (higher heating value)/$\sqrt{\gamma_g}$. Gases with the same Wobbe index deliver the same heat through the same burner orifice at the same pressure, so it measures interchangeability of fuel gases.
- **Associated gas:** produced with crude oil, either free (gas cap) or dissolved in the oil; usually wet (rich in C_2^+). **Non-associated gas:** from reservoirs containing gas only (dry gas wells, or gas-condensate wells), usually mostly methane.

Remember: Condensate = C_5^+ that drops out below the hydrocarbon dew point: inlet separator → LTS → stabilizer → splitter. Origin: thermogenic (most accepted), biogenic, abiogenic. Phase envelope: bubble line, dew line, C, cricondenbar (max p), cricondentherm (max T), retrograde zone.

B3 Gas-liquid separators: vertical, horizontal, double-barrel, spherical, three-phase; internals; sizing

Pass set · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2023 Q6(a)** [12] Explain the working mechanism of a horizontal separator.
- **2023 Q6(b)** [5] Functions of de-foaming plates.
- **2023 Q6(c)** [18] Separator at 1000 psig, 7 MMscfd, GLR (gas-liquid ratio) 40 bbl/MMscf: size vertical, horizontal single-tube and spherical separators (ρ_L 52 lbm/ft^3, γ_g 0.8, 110 °F, t = 3 min, 1/4 full).
- **2022B Q6(a)** [15] Design and operation of a double-barrel horizontal separator with a figure.
- **2022B Q6(b)** [10] Functions of de-foaming plates and vortex breaker.
- **2022 Q6(a)** [15] Design and operation of a vertical separator with a figure.
- **2022 Q6(b)** [10] Functions of centrifugal inlet diverter and wave breakers.
- **2021 Q3(b)** [15] Gas production fell; which separator would you choose to replace a horizontal one to cut operating cost? Schematic; advantages and disadvantages.
- **2021 Q3(a)** [10] What is vortex? Why is it undesirable in gas-liquid separation and how is it avoided?
- **2020 Q6(a)** [10] Briefly describe a conventional gas-liquid separator with a schematic.
- **2019 Q8(a)** [15] Schematic of a three-phase separator; its operation in petroleum production.

★ REMEMBER IT

- Every separator has four sections: **inlet diverter** (sudden change of direction does the bulk split) → **gravity settling** (liquid drops fall out of the gas) → **mist extractor** (wire mesh/vanes catch the fine mist) → **liquid collection** (gas bubbles rise out of the liquid).
- Vertical: small footprint, handles sand and slugs, good for low GLR (more liquid). Horizontal: large gas-liquid interface, cheaper for high gas rates and foaming. Spherical: compact and cheap, little surge capacity. Double-barrel: liquid drains to a lower barrel so it cannot be re-entrained.
- Internals: de-foaming plates (break foam), vortex breaker (stops a whirlpool sucking gas into the liquid outlet), wave breakers (stop sloshing in long horizontal vessels), centrifugal inlet diverter (spins the stream to throw liquid out).

Before you start: how a separator works

A well stream is a fast, turbulent mixture of gas, oil (condensate), water and solids. A separator is a pressure vessel that gives the mixture time and space: the stream is slowed down by hitting an inlet device, liquid falls by gravity while gas rises or flows over it, a mist extractor catches the tiny droplets, and controllers hold the pressure and the liquid level. Source: Muhit sir's lectures L3-L7 and Sanjay Kumar, *Gas Production Engineering*, ch. 4 (sizing).

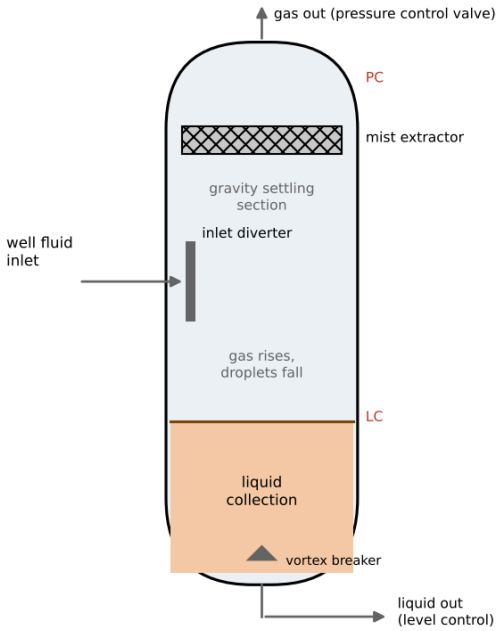
A conventional gas-liquid separator (2020 6a): sections and principles

1. **Primary (inlet) section:** an inlet diverter (baffle plate or centrifugal/tangential inlet) changes the momentum of the stream suddenly; the bulk of the liquid separates here.
2. **Gravity settling section:** large volume with low gas velocity and little turbulence, where droplets fall out of the gas.
3. **Liquid collection section:** gives retention time (1-5 min) so gas bubbles escape from the liquid; holds surges (slugs).

4. **Mist extractor:** wire mesh, vanes or plates near the gas outlet that coalesce droplets too small to settle (down to about 10 µm).
5. **Controls:** level controller and liquid dump valve; gas back-pressure valve; safety relief valve, pressure gauge, gauge glass.

Separation relies on **momentum** (change of direction), **gravity** (density difference) and **coalescence**. Factors: pressure, temperature and composition; higher pressure and lower temperature give more liquid up to an optimum. **Low-temperature separation** (LTS): the pressure drop across a choke cools the gas by the Joule-Thomson effect (operated 0 to 20 °F, not below −20 °F because carbon steel becomes brittle), recovering more condensate.

Vertical separator: design and operation (2022 6a)

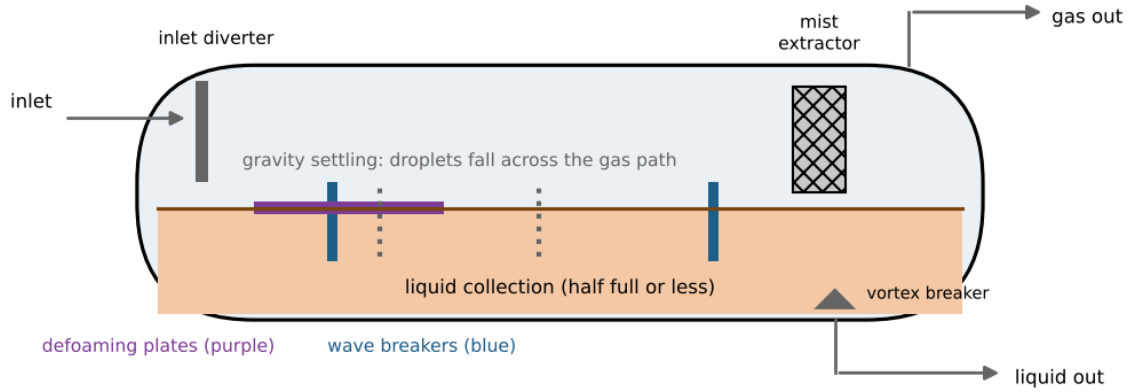


Vertical two-phase separator

The well stream enters tangentially through an inlet diverter, where gravity settling, centrifugal force and impingement give primary separation. Gas flows upward through the settling section; liquid falls to the collection section, separated from the inlet zone by a conical baffle that keeps a calm liquid surface for level control and lets dissolved gas escape. Centrifugal baffles near the top and the mist extractor at the gas outlet remove fine droplets. Liquid leaves at the bottom through the level-controlled dump valve; gas leaves at the top through the pressure control valve. A vortex breaker sits over the liquid outlet.

Advantages: handles large liquid slugs and sand without carry-over; better surge control; level control not critical; little re-vaporisation; small floor space (offshore). **Disadvantages:** more expensive to build and transport; larger diameter for the same gas capacity; harder to reach instruments on top.

Horizontal separator: working mechanism (2023 6a)

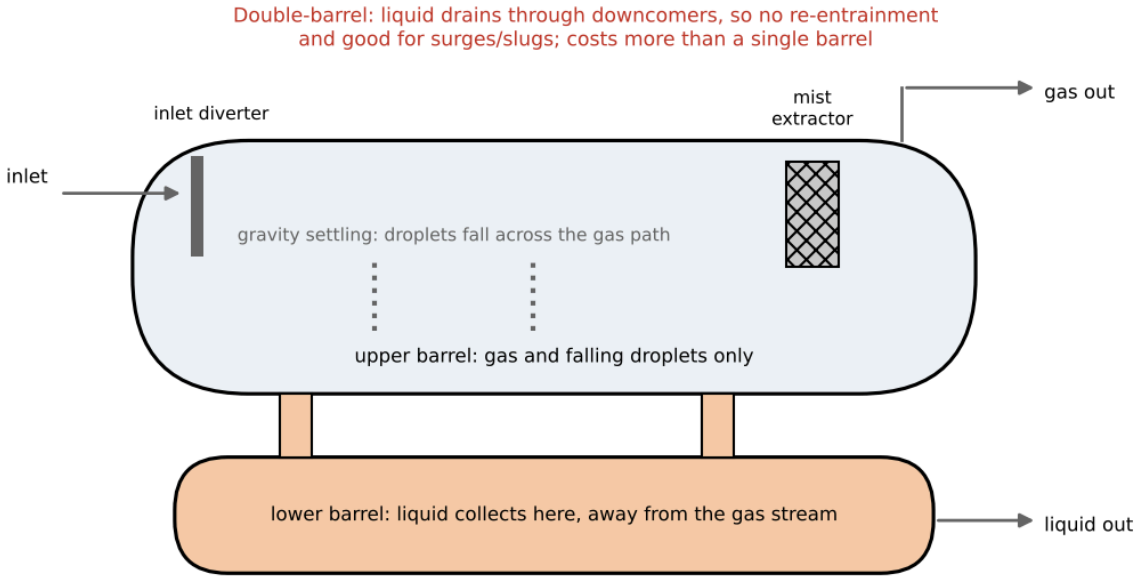


Horizontal single-tube separator with internals

1. The fluid hits the inlet diverter: sudden momentum change and gross separation of liquid and vapour.
2. Liquid falls to the bottom collection section, which gives retention time for entrained gas to escape and reach equilibrium; it leaves through the dump valve controlled by the level controller.
3. Gas and mist flow horizontally over the liquid through the gravity settling section; droplets fall to the gas-liquid interface.
4. Very small droplets are removed in the mist extractor (vanes, wire mesh, plates) before the gas leaves.
5. The pressure controller on the gas outlet opens or closes the pressure control valve to hold vessel pressure.
6. Usually operated half full of liquid to maximise interface area.

Advantages: smaller and cheaper than vertical for the same gas and liquid rate (high GOR (gas-oil ratio) gas, foaming crude); larger interface area so gas leaves the liquid easily; greater liquid capacity. **Disadvantages:** poor with solids (several drains needed); needs more plan area; less surge capacity; level control more critical.

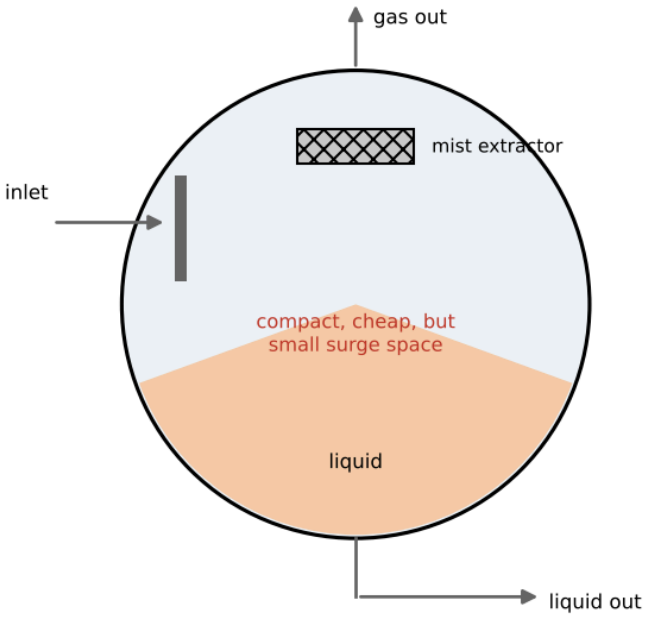
Double-barrel horizontal separator (2022B 6a)



Double-barrel horizontal separator

The stream enters the upper barrel and strikes the inlet diverter. Gas flows through the gravity settling section and baffle-type mist extractors to the gas outlet. Free liquid falls through flow pipes (downcomers) into the lower barrel. Small amounts of gas freed in the lower barrel rise back through the flow pipes. Because the liquid is held in a separate barrel, high gas velocities cannot re-entrain it at the interface. Good for high gas rates with liquid surges/slugs; costs more than a single-barrel unit.

Spherical separator



Spherical separator

Fluid enters through an inlet diverter that splits the stream; liquid falls through openings in a horizontal plate slightly below the interface into the collection section (thin liquid layer helps gas escape); gas rises through the mist extractor to the outlet. Level held by a float and dump valve,

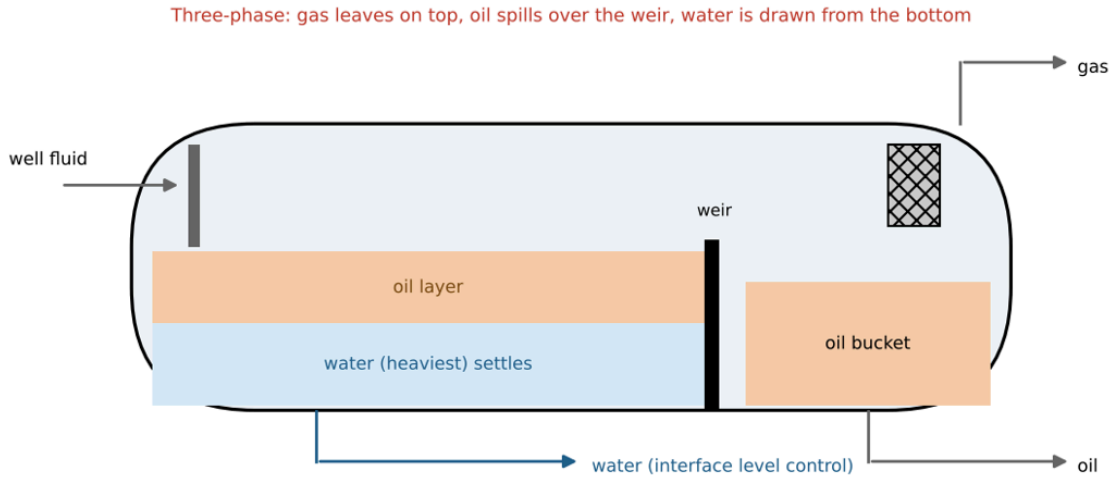
pressure by a back-pressure valve. Compact and cheapest, easy to transport, but very limited surge capacity and settling space: used for low liquid rates and moderate gas rates.

Which separator when gas production falls? (2021 3b)

When gas rate falls, a large horizontal separator is oversized and costly to maintain. Choose a **spherical separator** (or a small vertical one if the well makes sand or slugs): it is the cheapest to buy, install and operate for low-to-moderate gas and low liquid rates, compact and easy to move. Draw the spherical schematic above.

Spherical vs present horizontal: advantages	Disadvantages
lowest capital and operating cost; compact; easy to clean and transport	very little surge capacity: liquid slugs can flood it
good for low/moderate gas rates and high pressure	limited settling section and liquid capacity
fewer internals, simple to maintain	difficult level control (small liquid space)

Three-phase separator (2019 8a)



Horizontal three-phase separator with weir

Splits a well stream into gas, oil and water in three stages. **Primary:** fluid hits the inlet diverter; bulk gas separates and leaves through the mist extractor and pressure control valve. **Secondary:** liquid (oil, water, emulsion) settles in the collection section; free water drops to the bottom, oil and emulsion float above. A **weir** holds the oil level: oil spills over it into the oil bucket and leaves through a level-controlled dump valve; water is drawn from the bottom through a dump valve on an **interface level controller**. **Tertiary:** the emulsion layer is broken with heat and demulsifier chemicals so oil and water separate. In production it lets oil go to storage/sale, water to disposal and gas to processing, right at the well site.

Internals: de-foaming plates, vortex breaker, centrifugal inlet diverter, wave breakers (2023 6b, 2022 6b, 2022B 6b, 2021 3a)

Internal	Function
De-foaming plates	closely spaced inclined parallel plates (or tubes) at the gas-liquid interface; foam bubbles coalesce on them and break, so foam does not fill the settling space and carry liquid out with the gas
Vortex breaker	a cross or plate over the liquid outlet; stops a swirling vortex forming as liquid drains, so gas is not sucked into the liquid outlet
Centrifugal inlet diverter	the inlet stream enters a cyclone-like cylinder; centrifugal force flings liquid to the wall where it drains; strong primary separation with less agitation and foam than a baffle
Wave breakers	vertical perforated baffles across a long horizontal vessel, perpendicular to flow; damp waves and sloshing of the liquid (from slugs or motion offshore) that would upset level control and re-entrain liquid
Mist extractor	wire mesh or vane pack that coalesces fine droplets before the gas outlet

Vortex (2021 3a): a vortex is a whirlpool that forms at the liquid outlet as the liquid drains, like water going down a sink. It is undesirable because it pulls gas down into the liquid outlet (gas blow-by), which upsets level control, causes gas carry-under into the liquid line and pumps (cavitation), and can re-entrain liquid. Avoid it by fitting a vortex breaker, keeping enough liquid height above the outlet, and limiting outlet velocity (larger nozzle).

Separator sizing numerical (2023 6c)

Given: p = 1000 psig = 1014.7 psia; q_g = 7 MMscfd; GLR = 40 bbl/MMscf; ρ_L = 52 lbm/ft^3; γ_g = 0.8; T = 110 °F = 570 °R; t = 3 min; 1/4 full of liquid; take z = 1 (ideal gas, as in the textbook example). Method: Sanjay Kumar Example 4-1 (Souders-Brown with K values for each separator type).

Step (write this)	Why / what it means
$\rho_g = pM/(zRT) = 1014.7 \times 28.97 \times 0.8/(1 \times 10.73 \times 570) = 3.845$ lbm/ft^3	gas density
$D^2 = q_g Z(T + 460)\rho_g^{0.5}/[2.40 K_p(\rho_L - \rho_g)^{0.5}] = 7 \times 1 \times 570 \times 3.845^{0.5}/[2.40 \times K \times 1014.7 \times 48.155^{0.5}] = 0.4630/K$	gas capacity: D in ft, q_g in MMscfd
$V_L = Wt/1440 = 40 \times 7 \times 3/1440 = 0.583$ bbl	liquid volume to hold: W = 280 bbl/day liquid

(i) Vertical (K = 0.167):

Step (write this)	Why / what it means
$D = (0.4630/0.167)^{0.5} = 1.665$ ft ≈ 20 in	gas capacity sets D
$V_L = 0.1399D^2h \Rightarrow h = 0.583/(0.1399 \times 2.772) = 1.50$ ft	liquid height needed (0.1399 = π/4 ÷ 5.615)
rule: vertical separator at least 6 ft tall (feed 2 ft from bottom and 4 ft from top)	1.5 ft is too short
Vertical: 20 in × 6 ft (L/D = 3.6, within 3-8)	

(ii) Horizontal single-tube (K = 0.382):

Step (write this)	Why / what it means
$D = (0.4630/0.382)^{0.5} = 1.10$ ft $= 13.2$ in	gas capacity
minimum horizontal diameter 26 in = 2.17 ft (feed 10 in from bottom, 16 in from top)	so D = 26 in
1/4 full: $V_L = 0.1399D^2L/4 \Rightarrow L = 4 \times 0.583/(0.1399 \times 2.17^2) = 3.55$ ft	liquid capacity
$L/D = 3.55/2.17 = 1.6 < 3$, so use $L = 3D = 6.5$ ft	design rule L/D 3-8
Horizontal: 26 in × 6.5 ft	

(iii) Spherical (K = 0.35):

Step (write this)	Why / what it means
Gas: $D = (0.4630/0.35)^{0.5} = 1.15$ ft	gas capacity
Liquid, 1/4 full: $V_L = 0.0233D^3(D/2)^{0.5} \Rightarrow D^{3.5} = 0.583 \times 2^{0.5}/0.0233 = 35.4$	half the book's half-full coefficient 0.0466
$D = 35.4^{1/3.5} = 2.77$ ft ≈ 33-34 in	liquid capacity governs
Spherical: about 34 in (2.8 ft) diameter	

Flag: the textbook example uses 1000 psia and 1/2 full (answers 20 in × 6 ft, 26 in × 6.5 ft, 27 in sphere). The exam says 1000 psig and 1/4 full; with these the vertical and horizontal sizes stay the same, and the sphere grows to about 34 in because it must hold the same liquid in a quarter of its volume. If your teacher treats "1/4 full" like the textbook half-full formula, the sphere is 27 in; state the assumption.

Remember: Separator = inlet diverter (momentum) + settling (gravity) + mist extractor (coalescence) + controls. Vertical: slugs, sand, small footprint. Horizontal: cheaper, high GOR, half full. $D^2 = 0.463/K$ here; K 0.167 vertical, 0.382 horizontal, 0.35 sphere; $V_L = Wt/1440$.

B4 Water in natural gas: dehydration, water content (Campbell), hydrates

Pass set | Very likely · 100% | Medium

PAST-PAPER QUESTIONS

- **2023 Q7(c)** [15] Water content of a gas at 1000 psia, 100 °F by Campbell's method: CH_4 80, C_2H_6 5, C_3H_8 1.5, nC_4 0.5, CO_2 2.5, N_2 2.0, H_2S 8.5 %.

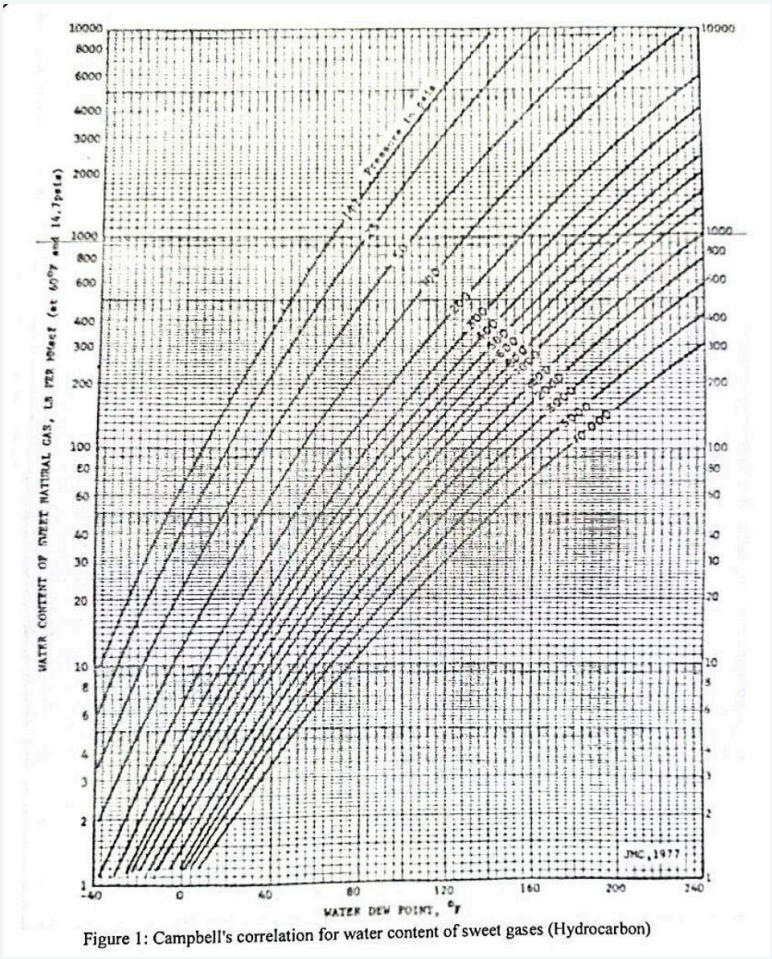


Figure 1: Campbell's correlation for water content of sweet gases (Hydrocarbon)

- **2022B Q5(b)** [5] Same as 2022 7(b).
- **2022B Q8(a)** [8] Reasons for removing water from natural gas; on what does water content depend?
- **2022B Q8(c)** [10] Same as 2023 7(c).
- **2022 Q7(b)** [8] Why is dehydration a must in natural gas processing?

- **2022 Q8(c)** [10] Same as 2023 7(c) (charts given).

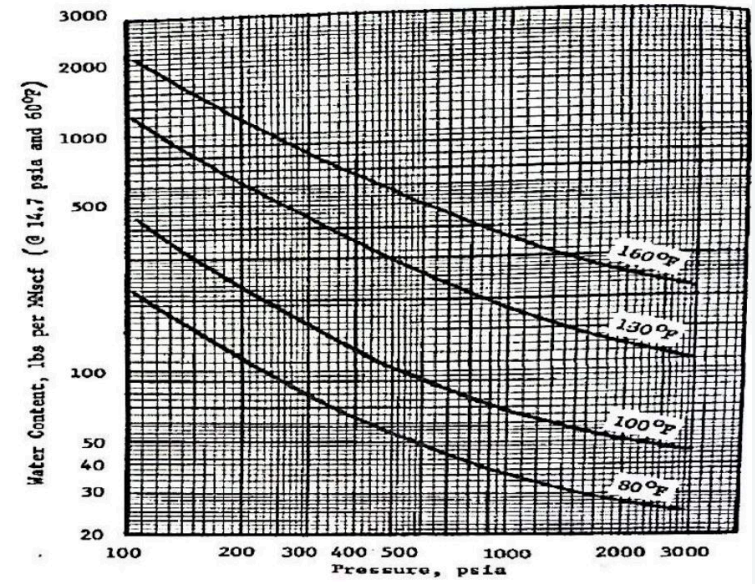


Figure 2: water content of CO_2 in saturated natural gas mixtures

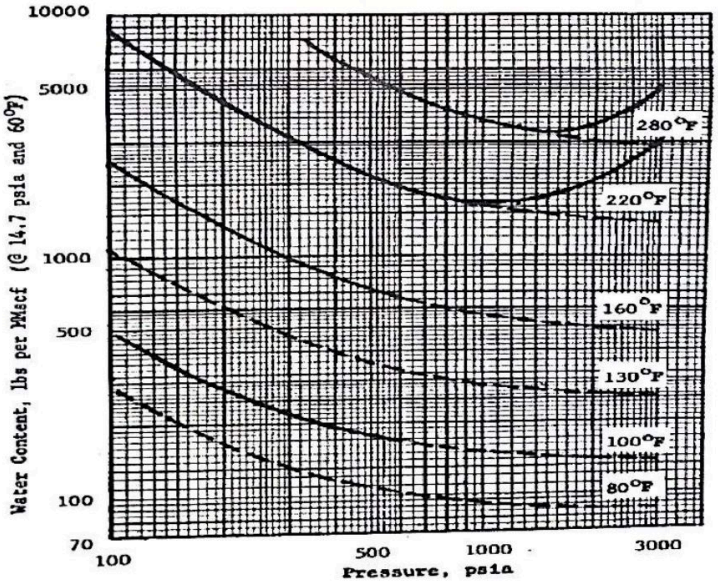


Figure 3: water content of H_2S in saturated natural gas mixtures

- **2021 Q4(a)** [15] Steps to avoid hydrate formation in pipelines, with the gas-water-hydrate equilibrium diagram.
- **2021 Q4(b)** [10] Alumina, silica gel, molecular sieve: which for sour gas, low cost, 12 ppm? Comparative analysis.
- **2020 Q5(c)** [6] Same as 2022 7(b).
- **2020 Q4(a)** [15] Explain natural gas hydrates with a diagram.
- **2019 Q7(c)** [10] Flow assurance in natural gas transport; what hydrates are and why they form.
- **2019 Q8(b)** [10] Natural gas dehydration by adsorption with a schematic.

★ REMEMBER IT

- Why dehydrate: water forms **hydrates** (ice-like cages of water around gas molecules that plug pipes at high pressure and low temperature), causes corrosion with CO_2/H_2S, cuts heating value and pipeline capacity. Spec: about 7 lb per MMscf.

- Water content rises with temperature and falls with pressure; H_2S and CO_2 hold more water than hydrocarbons (Campbell's method weights them separately).
- Avoid hydrates: keep T above the hydrate line (heating, insulation), lower pressure, remove water (glycol absorption TEG (triethylene glycol), solid desiccants), or inject inhibitors (methanol, MEG). Desiccants: silica gel (cheap), alumina, molecular sieves (lowest dew point, ppm level, tolerate sour gas best).

Before you start: why water matters in gas

Gas leaving the reservoir is saturated with water vapour (typically 400-500 lb per million standard cubic feet). As the gas cools in a pipeline, the vapour condenses. Liquid water plus gas at high pressure and low temperature forms **hydrates**, ice-like solids that plug lines; with CO_2/H_2S it forms acids. **Dehydration** removes the vapour so the dew point stays below the coldest pipeline temperature. Source: Muhit sir's lectures L8-L10 and Sanjay Kumar ch. 5 (water content charts).

Why dehydration is a must; what water content depends on (2022 7b, 2022B 5b, 2022B 8a, 2020 5c)

1. Liquid water and gas form **hydrates** that plug pipelines, valves and equipment.
2. Liquid water causes **slugging** flow and lowers pipeline flow efficiency.
3. Gas with CO_2 and/or H_2S is **corrosive** when liquid water is present (carbonic acid, sulfide stress cracking).
4. Water lowers the **heating value** of the gas.
5. Free water freezes in cryogenic (LNG, NGL (natural gas liquids)) plants.

So pipeline specs limit water to **6-8 lb/MMscf** (7 lb/MMscf). **Water content depends on:** (i) **pressure**: falls as pressure rises; (ii) **temperature**: rises with temperature; (iii) **salinity** of the reservoir water: falls as salt content rises; (iv) **gas composition**: higher-gravity gases hold less water, while CO_2 and H_2S hold more.

Water content by Campbell's method (2023 7c, 2022 8c, 2022B 8c)

Water content of a sour gas: $W = y_{HC}W_{HC} + y_{CO2}W_{CO2} + y_{H2S}W_{H2S}$, where each W is read from its own chart at the given p and T, and y_{HC} includes N_2 and all hydrocarbons.

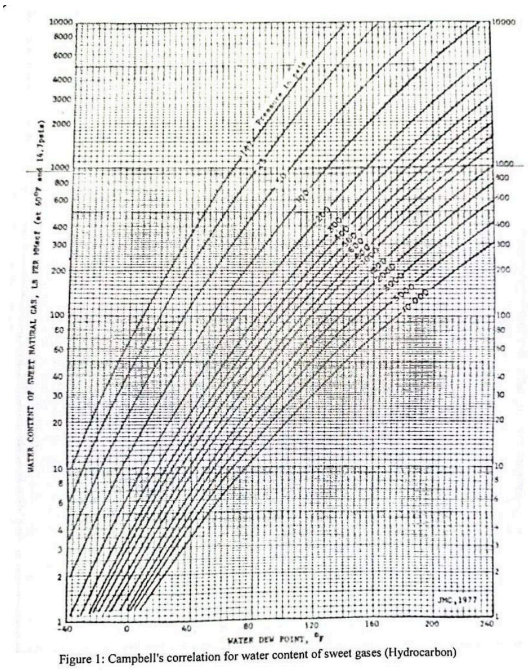


Figure 1: Campbell's correlation for water content of sweet gases (Hydrocarbon)

Figure 1 given with the paper: water content of sweet (hydrocarbon) gas vs dew point and pressure

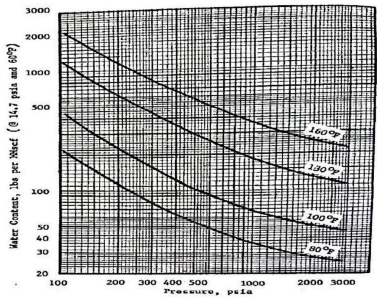


Figure 2: water content of CO_2 in saturated natural gas mixtures

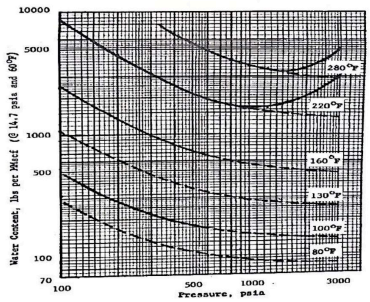


Figure 3: water content of H_2S in saturated natural gas mixtures

Figures 2 and 3 given with the paper: water content of CO_2 and H_2S in saturated gas

Step (write this)	Why / what it means
Gas at 1000 psia, 100 °F: CH_4 80, C_2H_6 5, C_3H_8 1.5, nC_4 0.5, CO_2 2.5, N_2 2.0, H_2S 8.5 %	given
y_{HC} = 0.80 + 0.05 + 0.015 + 0.005 + 0.02 = 0.89; y_{CO2} = 0.025; y_{H2S} = 0.085	N_2 is counted with the sweet gas
Fig. 1: dew point 100 °F on the 1000 psia line → W_{HC} ≈ 59 lb/MMscf	read along x = 100 °F up to the 1000 psia curve
Fig. 2: 1000 psia on the 100 °F curve → W_{CO2} ≈ 67 lb/MMscf	
Fig. 3: 1000 psia on the 100 °F curve → W_{H2S} ≈ 150 lb/MMscf	

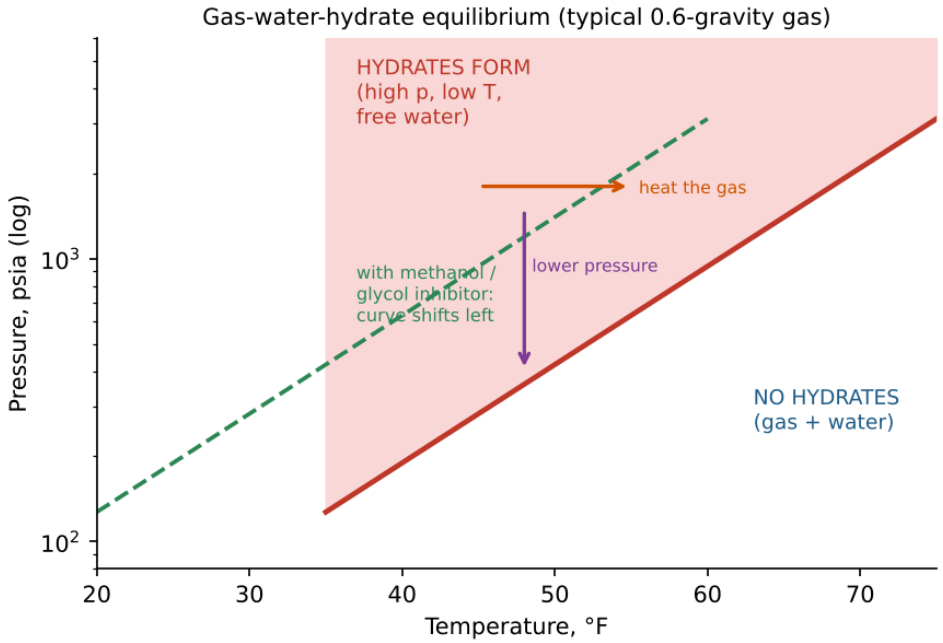
`W = 0.89×59 + 0.025×67 + 0.085×150 = 52.5 + 1.7 + 12.75`	weighted sum
`W ≈ 67 lb H2O/MMscf`	the same answer as Sanjay Kumar Example 5-2

> **Side note:** Chart readings can differ by a few lb/MMscf; show the lines you read on the chart. Note how 8.5% H_2S adds about 19% to the water content: sour gas holds more water than sweet gas.

Natural gas hydrates and why they form; flow assurance (2020 4a, 2019 7c)

Gas hydrates are solid crystalline compounds, like snow, in which water molecules (the hosts) form hydrogen-bonded cages that trap small gas molecules (guests: CH_4, C_2H_6, C_3H_8, iC_4, H_2S, CO_2, N_2). Van der Waals forces between gas and water make them stable even above 0 °C; they are less dense than ice. Formation is time-dependent: nucleation, then growth.

Conditions for formation: (i) free water; (ii) low temperature; (iii) high pressure; (iv) high velocity, agitation or pressure pulsation; (v) presence of H_2S and CO_2.



Gas-water-hydrate equilibrium curve and how inhibitors shift it

Phase diagram (lecture figure): line ABCD is the hydrate curve, HFCI the hydrocarbon vapour-pressure curve, EBFG the water freezing curve. Hydrates exist above and to the left of ABCD; none form below/right of it. Above HFCI the hydrocarbon is liquid; left of EBFG water is ice. The hydrate curve turns vertical at C, where it meets the vapour-pressure curve: C is the **maximum hydrate-forming temperature**.

Flow assurance means making sure gas (and liquids) keep flowing from reservoir to customer through the whole system, preventing blockages and flow problems: hydrates, wax and asphaltene deposits, scale, corrosion, slugging and liquid hold-up, sand erosion. For gas transport the main threat is hydrates and water.

Steps to avoid hydrates in a pipeline (2021 4a)

1. **Remove the water** (dehydration by glycol or desiccant) so the water dew point is below the lowest line temperature: no free water, no hydrate.

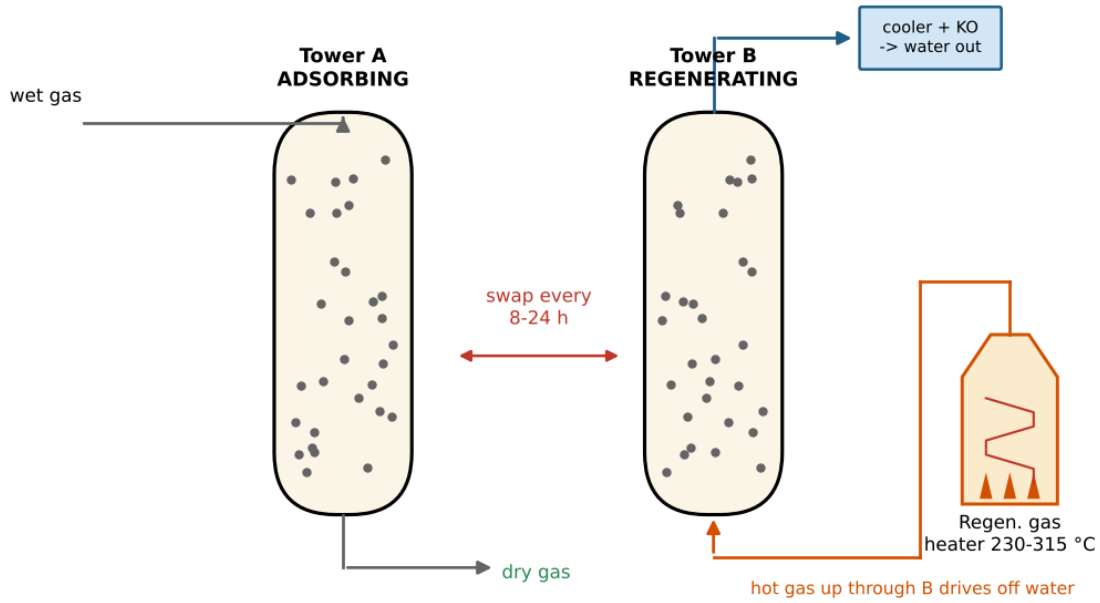
2. **Keep temperature above the hydrate temperature**: line heaters, insulation, buried or heat-traced lines (moves the operating point right of the curve).
3. **Lower the pressure** where possible (moves the point below the curve); depressurise to melt a plug from both sides.
4. **Thermodynamic inhibitors**: inject methanol or ethylene glycol (MEG), or salt solution; they shift the hydrate curve to lower temperature (Hammerschmidt equation).
5. **Kinetic hydrate inhibitors** (polyvinylpyrrolidone PVP, polyvinyl caprolactam, polyacrylamides): delay nucleation and growth at low dose.
6. **Anti-agglomerants** (quaternary ammonium or phosphonium salts): let small crystals form but keep them dispersed as a flowable slurry.
7. Avoid sudden pressure drops (Joule-Thomson cooling at chokes) and low spots where water collects; use pigging.

Choice of solid desiccant (2021 4b)

Property	Activated alumina	Silica gel	Molecular sieve (zeolite)
Lowest water content (dew point)	about −70 °C, ~ 1 ppm only when fresh	about −60 °C, 5-10 ppm	< 0.1-1 ppm, about −100 °C
Cost	cheapest	moderate	most expensive
Resistance to acid (sour) gas	poor: alkaline, reacts with acid gas	fair	good if acid-resistant grade (removes H_2S too)
Regeneration temperature	175-260 °C	170-200 °C	200-315 °C
Other	hard, resists liquid water, takes up heavy HC	breaks with free water	also adsorbs H_2S, CO_2, mercaptans; very high capacity at low humidity

Answers: (i) **Sour gas: molecular sieve** (acid-resistant type), which tolerates H_2S and can remove it with the water; alumina reacts with acid gases. (ii) **Inexpensive: activated alumina** (lowest cost per kg, long life). (iii) **Dehydrate to 12 ppm: silica gel or alumina is enough** (both reach a few ppm); silica gel is the economical choice, since 12 ppm does not need the deep drying (and cost) of molecular sieve. Molecular sieves are kept for < 1 ppm cryogenic duty.

Dehydration by adsorption with a schematic (2019 8b)

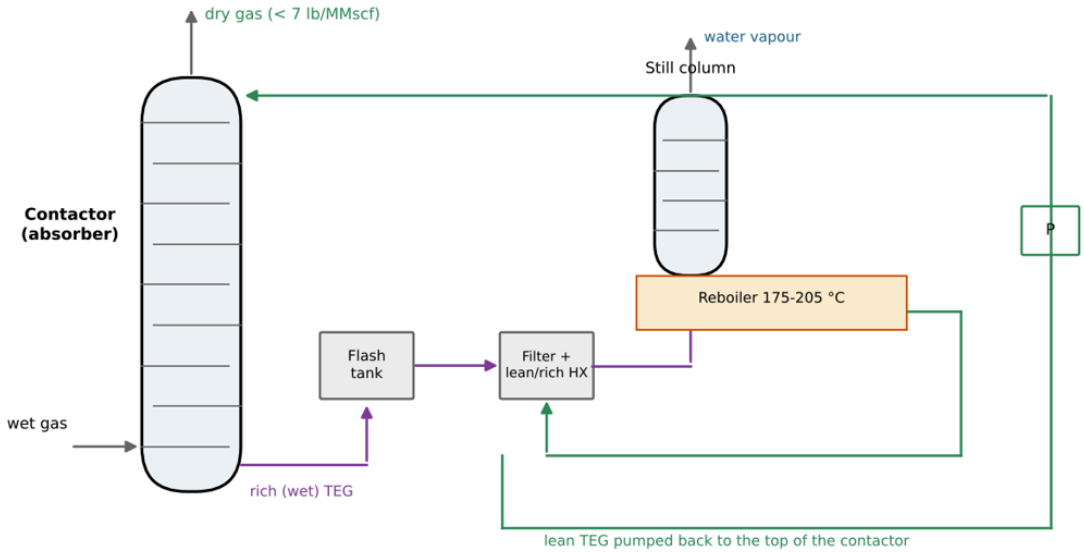


Two-tower solid desiccant dehydration plant

Adsorption is the holding of water molecules on the surface of a solid desiccant (alumina, silica gel, molecular sieve) by surface forces; these solids have huge surface areas per gram. Wet gas is first cleaned in an inlet filter-separator, then flows **downward** through tower A (down-flow avoids lifting the bed). Meanwhile tower B is regenerated: part of the gas is heated to 230-315 °C in the regeneration heater and passed **upward** through B, driving off the water (upward flow cleans the bottom of the bed, the last part dry gas touches). The bed is then cooled with unheated gas flowing downward. Hot wet regeneration gas is cooled, water knocked out, and the gas returned to the inlet. Timer-operated valves switch the towers every 8-24 h.

Advantages: very low dew point (< 1 lb/MMscf), tolerates higher temperatures and load swings, quick start-up, can also recover heavy hydrocarbons. **Limitations:** desiccant degrades and must be replaced, batch switching, sensitive to pressure surges (bed upset, channelling), attrition and fines at overload, higher cost than glycol.

Absorption (glycol) dehydration for comparison: wet gas flows up a tray contactor against lean triethylene glycol (TEG); dry gas leaves the top (below 7 lb/MMscf). Rich glycol passes through a filter, flash separator (releases dissolved gas for fuel), heat exchanger and a stripping still on a reboiler at 175-205 °C (350-400 °F), regenerating it to 99.5%; lean glycol is cooled and pumped back. Cheaper and continuous with low pressure drop, but cannot reach very low dew points.



TEG glycol dehydration unit

Remember: Dehydrate because of hydrates, slugging, corrosion, heating value; spec 6-8 lb/MMscf. Campbell: $W = y_{HC}W_{HC} + y_{CO2}W_{CO2} + y_{H2S}W_{H2S} = 67$ lb/MMscf. Hydrates need water + low T + high P (+ agitation, acid gas). Avoid: dry, heat, depressure, methanol/glycol, KHI (kinetic hydrate inhibitor), AA.

B2 Gas properties and rock properties: z-factor, pseudo-reduced properties, density, B_g, compressibility, viscosity, porosity and permeability

Pass set · Very likely · 95% · Hard

PAST-PAPER QUESTIONS

- **2023 Q5(c)** [10] Elucidate: gas formation volume factor, isothermal compressibility of gases.
- **2022B Q5(d)** [10] Porosity and permeability. Limestone: void 25 m^3, bulk 68 m^3: porosity.
- **2022 Q5(c)** [10] Porosity vs permeability. Sandstone: void 35 cm^3, bulk 83 cm^3: porosity.
- **2021 Q2(a)** [35] Gas C_1 80, C_2 3, C_3 2, nC_4 2, iC_4 1, nC_5 1, iC_5 3, nC_6 0.05, C_{7+} 1 (as nC_{10}), N_2 0.05, CO_2 2, H_2S 4 % at 180 °F, 1900 psia: z-factor and viscosity (no pressure correction).
- **2020 Q6(b)** [20] Pseudo-reduced p and T for C_1 90%, C_2 3%, C_3 2%, H_2S 2%, N_2 3% at 200 °F, 2000 psia; density and B_g with z = 0.92.

UNDERSTAND **Read this to understand; you do not need to write it**

Gases at reservoir pressure are not ideal, so we use pV = znRT. The **z-factor** comes from the Standing-Katz chart using **pseudo-reduced** properties $p_{pr} = p/p_{pc}$, $T_{pr} = T/T_{pc}$, where $p_{pc} = \Sigma y_i p_{ci}$ and $T_{pc} = \Sigma y_i T_{ci}$ (Kay's rule). The **gas formation volume factor B_g** = reservoir volume / surface volume = $0.02827zT/p$ (ft^3/scf, T in °R, p in psia).

CALCULATOR **Gas density and B_g after the chart (2020 6(b))**

Change any number; every step is recalculated and shown with the numbers put in.

1. T: **200** + 460 = **660** °R
2. p_{pr}: **2000** / **675.5** = **2.961**
3. T_{pr}: **660** / **358.9** = **1.839**
4. Gas gravity: $\gamma_g = M_a/28.97$ = **17.75** / 28.97 = **0.6125**
5. Gas density: $\rho = pM/(zRT)$, R = 10.73 = **2000** × **17.75** / (**0.92** × 10.73 × **660**) = **5.45** lbm/ft^3
6. B_g: $B_g = 0.02827zT/p$ = 0.02827 × **0.92** × **660** / **2000** = **0.00858** ft^3/scf

CALCULATOR **Porosity (2022 5(c), 2022B 5(d))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Porosity: $\varphi = V_{pore}/V_{bulk}$ = 100 × **35** / **83** = **42.2** %

Before you start: real gases and rock

An ideal gas obeys pV = nRT. Real natural gas at reservoir pressure is squeezed more than an ideal gas would be, so we write pV = **z**nRT. The **compressibility factor** z is read from the Standing-Katz chart using the **pseudo-reduced** pressure and temperature of the mixture. From z everything else follows: density, formation volume factor, compressibility. Source: Muhit sir's lecture L2 (gas properties) and the textbooks (Guo & Ghalambor ch. 2, Sanjay Kumar ch. 2, Ahmed). Field units: psia, °R = °F + 460, ft^3, lb.

$pV = znRT$, $R = 10.73$ psia·ft^3/(lb-mol·°R); $\gamma_g = M_a/28.97$; $p_{pc} = \Sigma y_i p_{ci}$, $T_{pc} = \Sigma y_i T_{ci}$; $p_{pr} = p/p_{pc}$, $T_{pr} = T/T_{pc}$

Gas formation volume factor and isothermal compressibility (2023 5c)

Gas formation volume factor B_g is the ratio of the volume a quantity of gas occupies at reservoir pressure and temperature to its volume at standard conditions (14.7 psia, 60 °F = 520 °R):

Step (write this)	Why / what it means
$B_g = V_{p,T}/V_{sc} = (zT/p)/(z_{sc}T_{sc}/p_{sc})$	same number of moles, real gas law
$= (14.7/520)\ zT/p =$ **$0.02827\ zT/p$** ft^3/scf	$z_{sc} = 1$; T in °R, p in psia
$= 0.005035\ zT/p$ bbl/scf	divide by 5.615 ft^3/bbl

B_g is much less than 1 (reservoir gas is compressed); its inverse $1/B_g$ is the gas expansion factor (scf per reservoir ft^3). It converts reservoir volumes to surface volumes in reserve calculations.

Isothermal compressibility c_g is the fractional change in volume per unit change in pressure at constant temperature: $c_g = -(1/V)(\partial V/\partial p)_T$. With $V = znRT/p$:

Step (write this)	Why / what it means
$\partial V/\partial p = nRT[(1/p)(\partial z/\partial p) - z/p^2]$	product rule on z/p
$c_g = -(p/z)[(1/p)(\partial z/\partial p) - z/p^2]$	divide by $-V = -znRT/p$
$c_g = 1/p - (1/z)(\partial z/\partial p)_T$ (psi^{-1})	for an ideal gas z = 1, so $c_g = 1/p$

It is used to describe how gas expands as reservoir pressure drops (pressure transient analysis, material balance). In reduced form $c_{pr} = c_g p_{pc} = 1/p_{pr} - (1/z)(\partial z/\partial p_{pr})$.

Pseudo-reduced properties, density and B_g (2020 6b)

Given: C_1 90%, C_2 3%, C_3 2%, H_2S 2%, N_2 3%; T = 200 °F = 660 °R; p = 2000 psia; z = 0.92.

Component	y_i	M_i	p_{ci}, psia	T_{ci}, °R	y_iM_i	y_ip_{ci}	y_iT_{ci}
C_1	0.90	16.043	667.8	343.1	14.439	601.02	308.79
C_2	0.03	30.070	707.8	549.8	0.902	21.23	16.49
C_3	0.02	44.097	616.3	665.7	0.882	12.33	13.31
H_2S	0.02	34.076	1306.0	672.4	0.682	26.12	13.45
N_2	0.03	28.013	493.0	227.3	0.840	14.79	6.82
Total	1.00				17.745	675.5	358.9

Step (write this)	Why / what it means
M_a = 17.745 lb/lb-mol; γ_g = 17.745/28.97 = 0.6125	apparent molecular weight and gas gravity
p_{pr} = 2000/675.5 = **2.961**	Kay's mixing rule
T_{pr} = 660/358.9 = **1.839**	
$\rho_g = pM_a/(zRT)$ = 2000 × 17.745/(0.92 × 10.73 × 660) = **5.45 lbm/ft^3**	z given as 0.92 (chart value at p_{pr} 2.96, T_{pr} 1.84)
B_g = 0.02827 × 0.92 × 660/2000 = **0.00858 ft^3/scf** (= 0.00153 bbl/scf)	1 scf of gas occupies 0.0086 ft^3 in the reservoir

Side note: Critical constants are from the standard table printed with the 2020 paper (GPSA values). H_2S is only 2%, so the Wichert-Aziz correction is small and the question gives z directly.

z-factor and viscosity of a sour gas (2021 2a)

Given: C_1 80, C_2 3, C_3 2, nC_4 2, iC_4 1, nC_5 1, iC_5 3, nC_6 0.05, C_{7+} 1 (as nC_{10}), N_2 0.05, CO_2 2, H_2S 4 mol%; T = 180 °F = 640 °R; p = 1900 psia. No pressure correction for viscosity.

Misprint flag: the mole percents add to 99.1, not 100. The answer below uses the fractions as printed; normalising to 100% changes z by less than 1% (0.826 instead of 0.832) and viscosity by 0.2%, shown at the end.

Component	y_i	M_i	p_{ci}	T_{ci}	y_iM_i	y_ip_{ci}	y_iT_{ci}
C_1	0.80	16.043	667.8	343.1	12.834	534.24	274.48
C_2	0.03	30.070	707.8	549.8	0.902	21.23	16.49
C_3	0.02	44.097	616.3	665.7	0.882	12.33	13.31
nC_4	0.02	58.124	550.7	765.4	1.162	11.01	15.31
iC_4	0.01	58.124	529.1	734.7	0.581	5.29	7.35
nC_5	0.01	72.151	488.6	845.4	0.722	4.89	8.45
iC_5	0.03	72.151	490.4	828.8	2.165	14.71	24.86
nC_6	0.0005	86.178	436.9	913.4	0.043	0.22	0.46
C_{7+} (nC_{10})	0.01	142.29	304.0	1111.7	1.423	3.04	11.12
N_2	0.0005	28.013	493.0	227.3	0.014	0.25	0.11
CO_2	0.02	44.010	1070.9	547.6	0.880	21.42	10.95
H_2S	0.04	34.076	1306.0	672.4	1.363	52.24	26.90
Total	0.991				22.971	680.9	409.8

Step (write this)	Why / what it means
$M_a = 22.97;\ \gamma_g = 22.97/28.97 = 0.793$	gas gravity
$p_{pc} = 680.9$ psia, $T_{pc} = 409.8$ °R	Kay's rule
Sour gas (6% acid gas): Wichert-Aziz correction. $A = y_{CO2} + y_{H2S} = 0.06$, $B = y_{H2S} = 0.04$	acid gases make Kay's rule over-predict z's deviation
$\varepsilon = 120(A^{0.9} - A^{1.6}) + 15(B^{0.5} - B^4) = 8.21 + 3.00 = 11.2$ °R	pseudo-critical temperature adjustment
$T'_{pc} = 409.8 - 11.2 = 398.6$ °R	corrected T_{pc}
$p'_{pc} = p_{pc}T'_{pc}/[T_{pc} + B(1 - B)\varepsilon] = 680.9 \times 398.6/(409.8 + 0.43) = 661.5$ psia	corrected p_{pc}
$p_{pr} = 1900/661.5 = 2.87;\ T_{pr} = 640/398.6 = 1.61$	pseudo-reduced
Standing-Katz chart (or Dranchuk-Abou-Kassem equation): **z = 0.83**	DAK gives 0.832; without the W-A correction z would be 0.814

Viscosity (Carr-Kobayashi-Burrows): read μ_1 (viscosity at 1 atm) from the chart of μ_1 vs M_a at 180 °F, then add the inset corrections for N_2, CO_2, H_2S. The chart fit (Standing) is used here:

Step (write this)	Why / what it means
$\mu_{1,uncorr} = (1.709\times10^{-5} - 2.062\times10^{-6}\gamma_g)T + 8.188\times10^{-3} - 6.15\times10^{-3}\log\gamma_g$	T in °F
$= (1.709\times10^{-5} - 1.635\times10^{-6})(180) + 0.008188 + 0.000620 = 0.01159$ cp	log 0.793 = −0.1008
N_2: $y(8.48\times10^{-3}\log\gamma + 9.59\times10^{-3}) = 0.0005 \times 0.00874 = 0.000004$ cp	

CO_2: $y(9.08\times10^{-3}\log\gamma + 6.24\times10^{-3}) = 0.02 \times 0.00533 = 0.000107$ cp	
H_2S: $y(8.49\times10^{-3}\log\gamma + 3.73\times10^{-3}) = 0.04 \times 0.00287 = 0.000115$ cp	
$\mu_g = \mu_1 = 0.01159 + 0.00023 =$ **0.0118 cp**	no pressure correction ($\mu/\mu_1 = 1$) as instructed

With the composition normalised to 100%: M_a 23.18, γ_g 0.800, p'_{pc} 667.6, T'_{pc} 402.2, p_{pr} 2.85, T_{pr} 1.59, z = 0.826, μ = 0.0118 cp. Either set of answers is acceptable if the assumption is stated.

Porosity and permeability; porosity numericals (2022 5c, 2022B 5d)

Porosity φ	Permeability k
fraction of the bulk rock volume that is pore (void) space: $\varphi = V_{pore}/V_{bulk}$	ability of the rock to let fluid flow through its connected pores
measures storage capacity	measures flow capacity
dimensionless (fraction or %)	darcy or millidarcy (md); from Darcy's law
depends on grain sorting, packing, cementation	depends on pore size and connection (connected pores only)
absolute (all pores) or effective (connected pores)	absolute, effective, relative

A rock can have high porosity but low permeability (shale: pores not connected).

> Sandstone (2022): φ = 35/83 = 0.4217 = **42.2%**
> Limestone (2022B): φ = 25/68 = 0.3676 = **36.8%**
> (Both are high for real rocks, typically 5-30%, but the method is what is marked.)

Remember: $B_g = 0.02827zT/p$ ft^3/scf; $c_g = 1/p - (1/z)dz/dp$; $\rho = pM/(zRT)$; Kay's rule for p_{pc}, T_{pc}; sour gas → Wichert-Aziz ε. Porosity = void/bulk.

B7 Reservoir flow and gas measurement: Darcy's law (linear and radial), orifice meter, choosing a meter Next to add Likely · 73% Medium

PAST-PAPER QUESTIONS

- **2023 Q8(c)** [15] Well at 600 STB/day, p_{wf} 1800 psi, k 120 md, h 25 ft, 40 acres, r_w 0.25 ft, B_o 1.25, μ_o 2.5 cp: pressure profile and drops across 1-ft intervals.
- **2022B Q6(c)** [10] Same as 2022 6(c).
- **2022B Q7(c)** [15] Same as 2023 8(c).
- **2022B Q8(b)** [7] Criteria for selecting a gas flow measurement device.
- **2022 Q6(c)** [10] Linear flow q = 0.001127kAΔp/(μL): L 2000 ft, h 20 ft, width 300 ft, k 100 md, φ 15%, μ 2 cp, p_1 2000, p_2 1990 psi: q, apparent and actual velocity.
- **2022 Q8(a)** [10] Explain the working principle of an orifice meter.

UNDERSTAND **Read this to understand; you do not need to write it**

Darcy's law is Ohm's law for rock: flow = (permeability × area × pressure drop)/(viscosity × length). In oilfield units q (bbl/day) = 0.001127kAΔp/(μL) for linear flow. Around a well the flow is radial, so pressure rises with the **logarithm** of distance: most of the pressure drop happens close to the well.

CALCULATOR **Linear Darcy flow (2022 6(c), 2022B 6(c))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Area: **300** × **20** = **6000** ft²
2. Flow rate: q = 0.001127kAΔp/(μL) = 0.001127 × **100** × **6000** × (**2000** - **1990**) / (**2** × **2000**) = **1.690** bbl/day
3. Flow rate: **1.69** × 5.615 = **9.49** ft³/day
4. Apparent velocity: **9.492** / **6000** = **0.00158** ft/day
5. Actual velocity: v_{act} = v/φ = **0.001582** / **0.15** = **0.0105** ft/day

CALCULATOR **Radial pressure around a well (2023 8(c), 2022B 7(c))**

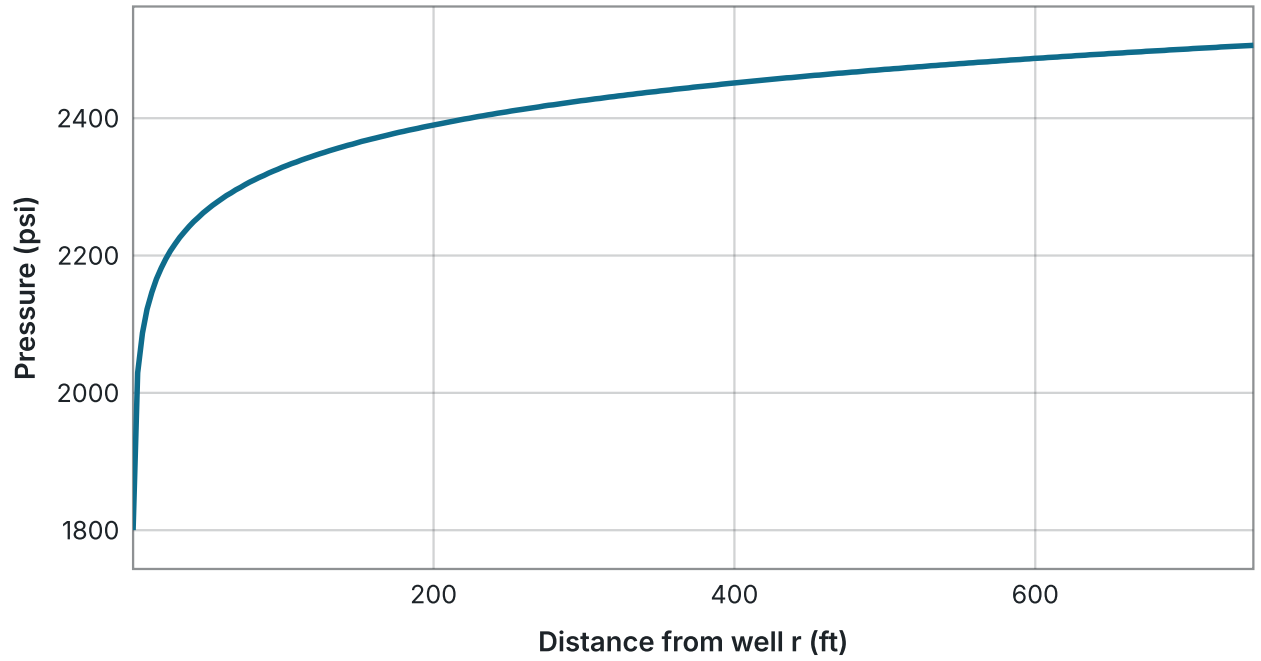
Change any number; every step is recalculated and shown with the numbers put in.

1. Coefficient: $Q_oB_o\mu/(0.00708kh)$ = **600** × **1.25** × **2.5** / (0.00708 × **120** × **25**) = **88.28** psi
2. Pressure at r: p = p_{wf} + coefficient × ln(r/r_w) = **1800** + **88.28** × ln(**100** / **0.25**) = **2328.9** psi
3. Drainage radius for 40 acres: √(40 × 43560 / π) = **745** ft

Try r = 1.25, 5, 20, 100 and 745 ft to fill the table.

INTERACTIVE **Pressure profile around the well**

Move the sliders and watch the curve change.

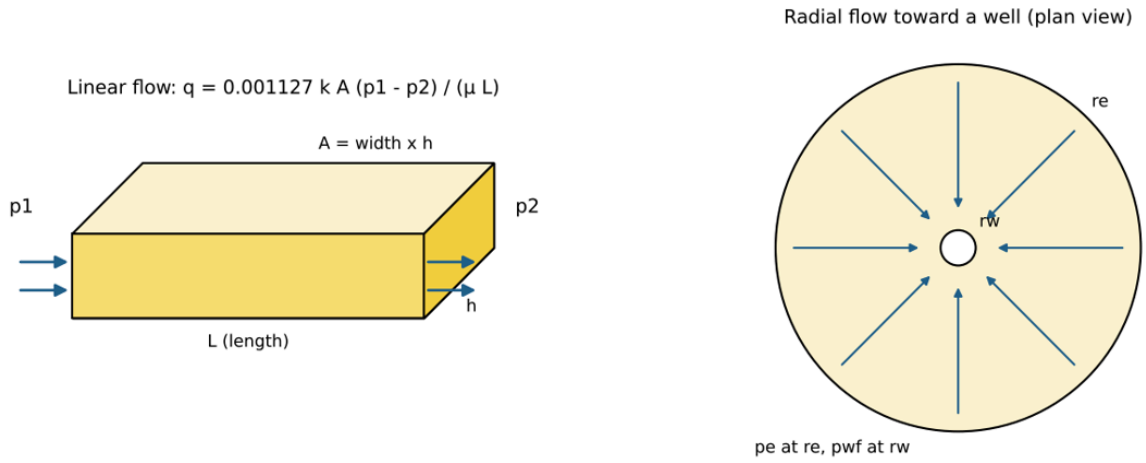


The curve rises steeply in the first few feet and is almost flat far away: damage near the well matters most.

Before you start: flow through rock

Reservoir rock is like a sponge: fluid flows through tiny connected pores. Darcy found that the flow rate is proportional to the pressure gradient and the rock's permeability, and inversely proportional to the fluid viscosity. In field units the constants 0.001127 (linear) and 0.00708 (radial) appear. Source: Muhit sir's lecture L17 (reservoir fluid flow) and Ahmed, *Reservoir Engineering Handbook* ch. 6; flow meters were "self study" in the lecture, so they follow Sanjay Kumar ch. 9 and the AGA-3 standard.

Darcy (linear, field units): $q = 0.001127\ k\ A\ (p_1 - p_2)/(\mu\ L)$ [q bbl/day, k md, A ft^2, p psi, μ cp, L ft]



Linear flow through a rock block and radial flow into a well

Linear flow numerical (2022 6c, 2022B 6c)

Given: L = 2000 ft, h = 20 ft, width = 300 ft, k = 100 md, φ = 15%, μ = 2 cp, p_1 = 2000 psi, p_2 = 1990 psi. Find flow rate, apparent and actual velocity.

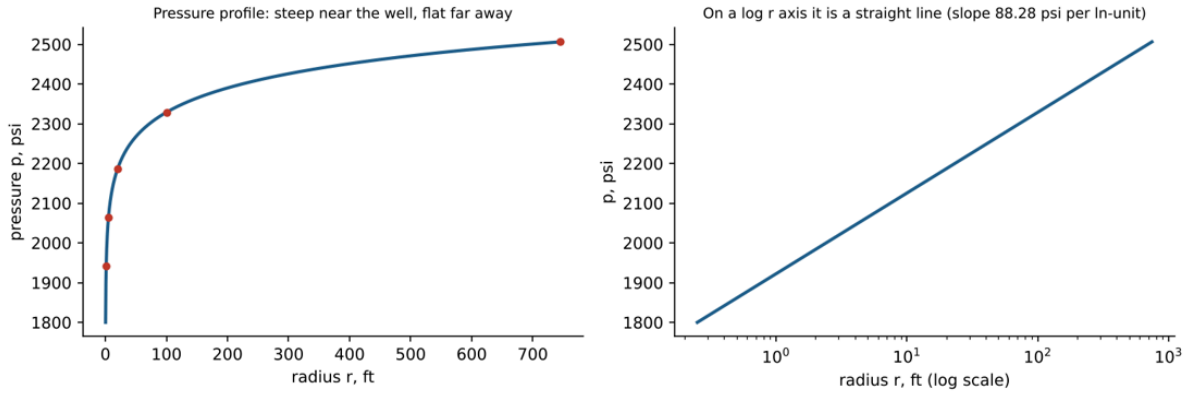
Step (write this)	Why / what it means
`A = width × h = 300 × 20 = 6000 ft²`	cross-section normal to flow
`q = 0.001127 × 100 × 6000 × (2000 - 1990)/(2 × 2000)`	substitute
`= 6762/4000 =` **`1.69 bbl/day`**	
`q in ft³/day = 1.6905 × 5.615 = 9.49 ft³/day`	1 bbl = 5.615 ft^3
`Apparent (Darcy) velocity v = q/A = 9.49/6000 =` **`0.00158 ft/day`**	as if the whole area were open
`Actual (interstitial) velocity` v_{act} `= v/φ = 0.00158/0.15 =` **`0.0105 ft/day`**	fluid only moves through the pores, so it moves faster

Radial pressure profile (2023 8c, 2022B 7c)

Given: Q_o = 600 STB/day, p_{wf} = 1800 psi, k = 120 md, h = 25 ft, drainage area 40 acres, r_w = 0.25 ft, B_o = 1.25 bbl/STB, μ_o = 2.5 cp. $p = p_{wf} + [Q_oB_o\mu_o/(0.00708kh)]\ \ln(r/r_w)$.

Step (write this)	Why / what it means
`Coefficient = 600 × 1.25 × 2.5/(0.00708 × 120 × 25) = 1875/21.24 = 88.28 psi`	constant in front of ln
`p(r) = 1800 + 88.28 ln(r/0.25)`	pressure profile
r_e`: π`r_e^2 `= 40 × 43,560 ft² ⇒` r_e `= 745 ft`	1 acre = 43,560 ft^2; the last interval 744-745 ft is at the boundary

r, ft	p, psi	Interval	Δp, psi
0.25	1800.0		
1.25	1942.1	0.25 → 1.25	142.1
4	2044.8		
5	2064.5	4 → 5	19.7
19	2182.3		
20	2186.8	19 → 20	4.53
99	2328.0		
100	2328.9	99 → 100	0.89
744	2506.1		
745	2506.2	744 → 745	0.12



Pressure profile around the well: linear and log radius scales

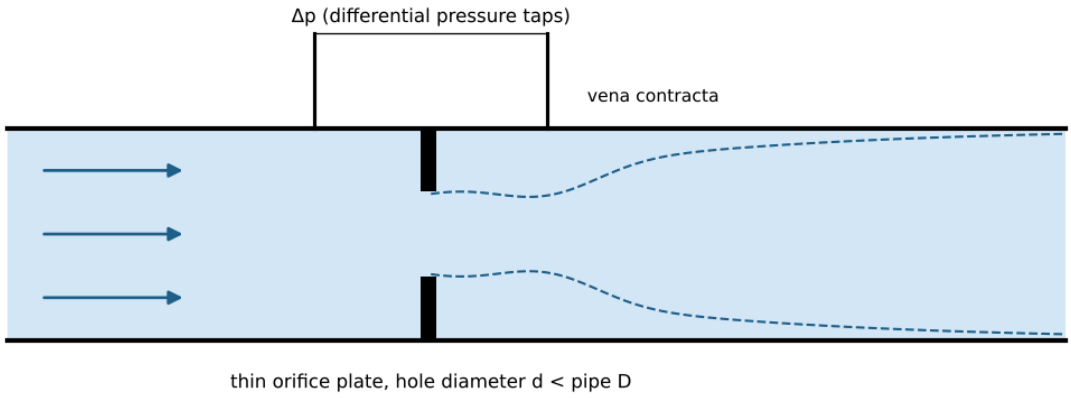
Conclusion: most of the pressure drop happens very close to the well: 142 psi in the first foot, but only 0.12 psi across a foot at the boundary. That is why damage (skin) near the wellbore matters so much and why stimulation (acidizing, fracturing) near the well pays off.

> **Side note:** The first interval is from r_w = 0.25 ft to 1.25 ft, as the question states; every other interval is 1 ft wide.

Reservoir flow classifications (theory from L17)

- **Fluids**: incompressible (density constant), slightly compressible (small change, liquids), compressible (gases).
- **Flow regimes**: steady state (∂p/∂t = 0 everywhere; strong aquifer or pressure maintenance), unsteady/transient (∂p/∂t varies with position and time), pseudo-steady state (∂p/∂t = constant everywhere; bounded reservoir).
- **Geometry**: radial (toward the well), linear (parallel paths, constant area, e.g. fractures), spherical/hemispherical (limited perforations, partial penetration).
- **Number of phases**: single, two or three-phase flow.

Orifice meter working principle (2022 8a)



Orifice meter with pressure taps

An **orifice meter** is a thin flat plate with a sharp-edged circular hole (diameter d, smaller than pipe D) clamped between flanges. As gas passes the hole, it accelerates; the stream keeps contracting just downstream to a minimum area, the **vena contracta**, where velocity is highest and pressure lowest. The pressure drop Δp between taps upstream and downstream (flange taps or pipe taps) is measured with a differential pressure transmitter. From Bernoulli and continuity between the upstream pipe and the throat:

Step (write this)	Why / what it means
$p_1 + \rho V_1^2/2 = p_2 + \rho V_2^2/2;\ V_1 A_1 = V_2 A_o$	energy and mass
$V_2 = \sqrt{[2\Delta p/(\rho(1 - \beta^4))]},\ \beta = d/D$	solve
$q = C_d A_o \sqrt{[2\Delta p/(\rho(1 - \beta^4))]}$	C_d about 0.6 corrects for vena contracta and losses

For gas, AGA Report No. 3 gives $q_h = C'\sqrt{(h_w p_f)}$, with C' combining basic orifice factor, expansion, temperature, gravity and supercompressibility factors. Flow is proportional to $\sqrt{\Delta p}$, so rangeability is only about 3:1 per plate. Advantages: simple, cheap, no moving parts, well standardised, easy to change plate. Disadvantages: high permanent pressure loss, wear of the sharp edge changes accuracy, needs long straight pipe runs.

Criteria for selecting a gas flow measurement device (2022B 8b)

1. **Accuracy and repeatability** required (custody transfer needs high accuracy).
2. **Flow range (rangeability/turndown)**: minimum and maximum flow expected.
3. **Fluid properties**: gas or liquid, clean or dirty, wet, corrosive, sour; density, viscosity, temperature, pressure.
4. **Permanent pressure loss** the system can afford (orifice high, ultrasonic almost none).
5. **Pipe size and installation**: straight-run requirements, space, line pressure rating.
6. **Cost**: purchase, installation, maintenance and calibration.
7. **Output required**: volume or mass, local reading or remote signal, totalising.
8. **Standards and regulations** (AGA-3 orifice, AGA-7 turbine, AGA-9 ultrasonic), reliability and maintenance needs.

Remember: Linear Darcy: $q = 0.001127kA\Delta p/(\mu L) = 1.69$ bbl/d; actual velocity = apparent/φ. Radial: $p = p_{wf} + 88.28 \ln(r/r_w)$, 142 psi lost in the first foot. Orifice: $q = C_d A_o \sqrt{(2\Delta p/\rho(1 - \beta^4))}$.